

# THE WALL STREET JOURNAL.

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Last week: DJIA 39142.23 ▼ 1070.48 2.66% NASDAQ 16286.45 ▼ 2.6% STOXX 600 506.42 ▲ 4.0% 10-YR. TREASURY ▲ 1 10/32, yield 4.325% OIL \$64.68 ▲ \$3.18 EURO \$1.1396 YEN 142.17

## What's News

### Business & Finance

- ◆ **Volatility in Treasury** markets and unexpected weakness in the dollar have raised fears of capital flight that would undermine America's financial primacy. **A1**
- ◆ **U.S.-China trade tensions** have resurrected within the Trump administration the idea of delisting Chinese stocks from U.S. exchanges, with potentially serious consequences for markets in both countries. **B1**
- ◆ **Trump's tariffs** are tipping the scales in the Pepsi-Coke rivalry because of where the soda makers source the concentrates that flavor their products. **B1**
- ◆ **The EU delayed** penalizing Apple and Meta Platforms just before trade talks began between the bloc and the Trump administration. **B1**
- ◆ **High-spending** U.S. travelers are helping to support United Airlines' revenue in what one consultant called "a tale of two business models." **B1**
- ◆ **Asset manager Castlelake** agreed to invest up to \$2 billion in residential mortgages through Invictus Capital Partners, a firm that specializes in residential credit. **B3**
- ◆ **Trump plans** to pull an additional \$1 billion of the school's funding for health research, showing his anger over escalating fireworks between the two sides. **A2**
- ◆ **Johnson & Johnson** has shifted its generative-AI strategy away from broad experimentation across the healthcare conglomerate to a more focused approach. **B4**

### World-Wide

- ◆ **Kyiv is under pressure** to respond to Trump administration ideas on how to end the war in Ukraine, including potential U.S. recognition of Russia's 2014 annexation of Crimea and barring Ukraine from joining NATO. **A1**
- ◆ **Defense chief Hegseth** created a Signal chat with his wife, personal lawyer and others and posted sensitive military information into it, a revelation that has added to the increasing scrutiny of the novice leader. **A3**
- ◆ **U.S. airstrikes targeted** Yemen's capital, killing 12 people and wounding 130 early on Monday. **A9**
- ◆ **Pope Francis met** briefly with Vice President JD Vance before the pontiff in his Easter message called for better treatment of migrants and other vulnerable and marginalized people. **A3**
- ◆ **Israeli soldiers erred** in killing 15 Gaza aid workers and violated orders when firing on a U.N. vehicle, the country's military said in its report on the incidents. **A6**
- ◆ **The Trump administration** further embraced the theory that the Covid-19 pandemic emanated from a laboratory in China, updating the government's main websites about the virus to make the argument. **A5**
- ◆ **Billionaire Gov. JB Pritzker** of Illinois is one of the top Democrats being watched as a potential presidential candidate as the party searches for a way out of the political wilderness. **A4**

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Alternative Energy: Where cleaner-energy policy goes now. **R1-6**

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## Pope Emerges From Convalescence on Easter



TIZIANA FABI/GETTY IMAGES

**PONTIFF'S BLESSING:** Thousands in St. Peter's Square on Sunday were treated to a surprise popemobile trip, drawing cheers and applause as Pope Francis continues his recovery from double pneumonia. He also met with Vice President JD Vance. **A3**

## Trump's Trade Offensive Hits America's Financial Primacy

Rising volatility and dollar weakness raise fears of a financial shift away from U.S.

Financial markets have flashed an ominous message this month about President Trump's ambitious and impro-

By Nick Timiraos, Jack Pitcher and Chelsey Dulaney

visational bid to reshape global trade: Be careful what you wish for.

Volatility in Treasury markets and unexpected weakness

in the dollar suggest that what began as a trade conflict could morph into a more dangerous "capital war." The clash threatens to raise U.S. borrowing costs by undermining Washington's financial primacy, which for years has drawn trillions of dollars of foreign funds into the country. "This idea that you can break trade, and not break the capital-flow side, is a fantasy," said Steven Blitz, chief U.S. economist at GlobalData TS Lombard.

Market volatility spiked after the president unspooled his fluid and unpredictable trade policy on April 2. The White House paused tariff increases on some of the na-

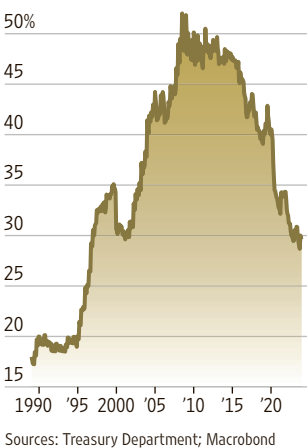
tion's largest trading partners hours after they took effect on April 9, but then ratcheted up tariffs on China.

A brutal bond-market selloff unfolded, with 30-year yields notching their largest weekly increase since 1987 despite decent bond auctions. The selloff occurred against the backdrop of a weakening dollar, which normally strengthens during bouts of stress. The abnormal combination prompted some analysts to warn about capital flight from the U.S.

Some of the market dislocation. Please turn to page A2

◆ **Greg Ip: Trump is absent from economic data..... A2**

**Estimated foreign ownership of U.S. Treasury securities as a share of publicly held debt**



Sources: Treasury Department; Macrobond

## U.S. Bids To Press Kyiv on Deal With Kremlin

Trump ideas include recognizing Crimea as Russian, barring Ukraine from NATO

By MICHAEL R. GORDON AND ALAN CULLISON

Ukraine is under pressure to respond this week to a series of far-reaching Trump administration ideas for how to end the war in Ukraine by granting certain concessions to Russia, including potential U.S. recognition of Moscow's 2014 annexation of Crimea and excluding Kyiv from joining NATO.

The ideas were outlined in a confidential document presented by senior Trump administration officials to their Ukrainian counterparts in Paris on Thursday, according to Western officials. They were also shared with senior European officials at the day-long meeting.

The U.S. is now waiting for Kyiv's response, which is expected to come at a meeting of U.S., Ukrainian and European officials in London later this week. Then, if there is a convergence among the American, European and Ukrainian positions, the proposals could be floated to Moscow.

To put pressure on Ukraine and Russia, Secretary of State Marco Rubio said Friday that the administration may pause its negotiating efforts if headway isn't made on core issues in the next several weeks.

"The Ukrainians have to go back home, they have to run it by their president, they have to take into account their views on all of this," Rubio said.

◆ **Lower oil prices to hit Russian growth..... A8**

## U.S., China Brace for Cold War As Tension Heats Up

By LINGLING WEI

For decades, no matter how relations between Beijing and Washington waxed and waned, trade and investment provided the glue that kept the two powers together.

Today, with economic relations between the two careening off the rails, China and the U.S. are headed toward what could be a Cold War that extends beyond trade—to deepening conflict or even military tension as both seek to form their own blocs.

During President Trump's first presidency, Washington and Beijing were both reluctant to throw their deep entanglements into complete disarray. Their first trade war played out over two years and involved frequent negotiations and fear of escalation on both sides.

This time, the two countries have effectively erected trade embargoes against each other in less than three months and are taking economic warfare into new territory. What is at stake is overall global security as well as economic stability.

"The U.S. and China are in a state of economic decoupling and there do not seem to be any guardrails to prevent escalations in trade tensions from spreading to other areas," said Rick Waters, a for-

◆ **China races robots in half marathon..... A9**

## INSIDE



### PERSONAL JOURNAL

You can—and should—wipe yourself off the internet, and here's how to do it. **A12**

### SPORTS

The two top seeds in the NBA playoffs didn't break the bank to get there. **A14**

## A Ceramics Maker Is Rarely Open. Obsessed Fans Trek There Anyway.

They hope to snag Torben Lov's dinnerware, used at Michelin-starred restaurants

By MATTIE KAHN

Frank Schubert was on vacation with his wife on the Danish island of Bornholm when he noticed the plates their dinner was being served on. Schubert, who was visiting from Germany, asked the servers where he could buy them.

The staff "had a laugh," he said. They wished him luck.

The dishes came from Lov i Listed, a ceramics store in town that is dark 348 days out of the

year, has no online shop and is never open past 2 p.m.

Dinnerware that emerges from this two-room store—and a studio next door—is in such high demand that fans share stories of making repeated trips to the island, hoping a visit will overlap with Lov i Listed's limited hours. An unadvertised wait list occasionally

opens up. Customers who land at the top have in some cases waited years for the privilege of placing an order.

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### SECOND OF TWO PARTS

## 'You're Taken Care of': Did Police Shield a Killer?

Jennifer Foley uncovered secret recordings between the man who shot her brother and the deputy chief in charge of investigations

By VALERIE BAUERLEIN

LORIS, S.C.—Weldon Boyd stopped his truck just off Highway 9 behind a black Chevy Silverado and the gun-toting hell-raiser who had run him off the road minutes earlier.

Boyd chased that truck about 9 miles before they ended up on Camp Swamp Road. Scott Spivey, an insurance adjuster who grew up nearby, got out of the Silverado holding a gun and yelling his last words.

Do not follow me anymore! Boyd and his friend Bradley Williams sat in their truck, armed and ready. They fired their 9mm handguns through the truck's

front windshield toward Spivey, stopping only to reload.

"At that point, I was mad," Boyd later told his parents.

Spivey, 33 years old, got back in his Silverado and fired several shots from inside the truck before going limp. He had been headed home after a Saturday afternoon drinking beer and whiskey at the bar of Boardwalk Billy's. Witnesses said they didn't see who started the gunfight.

When Horry County police arrived, Boyd said Spivey fired first, and that he and Williams shot in self-defense. Hours later, the lead detective in the case cemented Boyd's

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## Airbus Defers Its Dream Of No-Emissions Plane

By BENJAMIN KATZ

Five years ago, Airbus made a bold bet: The plane maker would launch a zero-emissions, hydrogen-powered aircraft within 15 years that, if successful, would mark the biggest revolution in aviation technology since the jet engine.

Now, Airbus is hitting the brakes. The company has cut the project's budget by a quarter, reallocated staff and sent remaining engineers back to the drawing board, delaying plans by as much as a decade.

Airbus joins companies recalibrating green efforts they embraced in recent years. Oil giant BP recently said it would

slash spending on renewable energy and pivot back to fossil fuels. Porsche has similarly scaled back its all-electric ambitions and is investing in new combustion-engine cars, citing lower-than-expected demand from consumers.

Airbus has spent more than \$1.7 billion on the project, people familiar with the matter said, but over the past year concluded that technical challenges and a slow uptake of hydrogen in the wider economy meant the jet wouldn't be ready by 2035.

The setback is a blow for the dream of clean aviation pushed by governments, in-

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U.S. NEWS



CAPITAL ACCOUNT | By Greg Ip

Trump Is Everywhere but in the Data

Imagine you didn't follow the news or social media and watched the world only through economic data. You would not have guessed the White House changed hands in January.

President Trump's deportations, tariffs, federal layoffs and funding suspensions have generated nonstop headlines and frayed confidence, yet left surprisingly little trace on the economy. Hiring, spending and inflation look a lot like they did under former President Joe Biden.

The disconnect is deeply disorienting. **United Airlines**, calling the economic environment "impossible to predict," issued two outlooks for earnings: one with recession, and one without.

Economists surveyed by The Wall Street Journal think that hiring will slow sharply this year and that inflation will shoot up. So far, there's little evidence of either. Job growth has averaged 173,000 over the past two months combined, almost the same as that of the prior six months. The unemployment rate has averaged 4.2%, a tenth of a

point higher than the prior six months. Both overall inflation and the Federal Reserve's preferred measure of underlying price pressures have averaged a tenth of a point less.

Elon Musk once said he could slash federal spending by \$2 trillion. His Department of Government Efficiency, or DOGE, now claims a more modest \$155 billion in savings. Yet in the first 80 days of Trump's administration, federal spending was \$154 billion higher than in the equivalent period a year earlier, a Journal analysis found.

DOGE aimed to get rid of up to 10% of federal employees—about 240,000. Through February and March, though, federal employment excluding the post office is down just 10,100.

Meanwhile, despite a border crackdown and deportation push, the number of foreign-born workers continued to grow in recent months, though the data, based on a monthly Labor Department sample, might not be fully reliable.

Trump's tariffs seem to have led to higher wholesale prices of steel, aluminum and

related products. But not much is happening at the consumer level. Gross domestic product, after growing 2.5% over 2024, is likely to be flat in the first quarter or even contract. But that seems to reflect unusual import behavior, and the effect of weather on consumption.

All of this should make us wonder: Is Trump really changing the economy as much as we—and he—think?

One obvious reason for the disconnect between the headlines and the data is time lags. Trump's first tariffs, on Canada, Mexico and China, took effect March 4; tariffs on steel and aluminum followed on March 12.

His much bigger 10% universal tariff and 145% tariff on China took effect in early April. High reciprocal tariffs on major trading partners have been paused. Tariffs on autos took effect in early April and will take effect on parts next month. Those tariffs won't show up until April data is released, in May. High reciprocal tariffs on major trading partners have been paused.

And since importers stocked up ahead of tariffs, it could be months before many newly levied tariffs get passed through to the final consumer. The foreign-born labor force could continue to swell with migrants who entered months ago, and federal employees who accepted deferred resignation could stay on the payroll until September.

Another explanation is that the U.S. economy is enormous. It is subject to countless influences, and actions by the president, even one as disruptive as Trump, aren't necessarily dominant. For example, while stocks sold off in response to Trump's tariff plans, they may have fallen anyway, having entered the year historically overvalued.

Finally, the president's policies trigger feedback mechanisms that often offset the initial impulse. For example, Trump's tariff threats have darkened the outlook for global growth. That has knocked down global oil prices, as has a decision by the Organization of the Petroleum Exporting Countries to increase output.

Airline fares have fallen

in the last two months in the face of softening demand, possibly reflecting DOGE restrictions on federal employees and foreigners uneasy about U.S. immigration border control policies. Gasoline and airfares contributed to a surprising drop in seasonally adjusted consumer prices in March from February.

Some of the feedback is behavioral. Trump had promised the largest deportation in history, but his daily removals through early March were actually running below fiscal 2024 levels, when Biden was president, according to the nonpartisan Trans-acti-onal Records Access Clearinghouse.

There may be fewer people to deport because illegal border crossings have plummeted under Trump and vulnerable migrants inside the country are trying harder to avoid the authorities.

Finally, there's political feedback. When a president acts, opponents pull every available lever to resist. Some of those levers aren't operating now: Trump has total control of Republicans and thus

Congress, while Democrats are in disarray. His exercise of presidential power has also blown through past norms, and possibly legal limits.

But courts are stepping in, staying some of his deportation efforts and federal layoffs.

Tariffs, Trump's most economically disruptive initiative, so far face no opposition from either courts or Congress. But they have scared investors and businesses who are making their views known. And the White House appears to be listening, prompting Trump to pause many tariffs, announce exemptions such as for smartphones, and initiate negotiations with trading partners.

That walkback, in turn, will likely change behavior: companies may now hesitate to change prices or sourcing if they think tariffs ultimately won't rise that much. Meanwhile, other countries won't retaliate while negotiations seem promising.

So Trump has not have left much impression on the economy yet. Don't assume that will last. Trump changes his mind a lot, but not his ultimate goals.

President Moves to Pull \$1 Billion More From Harvard

By Douglas Belkin  
And Liz Essley Whyte

The Trump administration has grown so furious with Harvard University after an escalating dispute that it is planning to pull an additional \$1 billion of the school's funding for health research, according to people familiar with the matter.

Trump administration officials, the people said, thought the long list of demands they sent Harvard last Friday was a confidential starting point for negotiations.

They were surprised on Monday when Harvard released the letter to the public. Before Monday, the administration was planning to treat Harvard more leniently than Columbia University, but now officials want to apply even more pressure to the nation's most prominent university, according to the people.

People familiar with Harvard's response say there was no agreement to keep the letter private, and that its contents—including requirements that Harvard allow federal-government oversight of admissions, hiring and the ideology of students and staff—were a nonstarter.

The letters to Harvard and other schools are coming from a new Trump panel called the Task Force to Combat Antisemitism.

In an open letter to the community, Harvard President Alan Garber said the list of demands made clear that "the



Harvard's release of a list of government demands has made the university a focus of resistance to the Trump administration.

intention is not to work with us to address antisemitism in a cooperative and constructive manner." And, he added, "we have informed the administration through our legal counsel that we will not accept their proposed agreement."

People close to Harvard say the task force is now escalating the fight to protect its own reputation. The gov-

ernment's demand letter to Harvard received blowback after the university released it, including from some on the right who publicly said it was overreach.

Harvard's Monday release of the letter positioned the university as a face of resistance against the Trump administration and has presented the task force with a playbook far different than

Columbia's, which quickly agreed to their demands to try to recover \$400 million in federal funding.

The task force had thought Harvard would also concede, according to people close to the matter.

Instead, the acrimony between the two sides deepened. Hours after Harvard released the demand list and announced its intention to

fight, the White House froze \$2.26 billion in funding. The administration also threatened Harvard's tax-exempt status and ability to enroll international students. These lines of attack have the potential to drain billions of dollars from the university.

Harvard reached out to the Trump administration in late March to try to avoid a confrontation, people familiar

with the matter said. Harvard had already spent months making what it said were good-faith efforts to quell campus antisemitism and implement structural changes. Harvard thought it was aligned with the general demands that arrived in an April 3 letter from the government task force, but then asked for more specifics.

People close to Harvard said the subsequent demands in the April 11 letter were far more intrusive than they could accept—and they perceived the correspondence to be a final offer.

The letter wasn't marked private, but members of the task force say that they had made clear in weeks prior that they wanted to keep their discussions private. Harvard disputes that there was any agreement about confidentiality.

Trump administration officials now doubt Harvard ever meant to negotiate and suspect the school aimed to fight the entire time, people familiar with the matter said.

The government's set of demands was mistakenly sent a day earlier than the task force intended, but its contents weren't an error, people familiar with the task force said. The administration stands behind the letter with the demands, a White House spokesman said. The New York Times earlier reported that a government official said the letter was sent mistakenly.

Financial Primacy Threatened

Continued from Page One

tion reflected hedge funds and other large investors forced to unwind certain trades that became unprofitable. Treasury Secretary Scott Bessent has said the selloff largely reflected technical factors—how investors' risk-management strategies forced sudden and one-time liquidations as correlations between different assets changed unexpectedly.

Some analysts have warned against rushing to sweeping conclusions given how foreign investors began the year overly exuberant about U.S. growth prospects, or American "exceptionalism." Trump's trade policy, by threatening to push up costs while uncertainty over the policy execution chills growth, has demolished that narrative.

The market volatility in the days after Trump's trade announcements would have been more worrisome if they had occurred against the backdrop of a continued slide in stocks, said Richard Clarida, who served as the Fed's vice chair from 2018 until early 2022.

The dollar began the year looking potentially as over-

valued as it had been since the 1980s, and Treasury yields remain in the middle of their trading range over the past two years, said Clarida, now a senior adviser at bond giant Pimco.

Bessent cautioned against rushing to judgment over one week's worth of market moves and reaffirmed Washington's support for a strong dollar. "We are still a global reserve currency," he said. "The dollar can go up and down."

But other analysts, economists, and former government officials warned that strained trading conditions might also reflect more-fundamental concerns. "The behavior of investors changed. Ordinarily they would've bought the dollar in this crisis time, and instead

they sold the dollar," said Steven Kamin, a former senior Fed economist who is now at the American Enterprise Institute. "It suggests that maybe investors are getting 'hinky' about the safety of Treasuries or the desirability of dollar assets."

Trump administration officials have previously argued that consumers wouldn't bear the cost of tariff increases because the dollar would strengthen. A weaker dollar makes it more likely that U.S. importers and retailers will have to pass along price rises.

In addition to Trump's erratic trade war, global investors are digesting possible increases in budget deficits in the U.S. and Europe and a potential erosion of U.S. monetary-policy independence.

Hedge-fund investors said an increasingly popular trade in recent weeks has been to bet on a widening in the spread between short-term and long-term rates on Treasuries because of concerns about higher budget deficits, pressure on the Fed to cut interest rates, and foreign-investor wariness about U.S. debt.

Foreigners have said they are unsettled by the prospect of a far-more-transactional approach to trade and currency policy, together with a preference for lower interest rates and higher budget deficits.

"The tariffs are the tip of the iceberg," said Ludovic Sub-

ran, chief economist and chief investment officer at Allianz, the German insurance giant. "If the dollar depreciates by 30%, what is the return on your investment in the U.S.? Everybody is doing the math."

In a call with analysts this month, BlackRock Chief Executive Larry Fink said the U.S. could face stiffer competition after benefiting from years of "over-allocation" and pointed to Europe as a potential better destination for foreign capital.

Ingo Mainert, a multi-asset chief investment officer at Allianz's asset-management arm, said the firm has been shifting to European assets from U.S. ones for the past month. He has fielded client calls over making more-significant reductions in U.S. exposure, including switching away from indices such as the MSCI World Index, which is more than 70% weighted to U.S. stocks.

That process could take six months to a year as clients complete formal reviews of their allocations, Mainert said. "We're now having a big discussion in the market—mainly outside the U.S.—about what would be better than these highly concentrated traditional benchmarks," he said.

Even if market conditions stabilize, a steady, gradual process that diminishes foreign appetites for U.S. assets could have serious ramifications

for American households and businesses. Foreign demand for Treasuries has boosted U.S. asset prices and lowered interest rates, including for 30-year mortgages, over the past two decades.

Roughly 30% of nearly \$29 trillion in Treasury debt held by the public is owned by foreigners, though foreigners' share of Treasury holdings has steadily declined since the 2008 financial crisis.

Normally, Treasuries have served as a hedge against stock-market risks because bond prices typically rise (and yields fall) as stock prices decline. That property has made them far more attractive for investors to own than would otherwise be the case, and any breakdown in that hedging property could be worrisome for the U.S. because it would mean one reason fewer for investors to own Treasuries.

In a world where the U.S. isn't viewed as a dependable partner, it makes sense for foreign investors to gradually diversify their holdings toward Swiss francs, gold, Japanese yen or euros, some analysts said.

"The longer this goes on, the more likely it is that we become less of a reserve currency than we have been. And that could start being quite expensive," said Eric Rosengren, who served as president of the Boston Fed from 2007 to 2021.

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**CORRECTIONS & AMPLIFICATIONS**

Due to a production error, several paragraphs were omitted from a Page One article in Saturday's edition about the police response to a killing in South Carolina. The full article is available online at WSJ.com/Corrections or via a QR code on A10.

Readers can alert The Wall Street Journal to any errors in news articles by emailing [wsjcontact@wsj.com](mailto:wsjcontact@wsj.com) or by calling 888-410-2667.



U.S. NEWS

Crackdown  
On DEI  
Meets K-12  
Resistance

By MATT BARNUM

The Trump administration recently sent a letter to states nationwide with a demand: Certify that your schools aren't engaging in what federal officials call illegal diversity, equity and inclusion programs—or risk funding.

Several states swiftly pushed back.

"We will not give in to misleading and damaging myths about diversity, equity and inclusion work," replied Vermont's education leader. The general counsel in Wisconsin's education department said the letter might be "unconstitutionally vague." A New York official asserted: "There are no federal or state laws prohibiting the principles of DEI."

President Trump's attempts to reshape American education have sparked high-profile pushback from Harvard University. Resistance also has emerged in K-12 schools, where practices have been slow to change and many Democratic officials are challenging Trump's edicts in court.

Republican leaders, meanwhile, are cheering Trump's crackdown.

DEI has "no place in education or society, and I have been working diligently to get it out of schools," said Tom Horne, Arizona's superintendent of public instruction, an elected Republican.

The Trump administration has moved quickly to dismantle the Education Department and restrict DEI and transgender-student accommodations using federal dollars as leverage. Public schools typically get only about 10% of their funding from Washington, but it is money they have come to depend on.

The administration has initiated a series of investigations, including into an all-gender bathroom at a Denver high school and a California statute that says teachers don't have to disclose students' gender identity to parents.

The education department also has created an online "End DEI" tip line. Submissions will help "identify potential areas for investigation," says the site.

Then came the department's letter directing states to certify, by April 24, that their schools aren't practicing DEI programs that "advantage one's race over another."

More than a dozen states have already declined to sign the document, according to an Education Week tracker. Many officials emphasize their states follow civil-rights laws, but won't certify compliance with what they see as a vague definition of illegal DEI.

Three lawsuits are ongoing over the administration's interpretation of civil-rights law in education, including one filed this past week by the NAACP.

The Los Angeles school system hasn't shifted its policies in response to Trump, superintendent Alberto Carvalho said. Caught between California law—which bans discrimination based on gender identity—and federal pressure, he is letting the state and Trump administration battle it out for now, he said.

"We continue to rely on the advice and guidance and the protective stance of the state of California," he said

In Washington state, superintendent of public instruction Chris Reykdal said, "We told districts: don't change your policies or procedures based on federal executive order...We don't think they have good legal arguments."

The Trump administration has notched victories. New Hampshire, Virginia and Wisconsin high school athletic associations have rolled back support for transgender athletes.

Alena Zachery-Ross, the schools superintendent in Ypsilanti, Mich., said she was rethinking programming that could be seen as racially targeted.

But Zachery-Ross doesn't plan to budge on the district's support for trans students' using the pronouns and bathrooms aligned with their gender identity. "That's something we value here and we're going to continue to support people to make sure they feel like they belong," she said.

Hegseth Had Second Sensitive Chat

Pentagon chief gave details on Yemen strikes to his wife, brother and lawyer

By NANCY A. YOUSSEF AND ALEXANDER WARD

WASHINGTON—Defense Secretary Pete Hegseth created a Signal chat with his wife, his personal lawyer and others, and posted sensitive military information into it, people familiar with the matter said Sunday, a revelation that has added to the increasing scrutiny of the novice cabinet member.

Hegseth was already facing questions for writing flight plans and other details about a military operation ahead of U.S. strikes on Houthis rebels in Yemen into a Signal chat with senior Trump administration officials. Hegseth posted nearly the same information into another chat featuring his wife and other aides who

didn't require real-time knowledge of the mission, a person familiar with the chat said.

The disclosure of the Signal chat comes after an unusual number of top political appointees have either been removed from the Pentagon or resigned just in the past few weeks, some with little explanation. President Trump's national security team, meanwhile, is attempting to broker sensitive deals with Russia, Ukraine and Iran, putting enormous pressure on a group that is largely inexperienced in sensitive foreign policy diplomacy.

The latest Signal chat group, a defense "Team Huddle," included 13 people, one person familiar with it said. The chat included Hegseth's brother, a Department of Homeland Security liaison who has traveled with the defense chief. The chat also included Hegseth's personal lawyer. Hegseth began the chat around the time of his confirmation hearing and it



Pete Hegseth

was used, in part, to craft strategies ahead of his appearance on Capitol Hill, the person said.

Shortly after the New York Times earlier reported on the new Signal chat, John Ullyot, a former top Pentagon spokesman working under Hegseth, wrote in Politico that the Pentagon is in "total chaos" and "disarray" under the secretary's leadership. Ullyot alleged that three fired Pentagon officials—all loyal to Hegseth—were wrongly smeared by anonymous offi-

cials as leakers who failed polygraph tests.

"While the department said that it would conduct polygraph tests as part of the probe, not one of the three has been given a lie detector test," Ullyot wrote. "Unfortunately, Hegseth's team has developed a habit of spreading flat-out, easily debunked falsehoods anonymously about their colleagues on their way out the door."

"From leaks of sensitive operational plans to mass firings, the dysfunction is now a major distraction for the president—who deserves better from his senior leadership," Ullyot wrote. He said it is hard to see Hegseth "remaining in his role for much longer."

Asked for further clarification about his claims, Ullyot, who resigned from the Pentagon last week, pointed The Wall Street Journal to his piece.

"There was no classified information in any Signal chat, no matter how many ways they try to write the story,"

Sean Parnell, Pentagon spokesman, posted on social media platform X.

In March, Director of National Intelligence Tulsi Gabbard said "any unauthorized release of classified information is a violation of the law and will be treated as such."

Among some Pentagon officials, the latest Signal chat revelation was more evidence of a Pentagon mired in unpredictability under Hegseth's leadership. He has fired at least 10 admirals and generals, changed longstanding practices and attacked diversity initiatives.

"The details keep coming out. We keep learning how Pete Hegseth put lives at risk. But Trump is still too weak to fire him," Senate Minority Leader Chuck Schumer (D., N.Y.) said on X. "Pete Hegseth must be fired."

White House spokeswoman Anna Kelly said Sunday night that "no classified information was shared" in the Signal chats.

Crossings at Mexican Border Grind to a Halt

CIUDAD JUÁREZ, Mexico—In this border city, large groups of migrants marching toward the U.S. have all but halted. Migrant encampments are disappearing, and few people are trying to sneak across remote desert areas to start new lives in the U.S.

"The door is closed," said Yorman Briceño, a Venezuelan migrant staying at a church-run shelter in Ciudad Juárez, just across from the border from El Paso, Texas.

By Kejal Vyas, Elizabeth Findell and Santiago Pérez

He had been waiting about six months for his Jan. 23 appointment to request asylum in the U.S. under a Biden-era program, until it was scrapped by President Trump on Jan. 20. Now, after watching how others have been rounded up in immigration sweeps and extradited by the U.S. to a notorious prison in El Salvador, he said he wasn't even sure he wanted to go anymore.

"There's no more hope for entering legally as long as Donald Trump is there," said Briceño.

During the presidential campaign, Trump promised to shut the U.S. border to illegal immigration after surreptitious crossings reached record levels under the Biden administration.

As the issue featured prominently in the campaign, migrant encounters tallied by U.S. authorities fell precipitously in the months leading to the election. Since taking office, Trump has delivered on his pledge as encounters have since fallen further to their lowest levels since the 1960s.

For the moment, tighter controls by the U.S. and Mexico, along with Trump's moves to shut down legal immigration pathways have brought migrant crossings to all but a standstill. The administration's targeted and dramatic expulsion of migrants to the prison in El Salva-



Illegal crossings at the U.S. border with Mexico are now at the lowest levels in decades. 'The door is closed,' a migrant said.

dor has also created a powerful deterrent.

Though Trump is struggling to deliver on another campaign promise—mass deportations—the border is an early political victory. According to a Wall Street Journal poll in March, 53% of voters approve of Trump's handling of border security, while 43% disapprove.

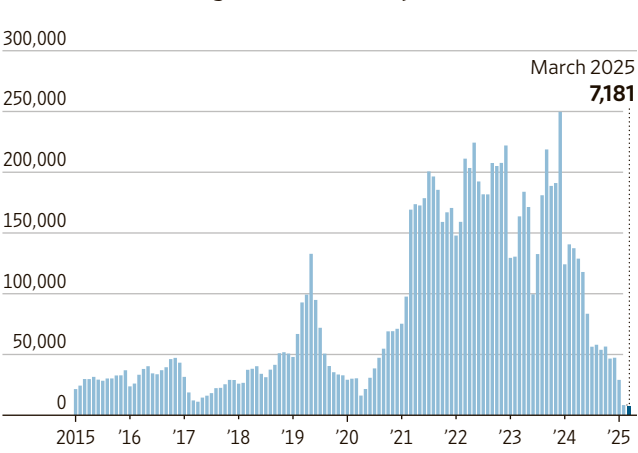
At an annual Border Security Expo this month in Phoenix, administration officials celebrated the low border crossing numbers to a crowd of hundreds.

Border czar Tom Homan said "gotaways"—an agency guess of how many people might have crossed the border without getting caught, based on clues such as footprints and camera sightings—had dropped to 41 the day before.

"Forty-one is pretty good," he said. "But we'll get to zero. And when we get to zero, we'll have operational control of the U.S. border for the first time in this nation's history."

Illegal border crossings declined significantly in the final year of the Biden administration, from a period of record highs that at one point saw 250,000 Border Patrol arrests in a single month. That number dropped to around 48,000 in December, the last full month of former President Joe Biden's

Southwest border migrant encounters by month



Source: Customs and Border Protection

term, and to just over 7,000 in March.

Border-security experts say that crossings often fall with changes of policy or administration, but also frequently rise as migrants and human smugglers adjust. Still, experts say the current reversal at the border is startling.

The administration has credited the drop in crossings to its decision to send thousands of active-duty troops to the border as a deterrent and its move to shut down a program known as CBP One, which migrants used to make appointments to ask for asylum. As a result, tens

of thousands of migrants who were awaiting a turn are now stranded in Mexico. Many are now heading back to their communities.

Trump's emphasis on closing the border has been effective, but some of Biden's later immigration policies—implemented after immigration policy became a top campaign issue—have given Trump helpful tools, said Andrew Selee, president of the Migration Policy Institute, a nonpartisan Washington think tank.

Biden officials pushed Mexico to step up immigration enforcement in early 2024, which

eliminated large concentrations of asylum seekers in border communities that would then overwhelm U.S. border towns. And an agreement last year to provide funding for Panama's government to deport migrants crossing the treacherous jungle paths of the Darién Gap has deterred asylum seekers from heading north.

While the Trump administration hasn't executed mass deportations, Selee said the U.S. government is sending a clear message with its harsh rhetoric and increased use of social media to show the incarceration of Venezuelan migrants in a prison for gang members in El Salvador.

The Mexican government has installed nine tent cities along the 2,000-mile-long border in anticipation of the mass deportations that Trump has promised. In Ciudad Juárez, a cluster of white tents resembling warehouses were erected on the site where Pope Francis gave an address in 2016 pleading for migrants' rights.

The facility, with capacity to hold 2,500 people, offers hot meals, beds and psychological counseling. It has sat largely empty since opening Feb. 20.

Migrants Issue Looms Large as Vance Meets Pope

By MARGHERITA STANCATI

ROME—Pope Francis met briefly Sunday with Vice President JD Vance, the Vatican said, before the pontiff in his Easter message called for better treatment of migrants and other vulnerable and marginalized people and urged a "just and lasting peace" in Ukraine.

The brief but symbolic meeting showed a desire by the Vatican and the White House to ease tensions after a public row earlier this year over the Trump administration's plans for large-scale deportations of immigrants.

Also on Sunday, the pope—still frail after a near-fatal bout of pneumonia this winter—called for a cease-fire in Gaza and for an end to the wars in Ukraine and Sudan.

"I would like us to renew our hope that peace is possible," Pope Francis said in a message read out by an aide to a crowd gathered in St. Peter's Square.



Vice President JD Vance and Pope Francis met at the Vatican Sunday. The Pope has been critical of U.S. immigration policies.

The meeting between Francis and Vance drew global attention because of the pope's outspoken recent criticism of U.S. immigration policies.

Francis said early this year that Trump's mass deportation plan "damages the dignity of many men and women." His remarks didn't go down well with the new administration, with Trump's border czar, Tom

Homan, inviting the pope to "stick to the Catholic Church and fix that and leave border enforcement to us."

The pope condemned hardline immigration policies again in his Easter message: "How much contempt is stirred up at times towards the vulnerable, the marginalized, and migrants!"

The 88-year-old pope's

meeting with Vance, a convert to Catholicism in 2019, lasted only a few minutes and they exchanged Easter greetings.

Vance had a longer meeting at the Vatican on Saturday with the pope's second in command, the Vatican Secretary of State Cardinal Pietro Parolin. The Vatican said they had a cordial "exchange of opinions" on immigration and other issues.

The pope has previously slapped down Vance for misreading Christian doctrine to justify the administration's immigration policies. Vance had spoken of concentric circles of love, with the family at the center. "Then you love your fellow citizens in your own country, and then after that, you can focus and prioritize the rest of the world," Vance said.

In a letter to U.S. bishops in February, the pope said Vance was wrong in his interpretation of the theological concept of "ordo amoris," or order of

love. "Christian love is not a concentric expansion of interests that little by little extend to other persons and groups," the pope wrote. "The true ordo amoris that must be promoted is that which we discover by meditating constantly on the parable of the 'Good Samaritan,' that is, by meditating on the love that builds a fraternity open to all, without exception."

Vance said his intention hadn't been to litigate with the pope. "I was certainly surprised when he criticized our immigration policy in the way that he has," he said at the time.

Vance received a warmer reception on Friday from Italian Prime Minister Giorgia Meloni, who is emerging as the Trump administration's favorite European interlocutor. "I'm grateful to be in the presence of a dear friend in a beautiful place," Vance said before he had lunch with the right-wing Italian leader.



# Another Billionaire Ramps Up His Profile

Speculation grows that Illinois Gov. Pritzker wants 2028 nod from Democrats

By JOHN McCORMICK

CHICAGO—If JB Pritzker runs for the Democratic presidential nomination, he will be betting his party’s best prospect is a political punch-throwing heavyset billionaire who inherited massive wealth. While that sounds like President Trump, the two-term Illinois governor would be wagering on himself.

Pritzker, an heir to the Hyatt hotel fortune, has become one of the most-outspoken critics of Trump as Democrats are struggling to counter him. Wealth has long opened doors for Pritzker and there are signs he wants the next one to be into the Oval Office.

The 60-year-old is visiting New Hampshire, traditional home of the nation’s first presidential primary, to speak on April 27 at a party fundraiser about what he sees as Trump’s authoritarianism and to call Democrats to action. The trip is likely to boost speculation that Pritzker, among those vetted by

Kamala Harris’s presidential campaign as a possible running mate, is interested in the 2028 nomination.

“There is no doubt that he is going to run,” said Chicagoan Bill Daley, who served as President Bill Clinton’s commerce secretary and President Barack Obama’s chief of staff. “The real question is whether he runs for re-election first or just runs for president.”

The governor, who declined an interview, has yet to say whether he will seek a third term. An announcement is expected in the next few months, with the March 2026 primary less than 11 months away.

Daley said he would recommend against another gubernatorial bid because a crisis or scandal can pop up at an inconvenient time. Pritzker, he said, has the financial wherewithal to do something most candidates couldn’t: announce a presidential bid in 2026 and lock down the best available campaign staff talent.

Pritzker has never shied from confronting Trump. “Take it from an actual billionaire, Trump is rich in only one thing: stupidity,” he said in his Democratic National Convention speech in August.

Trump’s estimated net

worth has grown significantly since 2024, according to Forbes, boosted by his social-media platform and cryptocurrency. He is now the nation’s wealthiest elected official.

The president and Elon Musk, Trump’s adviser and the world’s richest man, have thrown punches at Pritzker as well. Both have mocked his weight, with Trump suggesting last year the Illinois governor “wants to eat all the time.”

If Pritzker runs, Republicans will be certain to highlight the challenges Illinois faces. Private-sector jobs have been stagnant since he took office, and a massive unfunded pension liability hangs over the state like a black cloud. Taxes are among the highest in the nation, and the state’s population has shrunk since 2020. Violent crime and struggling schools plague Chicago, a favorite target for Trump and conservatives.

“People are voting with their feet,” said Paul Vallas, a conservative-leaning Democrat who is a former Chicago public-schools chief and past city budget director. “The state is an absolute disaster.”

Pritzker took over in 2019 amid a budget crisis. He has supported Illinois legislation



Illinois Gov. JB Pritzker

to protect abortion access, mandate paid time off for workers, raise the minimum wage to \$15 for nontipped workers 18 and older, tighten gun restrictions and invest in infrastructure.

He has aggressively recruited businesses, including in quantum computing, and worked to restore some order to state budgeting, helping prompt Moody’s to raise the Illinois credit rating three times during his tenure.

Along with Govs. Kathy Hochul of New York and Tim

Walz of Minnesota, Pritzker has been called to testify May 15 before a Republican-controlled U.S. House committee reviewing undocumented immigrant “sanctuary” status in states. The Democrats haven’t agreed to appear.

Other Democrats being watched for presidential ambitions include former Transportation Secretary Pete Buttigieg and at least five governors: Walz, Gavin Newsom of California, Josh Shapiro of Pennsylvania, Gretchen Whitmer of Michigan and Andy Beshear of Kentucky.

With Democrats divided over the war in Gaza, Daley said Pritzker, who is Jewish and running a state with a large Palestinian population, should likely expect protests at potential presidential campaign events.

Pritzker’s fortune has allowed him more independence than most politicians because he doesn’t have to worry about fundraising. With an estimated net worth of \$3.7 billion, he spent more than \$300 million to self-fund two gubernatorial campaigns.

The Illinois governor was his party’s second-biggest donor in this month’s Wisconsin Supreme Court election, based

on disclosure filings so far. He has long been one of America’s top campaign contributors, helping him make political friends nationwide.

The foundation the governor started with his wife in 2001 gave away more than \$50 million in 2023, the most recent tax records available show, with donations coast to coast that include a focus on early childhood development.

Anne Caprara, Pritzker’s chief of staff, said he is “deeply analytical” from his work as a venture capitalist and businessman. But, she said, he often tells business leaders “you can’t run government like business” because they have very different goals.

While he grew up in privilege in California, Pritzker hasn’t had an entirely easy life. As a 7-year-old, his 39-year-old father died while playing tennis in Hawaii. His mother, an alcoholic and Democratic activist, was killed a decade later when after driving drunk she jumped from a moving tow truck after her car broke down.

In 2023, as Pritzker ramped up his national political profile, he established a nonprofit called Think Big America with a stated mission of expanding reproductive rights.



## Government Jobs No Longer Cushion D.C. Economy

By PAUL KIERNAN AND RACHEL LOUISE ENSIGN

WASHINGTON—President Trump’s federal government shake-up is taking a toll on the U.S. capital’s economy.

Economists believe government layoffs and looming budget cuts will push the Washington, D.C., metro area into a recession. Cracks are already forming as federal workers pull back on spending, hotels and restaurants struggle and parents cancel plans for their kids’ summer camps.

“We see this as potentially catastrophic for the region,” said Julie Coons, president of the Northern Virginia Chamber of Commerce.

Federal tax receipts and Treasury borrowing have long buffered the nation’s seventh-largest metro area—which comprises the capital, northern Virginia and parts of Maryland—from gyrations in the business cycle. Greater Washington, D.C., has had lower unemployment than the rest of the country every

month since at least 1990, when the Labor Department’s current data series began.

The Covid pandemic hurt, emptying office buildings and shrinking foot traffic, but city officials hoped the Trump administration would turn things around by ordering federal workers back to the office five days a week. They didn’t fully anticipate that Elon Musk’s Department of Government Efficiency would so quickly and broadly take aim at the local workforce.

In Arlington’s Rosslyn neighborhood, bookings at the Residence Inn are 10% to 15% below target for the coming months, according to general manager Flavia Sampaio, who said local hotels rely heavily on business from government agencies. Across the Potomac River in D.C., Bluebird Sky Yoga co-owner Kristine Erickson has seen a slowdown.

“A lot of times they’ll say, ‘My spouse or I were just let go from jobs and we’re cutting down on extra expenses and that means yoga,’” she said.



Co-owner Kristine Erickson says business is down at Bluebird Sky Yoga; the story is similar at Cork Wine Bar and Market.

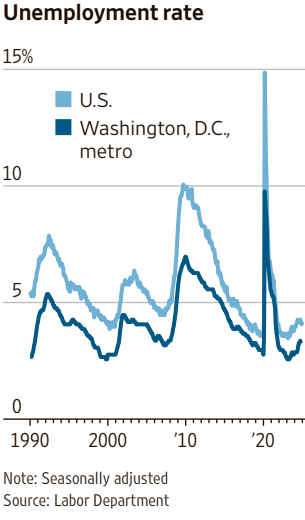
February sales at Cork Wine Bar & Market, a restaurant on 14th Street, were down about 15% to 20% from a year earlier, said co-owner Diane Gross. March sales were helped by a “tariff sale” on wine but still down around 10%.

“Eating out is the first thing to go,” Gross said.

The federal government employs around 17% of full-time workers in the Washington metro area, or around 400,000 people. An additional 500,000 or so are employed in “professional, scientific and technical services,” a category that includes government contractors and consultants.

Enrico Moretti, an economist at the University of California, Berkeley, said each federal job in the capital region likely creates 1.6 additional jobs in the local economy over a decade, an effect that would be reversed in the event of job cuts.

Oxford Economics projects gross domestic product in the Washington, D.C., metro area will fall 0.5% over the course of this year, worst among the 50



largest U.S. metro areas other than New Orleans, said Barbara Denham, lead economist for cities and regions.

The DOGE hit could be especially intense within the district’s boundaries. Glen Lee, D.C.’s chief financial officer, estimated in late February that federal jobs there will fall by 40,000, or 21%, through the 2029 fiscal year.

The Edlavitch Jewish Community Center of Washington, D.C., is used to its annual summer camp’s being in such high demand that parents sit at their computers refreshing the webpage on enrollment day. “It’s like rock-concert tickets,” said Jesse Bordwin, the center’s chief experience officer.

This year, dozens of families either declined to enroll or canceled. Center staff reached out to understand why, and the parents of 24 campers cited either layoffs or uncertainty related to DOGE.

The regional economy is showing some resilience. Metro ridership, office occupancy and foot traffic in downtown Washington have picked up in recent months, after Trump’s return-to-office orders. Data from Zillow suggest rents in D.C. continued rising in March. Bank of America credit- and debit-card data show a drop in local spending in February but a rebound in March.

Sandra Lin, who owns a sandwich and ice-cream shop

at the Ronald Reagan federal building, said business is up as some customers return for the first time in five years.

Still, unemployment claims in the area are starting to tick up. And laid-off workers, or those worried about their jobs, are cutting expenses.

Michael Matheke-Fischer, a former USAID contractor specializing in global health, was laid off in January. His wife, Nandini Pillai, also worked in global health for a USAID grant recipient and was furloughed around the same time.

The family of four no longer orders in for dinner, is driving as little as possible and is eating less meat, Matheke-Fischer said. They also canceled day-camp plans. Matheke-Fischer, 43 years old, expects to be able to cover mortgage and basic expenses for a few months but is worried about what comes next.

“I lived in India for a long time and my wife is of Indian descent,” he said, “so thankfully, we know what to do with vegetables.”

## Airbus Puts Off Green Plan

Continued from Page One

vestors and customers.

Airbus says the work and money haven’t been wasted. The company has established that hydrogen is technically feasible and delaying the project will give it more time to fine-tune, executives said.

“Our destination is not changing,” Bruno Fichfeux, Airbus’s head of future programs, said in an interview. “To get there, we need to adjust to reality.”

Airbus first unveiled a trio of proposed hydrogen-powered planes toward the end of 2020. The most ambitious would carry as many as 200 passengers and have a range

of up to 2,000 nautical miles—enough to fly from New York to Las Vegas.

“We intend to play a leading role in the most important transition this industry has ever seen,” Airbus Chief Executive Guillaume Faury said at the time. Airbus recruited a dozen airlines, from Delta to Air New Zealand, and more than 200 airports to explore how hydrogen could be built into their operations.

Executives at airlines and suppliers privately questioned whether the 2035 goal was attainable, given hydrogen technology was still in its infancy.

There were a raft of technical challenges, not least the safety concerns exemplified by the 1937 Hindenburg disaster. Engines would need to be reconfigured to run on a different fuel. Aircraft would need to store the hydrogen in liquid form at minus-423 degrees Fahrenheit. The heavier fuel load and equipment would weigh on seat capacity and

range. And a new supply chain would be needed to produce the fuel in sufficient quantities, transport it, and safely store it at airports.

“We don’t think hydrogen’s the answer,” then-Boeing CEO David Calhoun said when asked in late 2022 what the company was doing on the technology.

But Faury was determined. An engineer by training, the Frenchman had tried to persuade Airbus’s leadership to take hydrogen seriously even before becoming CEO. Part of his motivation came from his time as head of research and development at French carmaker Peugeot in the mid-2010s. The auto industry had been caught off guard by the rise of electric vehicles, and Faury was wary of a similar battle for Airbus, people familiar with his thinking said.

Airbus also had a commitment to a major shareholder—the French state. The company had been a major beneficiary of

a Covid-era government support package for the aviation and aerospace sector of 15 billion euros, equivalent to roughly \$16.6 billion. The deal required Airbus to spend part of the money on bringing green aircraft to market by the 2030s.

The hydrogen project helped Airbus access additional government funding, as well as private green financing. It also helped attract new engineers at a time when green activist Greta Thunberg had helped popularize “flight-shaming” in Europe.

Airbus ultimately assigned the project an annual budget of about €400 million, primarily funded through its own coffers, people familiar with its financing arrangements said. The company, which doesn’t detail its R&D spending, declined to provide a breakdown of its spending.

Last year, it started to become clear that Airbus might need to backpedal on the 2035 promise. The company’s initial

200-seater concept, which relied on feeding hydrogen directly into a regular jet engine, was fundamentally flawed: The combustion would still produce nitrogen-oxide emissions. Engine makers were also hesitant about investing heavily in the project, people familiar with it said.

Instead, Airbus shifted focus to hydrogen-fuel cells, which use a chemical reaction to generate energy for an electric motor. It would produce only water vapor—but require a more radical redesign of the airframe and propulsion system. The plane would carry only 100 passengers about 1,000 nautical miles.

Over time, even that proved challenging because of the extra weight of the fuel cells and their limited electricity generation. Instead of a short-haul narrow-body—the workhorse of the aviation industry—at best the aircraft would be more akin to a less-appealing regional turboprop.

Meanwhile, Airbus executives watched as broader enthusiasm for hydrogen withered away. BP and Finland’s Neste, for example, are among companies to have pulled plans for new hydrogen plants.

In February, Airbus told workers that the hydrogen project’s budget was being cut, and its timeline delayed.

Later that month, Faury said the project hadn’t yet led to a commercially viable aircraft that could compete on price and performance. Engineers would start a second “development loop” to figure out what seat capacity and range the hydrogen aircraft might have, he said on an earnings call.

At a company event in March, the CEO compared the project to Concorde, a supersonic jet relegated to the annals of aviation history because it was so costly to operate. “We have come to the conclusion that it would be wrong to be right too early,” Faury said.



U.S. NEWS

Trump Puts Covid ‘Lab Leak’ Theory on Sites

By Christopher M. Matthews

The Trump administration further embraced the theory that the deadly Covid-19 pandemic emanated from a lab in China, updating the government’s main websites about the virus to detail that argument. The White House revamped the primary websites for information about Covid and for ordering virus tests, Covid.gov and Covidtests.gov, to lay out its contention that the virus leaked from a lab in Wuhan, China. Under a banner titled “Lab Leak” that features an image of President Trump, the new

page says that prior statements by government officials, including Dr. Anthony Fauci, that the virus emerged from a wild-animal market in China are false. The page says that Wuhan is home to China’s foremost research lab on the SARS virus and has “inadequate biosafety levels.” Researchers there had Covid-like symptoms before the virus was discovered at a wet market, it says. “‘The Proximal Origin of SARS-CoV-2’ publication—which was used repeatedly by public health officials and the media to discredit the lab leak theory—was prompted by

Fauci to push the preferred narrative that COVID-19 originated in nature,” the website says. Trump has repeatedly endorsed the lab leak theory and attacked Fauci, who served as the director of the National Institute of Allergy and Infectious Diseases until 2022 and became the face of the government’s Covid response. Covid-19 emerged in Wuhan in late 2019 and then spread

rapidly through the world in 2020 and 2021 before the development of vaccines helped limit deaths. It marked one of the worst pandemics in modern history. The origin of the virus still divides the U.S. intelligence community, in large part because the Chinese government hasn’t cooperated with international investigations. Four U.S. intelligence agencies have favored, with low confi-

dence, the animal transmission theory. So has the National Intelligence Council, a body of senior intelligence officers that reports to the Office of the Director of National Intelligence. The Central Intelligence Agency has concluded that the pandemic most likely arose from a laboratory leak, lending credibility to that view. The Federal Bureau of Investigation and the Energy Department have also identified a laboratory mishap in Wuhan as the probable source of the Covid virus, which killed more than 1.2 million Americans and more than seven million

people worldwide. The CIA said in January that it had reached the assessment with “low confidence” and that it would continue to evaluate “any available credible new intelligence reporting or open-source information that could change CIA’s assessment.” The administration’s revamped website also attacks the World Health Organization for its handling of the pandemic as well as former New York Gov. Andrew Cuomo for directing people with Covid to nursing homes, which it calls “medical malpractice.”

The Covid-19 virus killed more than 1.2 million Americans.

U.S. WATCH

FLORIDA  
Florida State  
Classes to Resume

Classes will resume at Florida State University on Monday, four days after a deadly shooting on campus left two people dead and six others injured, school officials said. Students and instructors will have the option of holding classes remotely or in person, depending on the class. The school has waived all mandatory attendance policies that could affect grades, so students won’t be punished if they choose not to go to class in person, FSU President Richard McCullough said in a letter to students and faculty. Students also can request an incomplete grade for their class if they feel they are unable to complete a course, McCullough said. The gunman, identified as the stepson of a sheriff’s deputy, arrived on campus an hour before the shooting Thursday and stayed near a parking garage before he walked in and out of buildings and green spaces while firing a handgun just before lunchtime, police said. In roughly four minutes, officers confronted 20-year-old Phoenix Ikner, a Florida State student, and shot and wounded him, Tallahassee police said. The two victims who died were Robert Morales, a university dining coordinator, and Tiru Chabba, an executive for food service vendor Aramark, according to family members and attorneys for the families. —Associated Press

WEATHER  
Oklahoma, Texas  
Hit by Storm, Hail

A slow-moving, active storm system brought heavy rain, large hail and tornadoes to parts of Texas and Oklahoma and left two people dead as severe weather warnings Sunday continue to threaten parts of the south-central and Midwest U.S. On Easter Sunday, communities in Texas and Oklahoma were beginning to assess the damage wreaked by tornadoes. There were 17 reported events Saturday, according to Bob Oravec, lead forecaster with the National Weather Service’s Weather Prediction Center. Five were confirmed in south-central Oklahoma, including one that inflicted at least EF1 damage on a small town that was still recovering from a March tornado. The storm also brought heavy rain to a broad swath of north-central Texas across central-eastern Oklahoma, which saw 2 to 4 inches accumulate Saturday into Sunday. Police in Moore, about 10 miles south of Oklahoma City, received dozens of reports of “high-water incidents” over the weekend, including two cars stranded in floodwaters Saturday evening. One car was swept away under a bridge, and police said they were able to rescue some people, but a woman and 12-year-old boy were found dead. Oravec said the system wasn’t moving much over Texas and Oklahoma Saturday, leaving the area stuck under a very active thunderstorm pattern that produced large hail, flash flooding and tornadoes. The whole storm system is moving northeast into Arkansas, Missouri and southeastern Iowa, Oravec said. —Associated Press



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WORLD NEWS

Israeli Soldiers Violated Orders, Erred

Probe finds failures in killing of 15 Gaza aid workers; military fires field commander

By DOV LIEBER

Israeli soldiers erred in killing 15 Gaza aid workers and violated orders when firing on a United Nations vehicle, the country's military said Sunday in its report on the incidents. The military fired the unit's field commander for giving an inaccurate and incomplete account when debriefed and reprimanded a brigade commander for his responsibility over the troops. The conclusions were the result of an internal investigation into last month's shootings, which offered a measure of vindication for Palestinian authorities in Gaza, who said from the outset that those who were

killed were legitimate emergency responders traveling in well-marked vehicles. The incident occurred before dawn on March 23, a few days after Israel ended a two-month cease-fire and began striking targets in Gaza. It began around 3:30 a.m., when an ambulance looking for people injured by an Israeli airstrike came under fire in a neighborhood in the town of Rafah. When other ambulances arrived to search for the missing crew, they, too, came under fire by Israeli forces. In total, eight Palestine Red Crescent personnel, six emergency rescue workers, and a U.N. staffer were killed. A surviving medic was released after being briefly detained, while another continues to be held by Israel. Israeli troops used bulldozers to bury the bodies and vehicles on the side of the road. Several days passed before Israel's military helped U.N. per-



The body of a Palestinian paramedic who was killed with other first responders in the southern Gaza Strip in March.

sonnel locate the bodies. "The examination determined that the fire in the first two incidents resulted from an operational misunderstanding by the troops, who believed they faced a tangible threat from enemy forces," the Israeli military said. According to the Israeli mili-

tary investigation, soldiers fired on the first ambulance, believing incorrectly that it was a Hamas police vehicle rather than an ambulance. A survivor of the initial shooting was detained, and the soldiers reported after a field interrogation that they had a Hamas operative in hand, which was later

discovered to be incorrect. When the second convoy approached, the Israeli commander opened fire, leading the other soldiers to open fire as well. The military said he told investigators that he was wearing night-vision goggles and couldn't see the flashing emergency lights because of the narrow field of view. When the U.N. vehicle approached, the same commander fired warning shots, prompting other soldiers to fire at the vehicle, killing its passenger. Firing on the U.N. vehicle "involved a breach of orders during a combat setting," the military said. It said additional details emerged after the troops' commander reported the event. Israel said it dismissed the officer, the deputy commander of the Golan Reconnaissance Battalion, for providing "an incomplete and inaccurate report during the debrief." Israel also reprimanded the

commanding officer of the 14th Brigade, citing his overall responsibility. Jonathan Whittall, head of the United Nations' humanitarian office in Gaza and the West Bank, said the investigation lacked accountability. "Without accountability, we risk continuing to watch atrocities unfolding, and the norms designed to protect us all, eroding," Whittall said. "Too many civilians, including aid workers, have been killed in Gaza." The Israeli military's initial explanation that the emergency workers had sped toward troops with their emergency lights off and that the convoy included at least six Hamas militants began to show cracks almost immediately. Interviews with the survivor and a cell-phone video made clear their emergency lights had been on, and the one Hamas militant named by Israel's military wasn't among the dead.

Talks With U.S. to Continue, Iran Says After Latest Round

ROME—Iran and the U.S. concluded a new round of talks Saturday in Rome in what Iranian officials described as a step forward, and

By Benoit Faucon, Michael R. Gordon and Laurence Norman

opening the way for further talks this week. Iranian Foreign Minister Abbas Araghchi offered a guarded assessment after exiting the indirect negotiations held with U.S. special envoy Steve Witkoff. "There is no reason for much optimism, and there is no reason for much pessimism either," Araghchi told Iran state media. He said technical teams and higher-level officials would meet this week. The talks would continue to be mediated by Oman, which will also host, he said. Saturday's talks took place in the Omani embassy in Rome. U.S. officials said the two sides would meet this week, but didn't provide details. Oman's Foreign Ministry said the U.S. and Iran are ready to move into a new phase of their discussions. "This phase aims to reach a permanent and binding agreement that guarantees Iran's full renunciation of nuclear weapons, the complete lifting of sanctions and the safeguarding of its right to develop nuclear energy for peaceful purposes," it said. Iran was planning before

the Saturday meeting to set out proposals, including that the U.S. guarantee not to leave a future accord, people familiar with the matter said. The previous weekend, Araghchi negotiated with Witkoff, the highest-level U.S.-Iran talks in years. The talks began with indirect discussions mediated by Omani officials in which the two sides were in separate rooms, but culminated in a 45-minute direct meeting between Araghchi and Witkoff, a person familiar with the talks said. The Iranians were expected to discuss ways that their stockpile of enriched uranium could be managed under a deal, a process for lifting economic sanctions and their hope to organize a high-level visit in Washington, Iranian and Arab officials say. A spokesman for Witkoff declined to comment on any demands the Iranians might make. "The president has been clear: Iran cannot have a nuclear weapon or enrichment program," the spokesman said. Iran insists its program is for peaceful purposes, but the U.S. has long accused Tehran of preparing the option to develop a nuclear weapon and has insisted the program be dismantled or strictly limited.

Tehran wants Washington to cover its losses if the U.S. pulled out of a deal.

Ali Shamkhani, a top aide to Supreme Leader Ayatollah Ali Khamenei, wrote on X as Saturday's talks began that an agreement needs to include "guarantees, balance, no threats, speed, lifting sanctions...containing troublemakers (like Israel), and promoting investment." He also said Iran won't accept the dismantlement of its nuclear program, an approach Israeli Prime Minister Benjamin Netanyahu has championed. Nor would Iran accept an approach adopted by the United Arab Emirates, which operates nuclear reactors but forgoes enriching uranium and depends on imported fuel, he said. Iran's proposal on guarantees reflects its experience with the 2015 nuclear deal negotiated by the Obama administration, whereby Iran curbed its nuclear enrichment in exchange for sanctions relief. President Trump withdrew the U.S. from that deal in 2018, calling it "one-sided." The deal was in the form of a "plan" instead of a treaty, which the White House calculated might not receive the two-thirds Senate vote that would be needed to ratify it. A push for guarantees became the major Iranian ask during talks with the Biden

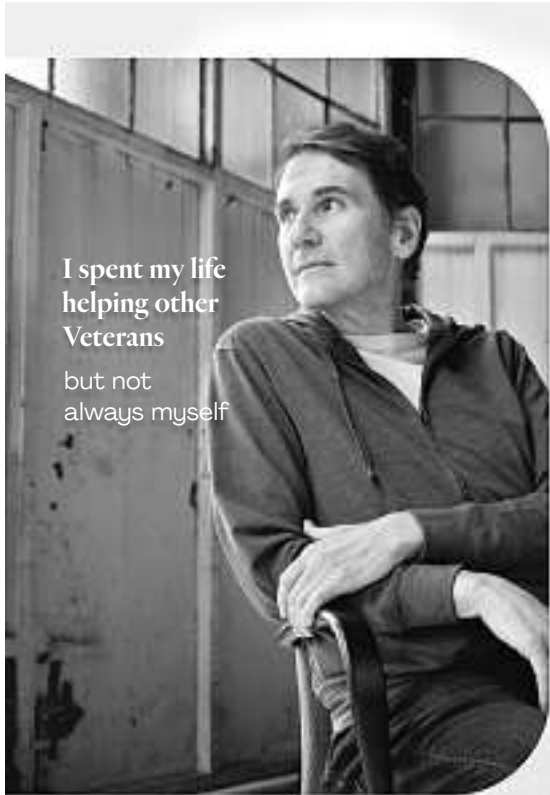


Italian Deputy Prime Minister and Minister of Foreign Affairs Antonio Tajani, left, welcomed Iran's Foreign Minister Abbas Araghchi in Rome on Saturday.

administration on reviving the 2015 deal, which eventually failed. The U.S. was willing to offer words pledging not to leave the deal but told the Iranians repeatedly that no administration can bind a future president and Congress. Richard Nephew, a U.S. negotiator with Iran during the Biden administration, said that issue kept the two sides from going back into the 2015 accord. "Even treaties have withdrawal provisions for national-security reasons." Officials also argue that Washington can't offer a guarantee because it needs the flexibility to withdraw if Iran takes hostile action against American interests and allies or violates the deal. The best hope Iran has for preserving long-term U.S. support for an accord, they say, is for Tehran to comply with a deal that strictly limits its pro-

gram while avoiding aggressive policies in the region. Iran wants the U.S. to cover Tehran's losses if Washington were to pull out of a deal, said European and Iranian officials and another person familiar with the Iranian position. The idea of a financial penalty was floated by the Iranians to the Biden administration, but went nowhere. "Iran's worry is not just to sign a deal but one with conditions that would prevent it from being unilaterally unwound," said Mohamed Amersi, a Middle East-focused member of the advisory board at the Wilson Center, a Washington think tank. Iran stepped up its nuclear program after Trump quit the nuclear pact and is now producing 60% high-enriched uranium, which can be swiftly converted into the 90%-grade material needed for a bomb.

To stockpile the surplus fuel that would arise from a reduction of its nuclear enrichment, Tehran is open to having it managed by Russia—with whom Trump is pursuing a rapprochement, the people said. The stockpiles are currently supervised by the United Nations atomic agency. Russia has emerged as something of a go-between. Before heading to Rome, Araghchi visited Moscow to meet Russian President Vladimir Putin. Trump has discussed Iran with Putin, and Witkoff traveled to Moscow before the first round of talks as part of his Ukraine diplomacy. Last Monday, Witkoff suggested the administration would allow Iran to keep its enrichment at no higher than 3.67%. But on Tuesday he said Iran has to agree to "eliminate its nuclear enrichment and weaponization program."



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Cold War Is Looming With China

Continued from Page One mer senior U.S. diplomat who now runs the China center at the Carnegie Endowment for Global Peace. "It's becoming more difficult to argue that we're not in a new Cold War." For Chinese leader Xi Jinping, it is a moment for which he and his inner circle have been preparing since Trump's first term. After the initial shock from Trump's recent tariff hits, Beijing is in full-blown retaliatory mode, vowing to "fight to the end." Its tools to hit back at the U.S. aren't limited to economic weapons such as retaliatory tariffs, blacklists targeting U.S. companies and restrictions on its exports of critical minerals. Beijing's recent statement that it is done with tit-for-tat tariff responses signals that it might be moving to other, noneconomic methods, people who consult with Chinese officials said. One option Beijing has, the people said, involves leveraging the data, call logs and other information it gathered from years of intrusions into computer networks at U.S. ports, water utilities, airports and other targets. The Wall Street Journal reported this month that in a secret Geneva meeting in De-

cember with the departing Biden administration, Chinese officials linked a series of cyber assaults on U.S. infrastructure to Washington's support for Taiwan, the self-ruled island Beijing has pledged to bring under its fold. Beijing could also step up strategic coercion against U.S. partners, especially in the Indo-Pacific, when the Trump administration's commitment to providing security in the region appears to be in doubt. Alarms about the security threat from China, which had been rising within the U.S. political and military establishment, have ramped up further. At an April 10 Senate hearing, Adm. Sam Paparo, head of U.S. forces for the Indo-Pacific, called attention to China's increased military activities near Taiwan, which he said threatened the security of the U.S. and its allies. On the U.S. side, Trump has signaled he is willing to withdraw security guarantees for countries depending on American support unless they make economic concessions. Behind the heightened risk of escalation beyond trade is a lack of effective communication between senior officials on both sides of the Pacific. Initially, Beijing hoped for dialogue, but its insistence on formal diplomatic protocol proved to be a mismatch with a Trump team willing to engage only with those closest to Xi, in particular the Chinese leader's chief of staff, Cai Qi, whose portfolio includes cybersecurity—a core concern for Washington.

Amid Trump's tariff assault, Beijing has clammed up. Most recently, as trade tensions spiral, Trump has indicated that he would like Xi to call him. Trump officials have also suggested to Chinese diplomats that Foreign Minister Wang Yi reach out to Secretary of State Marco Rubio, people familiar with the matter said. Beijing has refused to engage on either front. Instead, policy advisers in Beijing are floating names including former Singaporean Prime Minister Lee Hsien Loong and Mohammed bin Salman, crown prince of Saudi Arabia, as potential intermediaries with Trump, the people said. But those discussions haven't gone far enough in either capital, as neither side appears to be in a rush to negotiate. In remarks to reporters on Thursday, Trump reiterated that the U.S. will "make a deal" with China. "We've had some very good talks," he said. "I think we have a lot of time." As the communication stalemate continues, both powers are seeking to recruit allies in their battle. The Trump administration is seeking to cut deals with dozens of countries to cooperate in isolating China. Meanwhile, Xi and his senior lieutenants have fanned out, trying to pull trading partners away from the U.S. In exchange for reductions in tariffs imposed by the U.S., the Trump administration is planning to pressure more than 70 nations to bar China from shipping goods through their countries to the American market, restrict Chinese

investments and prevent cheap Chinese products from flooding their markets. Building coalitions is unlikely to be easy for either side. Even though Beijing's manufacturing overdrive has antagonized many countries, some of them, especially those in Asia that count China as one of their biggest trading partners and sources of investment, are finding it very hard to completely pivot to the U.S. Another focus for Beijing's charm offensive is Europe, which has been alarmed by Trump's handling of Russia's war in Ukraine. Meanwhile, Washington is likely to further restrict Chinese companies' access to American technology, making it even more difficult to unwind the trend of economic separation. Ryan Fedasiuk, a former China policy adviser at the State Department and currently an adjunct professor at Georgetown University, said the Commerce Department may significantly increase its use of export controls by blacklisting subsidiaries of the Chinese companies already on its trade list. If the economic warfare keeps accelerating, foreign-policy experts say, both sides may expand their toolboxes to strike back at each other. "What we're seeing now is the biggest trade war in history," said Yun Sun, director of the China program at the Stimson Center, a Washington think tank. "The risk of the trade war expanding to other domains is quite high."







WORLD NEWS

Lower Oil Prices Likely to Hit Russian War Budget

By GEORGI KANTCHEV  
And ALAN CULLISON

Russia dodged “Liberation Day” tariffs, but Moscow is still perilously exposed to President Trump’s trade war for one reason: oil.

The industry is both the engine of the Kremlin’s economy and its primary source of vulnerability; oil and gas make up around a third of state-budget revenues.

Global crude prices slumped this month and remain volatile after tariff-fueled recession fears gripped the market. Russia’s benchmark Urals blend is hovering under \$55 a barrel, far below the target in this year’s budget of around \$70. Some analysts say that if prices remain low, the Russian economy will face a hard landing, in which case they project the budget deficit would nearly double this year.

The downturn comes at a

precarious moment for Moscow: Its economy was already slowing before the collapse in the price of oil, and the Kremlin is engaged in negotiations with the U.S. over a cease-fire in Ukraine. Moscow has been slow-walking the peace talks, calculating that battlefield gains give it more leverage for maximum concessions in those discussions.

While current oil prices aren’t likely to push Russian President Vladimir Putin to abandon his military campaign, a deeper fall could alter his calculus. Meanwhile, the oil squeeze will leave the Kremlin with painful choices, some analysts say.

“If the oil price stays down, they will feel the pinch, and they are already beginning to feel the pinch,” said Elina Ribakova, a nonresident senior fellow at the Washington-based Peterson Institute for International Economics. “If



Global crude prices slumped this month and remain volatile.

this continues, they will face a guns versus butter situation.”

Analysts at J.P. Morgan wrote in a note to clients that despite Moscow becoming more insulated from global trends because of sanctions and severed international ties, “the tsunami generated by U.S. trade policy is unlikely to leave Russia unscathed.”

national-security threat.

Oil’s central place in the Russian economy extends beyond the state coffers. In what economists call a multiplier effect, a booming energy industry cascades to adjacent sectors: steel for the pipes used by oil drillers, for instance, or construction for the infrastructure around refineries. A contraction would have knock-on effects across whole communities that are organized around and supported by the extraction and refining of crude.

Sergey Vakulenko, a former Russian energy executive and senior fellow at the Carnegie Russia Eurasia Center, estimates that Russia loses about \$25 billion a year for every \$10 drop in the oil price. “So this does affect revenues, and the economy as a whole is deprived of money,” he said.

Still, some analysts say oil prices would need to stay lower

for a prolonged period to have an effect on Moscow’s war plans. Moscow sources heavy munitions from North Korea and produces its own steel for tanks and machinery. China has shored up Moscow economically throughout the war.

Soon after Trump took office, his administration signaled that it might try to bludgeon Moscow into making peace by calling for a boost in oil production in the U.S. and Saudi Arabia, thus lowering prices. Trump’s special envoy to Ukraine, Keith Kellogg, suggested in January that if oil fell to \$45 a barrel it might be enough to end the conflict.

When Trump unveiled a global list of targets for higher tariffs, Russia, along with only a handful of other countries, was missing. The administration said the numerous sanctions on Russia had precluded meaningful trade with Moscow.

U.S. Bids To Pressure Ukraine

Continued from Page One

said. “But we need to figure out here now, within a matter of days, whether this is doable in the short term. Because if it’s not, then I think we’re just going to move on.”

The U.S. diplomatic push is intended to set the stage for a cease-fire, which would be broadly along current battle lines, and an eventual settlement. Accepting some of the Trump administration’s ideas could prove very difficult for Kyiv since Ukraine has refused to accept that Russia has a legal claim to any of its territory.

A senior State Department official on Sunday cast the ideas presented to the Ukrainians as options for Kyiv to weigh and not a take-it-or-

leave-it proposition. The official said a “list of potential options” was shared “for discussion and feedback.” A spokesman for the National Security Council didn’t respond to a request for comment.

Rubio, Special Envoy Steve Witkoff and Keith Kellogg, the retired Army lieutenant general who serves as an envoy to Ukraine, met on Thursday in Paris with senior Ukrainian officials, including Andriy Yermak, a top aide to President Volodymyr Zelensky, Defense Minister Rustem Umerov and Foreign Minister Andrii Sybiha.

Rubio, Witkoff and Kellogg are planning to attend the coming London meeting, according to U.S. officials. After that, Witkoff might make another trip to Russia, though no travel has been announced.

Witkoff, a real-estate executive who is close to Trump, has met with President Vladimir Putin of Russia three times and reported that he has made progress in his discussions with the Kremlin leader. Other

U.S. officials have advised Trump to be more skeptical of Putin’s intentions.

A U.S. move to recognize the Russian seizure of the Crimean Peninsula in 2014 would reverse a more than decade of American policy from both Democratic and Republican administrations. In 2018, Secretary of State Mike Pompeo, who served during the first Trump administration, denounced Russia’s annexation of Crimea as a threat to “a bedrock international principle shared by democratic states: that no country can change the borders of another by force.”

The U.S. Congress has passed legislation opposing U.S. recognition of Russia’s annexation of Crimea.

The ideas presented by the U.S. in Paris also include ruling out Ukraine’s membership in NATO. “NATO isn’t on the table,” Kellogg said on Fox News Saturday.

Yet another U.S. idea, Western officials said, calls for designating the territory around the nuclear reactor in

Zaporizhzhia as neutral territory that could be under American control.

In a March call with Zelensky, Trump raised the possibility that the U.S. would acquire Ukrainian power plants, including nuclear facilities, calling it the “best protection for that infrastructure.” The Zaporizhzhia plant, the largest in Europe, would then presumably feed power to both Ukrainian territory and regions that Moscow has conquered since its invasion of 2022 and that remain under its control.

The ideas put forth by the Trump administration fall short of some Russian demands, according to Western officials. They wouldn’t concede that Russia has a legal right to control four regions in the eastern part of Ukraine that it claims to have annexed,

though they don’t require the Russian military to leave those areas. The U.S. willingness to consider recognizing Russia’s annexation of Crimea was earlier reported by Bloomberg.

The U.S., Western officials say, is also not proposing a cap on Ukrainian forces and isn’t precluding Western military support for Kyiv or the deployment of European troops there, a sticking point with Moscow.

“Every sovereign nation on Earth has a right to defend itself,” Rubio said Friday. “Ukraine will have a right to defend itself and to enter into whatever agreements it wants to enter into on a bilateral basis with different countries and so forth.”

While the Trump administration temporarily withheld arms and intelligence from Ukraine to pressure it to seek

a diplomatic compromise with Russia, it hasn’t imposed economic sanctions or taken any concrete steps to pressure Putin.

Ukraine has said it is willing to agree to a comprehensive 30 day cease-fire if the Kremlin would follow suit. Putin declared a short cease-fire for the Easter holiday but Ukrainian officials said Moscow continued to carry out attacks during the holiday.

There is still no resolution regarding what types of security guarantees Ukraine might receive if they agree to a peace settlement. The Trump administration hasn’t said if it is willing to provide any military support to European nations that send troops to Ukraine as part of a “reassurance force” to deter future Russian aggression.

The Kremlin has sought a lifting of U.S. sanctions and rekindling economic ties between Russia and the U.S. during talks in Moscow and Saudi Arabia that Putin’s special envoy, Kirill Dmitriev, has taken part in.

The possible concession on Crimea would overturn 11 years of U.S. policy.

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WORLD NEWS

China Races Humanoid Robots in Half-Marathon

Humans ran with 21 robots in a showcase of Beijing’s cutting-edge technology

By Yoko Kubota  
And Raffaele Huang

BELJING—Metal met asphalt in a half-marathon that featured thousands of human runners—and 21 Chinese humanoid robot models.

Saturday’s road race involving human runners and a score of robots in Beijing has been billed as a showcase of China’s cutting-edge technology.

Indeed, the robots, and their developers, had much at stake in the high-profile event. If the robots fell, froze or got lost during the widely promoted event, it would be an embarrassment for their developers.

The 13-mile race—which its organizers said was the first of its kind—was a chance for China to show off the latest state of its humanoids, one of several technology areas at the forefront of the U.S.-China tech rivalry. China has said it wants the country to be a world leader in humanoid robots by 2027. Chinese authorities have lavished support such as subsidies, talent bonuses and tax breaks on robotics companies.

In reality, the race showed both how quickly and smoothly some robots are able to run, but also how far away humanoids still are from being able to mimic human activity.

“Running is a very basic ability of human beings,” said Tang Jian, the chief technology officer of Beijing Humanoid Robot Innovation Center, which developed one of the robots, Tien Kung Ultra. “Since we are making human-like robots, we want to give them as many basic human abilities as possible,” which would be critical in eventually deploying them for industrial purposes.

China is making a major push to produce more and



Tien Kung Ultra, a humanoid robot, runs across the finish line. Below, a robot, developed by Noetix Robotics, in sneakers.



more sophisticated robotics in part to raise the automation level of its factories.

Before the race, thousands of human runners crowded in one lane behind the starting line, many jogging in place to keep warm. The humanoids, separated by fences into another lane from the human runners, stood largely still as they waited for the start.

When a siren marked the start of the race at 7:30 a.m., the human runners dashed and spilled onto the course. But the robots’ starts were staggered by a minute or two each to prevent them from ramming into each other.

The first robot off the mark was Tien Kung Ultra, a 5-foot-9 inch, 115-pound humanoid, featuring a pitch-black head and sporting an orange tank top. Three people accompanied it to help control the robot.

The race was the culmination of months of training for the humanoids, which had to

master skills that are innate to humans. They had to navigate the course’s flat and hilly roads and maneuver around six left turns and eight right turns, according to the organizers. Developers had to train the robots to keep their stability and balance to avoid falling over a half-marathon distance.

their batteries. The faster they run, the shorter the distance they can cover.

Components and parts could easily break while running, so developers replaced plastic parts with metal and used extra-strong—but costly—materials.

Tien Kung Ultra was developed by Beijing Humanoid Robot Innovation Center, a research institute also called X-Humanoid and formed by robotics firm UBTech, electronics and electric-vehicle maker Xiaomi and the local Beijing government. It could run on average 6 miles an hour and could handle hills, stairs, grass and sand, a profile of Tien Kung Ultra posted online showed.

X-Humanoid revamped an existing robot model for the half-marathon. Tien Kung Ultra has longer legs than its original model and is taller by around 4 inches, helping it run faster, state media re-

ported. The developers also used stronger materials to make the hip joints of the robot and installed a pair of sneaker soles on the robot’s feet to absorb shocks and reduce damage to its legs and feet.

Tien Kung Ultra largely breezed ahead of other robots on the route. It fell once when the battery failed, Tang said. The robots were allowed to swap batteries, and Tien Kung Ultra changed its battery three times.

While many robots were directed by a remote control, Tien Kung Ultra used wireless tracking technology that allowed it to communicate with a person running in front of it.

Besides Tien Kung Ultra, there was Xiaowantong N2, or Little Rascal N2, a 4-foot, 66-pound childlike robot developed by Noetix Robotics. At 5 feet 6 inches, Huanhuan was the only one with a head resembling a human. Another robot called Shennong had propellers.

Some were steady and well balanced while others walked—barely. Some never made it much further beyond the starting line. Huanhuan wobbled, at times heading in the wrong direction, then ended its race quickly by sitting down on the course and refusing to go further.

Shennong was particularly disastrous, swinging wildly just after the start, causing the human support runner who was trying to control it to trip. The robot slammed into a fence, breaking into pieces. Shennong’s arm dangled as it left the course.

The human male champion completed the race in one hour, two minutes and 36 seconds.

After two hours, 40 minutes and 42 seconds, Tien Kung Ultra was the first robot to reach the finish line. A large crowd of spectators, including government officials, was eagerly awaiting the robots. Many surrounded Tien Kung Ultra to take photos. The robot maintained its blank expression.

WORLDWATCH



MILK RUN: A not-yet-named Asian elephant calf born early Saturday to 19-year-old Farha—already the mother of three—stuck close to mom at Switzerland’s Zurich Zoo on Sunday.

YEMEN

U.S. Airstrikes Target Capital

U.S. airstrikes targeting Yemen’s capital killed 12 people and wounded 30 others, the Houthi rebels said early Monday.

The deaths mark the latest in America’s intensified campaign of strikes targeting the rebels. The U.S. military’s Central Command didn’t acknowledge the strikes.

The Houthis described the strike as hitting the Farwa neighborhood market in Sanaa’s Shuub district. That area has been targeted before by the Americans.

Strikes overnight into Monday also hit other areas of the country. The strikes come after U.S. airstrikes hit the Ras Isa fuel port in Yemen last week, killing at least 74 people and wounding 171.

—Associated Press

LEBANON

Suspected Rocket Plotters Detained

Lebanese authorities detained several people who were allegedly planning to launch rockets into Israel and seized the weapons, the military said Sunday. The army said information on this planned attack was obtained as military intelligence investigated following the detention last Wednesday of several people, including a number of Palestinians, allegedly involved in firing rockets toward Israel in two separate attacks in late March that Israel answered with airstrikes.

The army said troops raided an apartment near the southern port city of Sidon and seized some of the rockets and the launchers and detained several people involved in the operation.

—Associated Press

EL SALVADOR

Bukele Seeks Swap With Venezuela

Salvadoran President Nayib Bukele on Sunday proposed exchanging 252 Venezuelan deportees from the U.S. that his government has kept imprisoned for an equal number of what he called “political prisoners” in Venezuela. In a post on X directed at Venezuelan President Nicolás Maduro, Bukele listed a number of family members of high-level opposition figures, journalists and activists detained during an electoral crackdown in Venezuela last year.

“The only reason they are imprisoned is for having opposed you and your electoral fraud,” he wrote to Maduro. Among those he listed were what he said were 50 detained citizens from a number of countries around the world.

—Associated Press

Fans Covet Hard-to-Get Ceramics

Continued from Page One

The transformation of a line of humble ceramics into quiet status symbol is owed to a combination of location, a pivotal alliance with a Michelin-starred chef and its creator’s refusal to keep pace with soaring demand.

Bornholm, which is closer to Sweden than mainland Denmark and has fewer than 40,000 residents, is home to a disproportionate number of famous ceramics studios. Oh Oak supplies tableware for Michelin-starred KOKS in the Faroe Islands, while Anne Mette Hjortshøj’s beakers have been on FX’s hit show “The Bear.” And Lov i Listed has become a bona fide tourist attraction.

No one is more surprised than Torben Lov.

The 73-year-old grew up in Bornholm and completed a ceramics apprenticeship on the island. He and his wife Susanne Lov, 68, launched Lov i Listed in 2002 with a set of dinnerware—20 simple components in five serene colors. He made enough to fill the shelves of his store, crafting the dishes in his modest studio with old-fash-

ioned metal weights, hand-scribed notes and a foot pedal-operated wheel.

Even in the beginning, the product moved fast. Reasonable prices help. Plates cost between \$30 and \$75; a salt cellar is \$17. Not West Elm, but not Ginori either. If there’s a secondhand market, Lov doesn’t know about it. It seems those who’ve secured a set aren’t keen to part with it.

The potter uses clay imported from Germany, which he measures, shapes and fires himself. He has at various points tried to increase production with hired hands. No one has ever performed to his standards. “Things take time,” Lov said, shrugging.

The people of Bornholm—steeped in the “age-old tradition” of ceramics on the island, as Susanne puts it—paid attention from the start.

That included Nicolai Nørregaard, the chef and owner of Michelin-starred restaurants Kadeau Bornholm and Kadeau Copenhagen. Nørregaard was born in Bornholm, and his father and Lov lived across the street from each other as children. Both dabbled in ceramics. Nørregaard’s father gave up the practice. Lov stuck with it.

When Nørregaard opened Kadeau Bornholm in 2007, it was a foregone conclusion that his table would be laid with local dinnerware.

“In the town where I grew

up, I think around 1,100 people lived there, and there were like seven or eight ceramics shops,” Nørregaard said. “No one had white porcelain plates. That was just not natural. Ceramics was something we would trade with. The potters paid the dentists with it.”

For the first few months of Kadeau’s existence, Nørregaard cobbled together a set of plates and bowls, saving up for finer pieces as the restaurant found

its footing. But within the year, he reached out to his father’s old childhood friend.

Now, almost every single dish that comes out of Nørregaard’s kitchens is served on a plate or bowl that Lov has made. The potter redesigns the service each year, building out what is in essence a custom line created to complement the restaurants’ latest seasonal menu.

Nørregaard estimates he purchases about 500 pieces for

each new season. Archived designs sit in a storeroom; Nørregaard knows he could sell them—and has unloaded some in the past—but he likes his collection. He now owns more of Lov’s dishes than Lov does.

While Nørregaard never meant to monopolize his friend’s time, he knows the line’s elusiveness is part of its appeal.

To maintain it, the trio now have an unofficial exclusivity agreement. Lov doesn’t take

commercial commissions, and Nørregaard respects Lov’s process. When Nørregaard’s own friends ask him to call in favors, they receive the same advice the clamoring public does: Better get in line.

That’s painful guidance for the swarms of visitors from as far as the U.S. and Japan who have tried.

On Instagram, the comments section beneath posts sharing the store’s hours are filled with disappointed shoppers. “Ouch, ouch, ouch,” one woman wrote. She was on her third visit to the island; it would be her third thwarted attempt.

Schubert, the German diner, tried to visit Lov i Listed after his dinner at a restaurant Nørregaard used to operate at the Hotel Nordlandet. Predictably, it was closed. Later, he attempted to reach out to Torben and Susanne personally, but was told they were too busy to take new commissions.

More than five years after his initial trip to Bornholm, he received an email from Susanne. He still couldn’t place an order, but he could tell her what he wanted and she would see what was possible.

In the end, Schubert was granted a set of plates, bowls and mugs, 26 pieces total for which he paid about 1,000 euros, or around \$1,100.

“A lot of money for simple ceramics,” he said. “But worth every cent.”



Torben and Susanne Lov opened their business in Bornholm, Denmark, in 2002, and locals quickly cleaned out their shelves. Right, Torben Lov makes all the pieces.





FROM PAGE ONE

Shooter  
Asked Police  
For Help

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version of events, citing interviews with witnesses and the two shooters.

Spivey's actions on Sept. 9, 2023, driving erratically and pointing a gun at people and vehicles along Highway 9, had led to his death, Detective Alan Jones wrote in his initial report. It looked like a case of road rage, he said.

Prosecutors later came to the same conclusion. Under a South Carolina statute known colloquially as the stand-your-ground law, Boyd and Williams were permitted to shoot in self-defense after Spivey brandished a gun. Boyd said one of his bullets killed Spivey.

A prosecutor from the state attorney general's office closed the case a year ago.

Spivey's sister, Jennifer Foley, didn't believe any of it—not that Spivey shot first or that Boyd had a right to kill him. She suspected the road rage was Boyd's, angry at her brother, who had waved a gun out the window, then pulled his truck in front of Boyd's and braked.

Boyd had called 911 to report Spivey, but he chased him instead of letting police handle it, Foley said. She believed Boyd intended to capture or punish her brother himself. Boyd told police he followed Spivey to help officers and protect other drivers. He and Williams have denied any wrongdoing.

Boyd and Williams had only to say they feared for their lives when confronted by an armed man to be shielded under the state's stand-your-ground law, one of the most permissive in the U.S. That standard makes testimony difficult to refute without substantive evidence to contradict it.

After feeling stonewalled by Horry County police for more than a year, Foley got an electronic copy of the police file in February. Within hours, Foley, a 37-year-old high-school biology teacher, learned Boyd had more than the law on his side.

In a phone call minutes after the shooting, Boyd asked for help from Brandon Strickland, deputy chief of the Horry County Police Department. Strickland told Boyd that he would protect him—handpicking the investigators to send to the scene.

"My people," Strickland called them. In nine calls recorded over four days, Strickland, who was in charge of criminal investigations, advised Boyd and kept him apprised of the case.

Their conversations are among more than seven hours of recordings made by an app on Boyd's phone before, during and after the fatal shooting. Boyd's calls with Strickland and others are included in the 70-gigabyte police file.

Foley and her family were denied access to the file because Spivey wasn't classified as a victim under the stand-your-ground law. In the state's view, Spivey was responsible for his own homicide.

Boyd himself opened the police file to scrutiny. After the Spivey family filed a wrongful-death lawsuit against Boyd last year, his lawyer pressured police to turn over the file to help in Boyd's defense. That allowed Foley and her family their own copy.

This account is based on interviews, 911 calls and the police file, which includes body-camera and dash-camera footage from the night of the shooting, commercial surveillance-camera footage of the chase, text messages and Boyd's phone recordings. Boyd and his lawyer, Kenneth Moss, declined to comment.

Foley first listened to the recordings over Super Bowl weekend.

"I got the right people. I got the people coming that need to come," Strickland told Boyd minutes after Spivey was killed. The following day, Strickland said to Boyd, "You're taken care of."

Boyd relayed Strickland's assurances to Williams, the other shooter. "When I tell you we had a lot of my friends work together on this for us? We definitely did," Boyd said.

The recordings led Foley and her family to believe



Clockwise from top, the windshield of Weldon Boyd's truck, showing a dash-mounted Samsung tablet; former Horry County Police Deputy Chief Brandon Strickland; Weldon Boyd in February during a deposition for a wrongful-death lawsuit.



CLOCKWISE FROM TOP: SHELLENE TAMAS/HORRY COUNTY POLICE DEPARTMENT; HORRY COUNTY POLICE DEPARTMENT (BRANDON STRICKLAND)



Scan this code to hear recordings from the second part of this series.

*'I chased him. Oh, I was on his ass, and his truck couldn't outrun my truck, and he knew it. So, yeah, he was terrified.'*

Weldon Boyd in a conversation with his mother about Scott Spivey.

Spivey's homicide investigation was tainted from the start. She didn't know if police, prosecutors, Boyd or Boyd's lawyer had listened to the recordings—or whether Boyd's conversations were missed, ignored or concealed.

Despite promises of protection from a high-ranking friend, Boyd had a nagging worry.

Spivey's family might say that "I knew that the guy had a gun, but yet I still pursued him as if I was looking for a fight, and that Scott was scared and got out and thought he was defending himself against me," Boyd told his mother two days after the shooting.

"They're gonna make me out to be a murderer," he said.

Kill shot

Boyd and Williams were en route to Bonnie Bay farm, a pick-your-own-blueberry attraction and firing range that Boyd owned, when Spivey pulled his Silverado truck alongside them and waved a .45 caliber pistol. Spivey pulled in front and braked, forcing Boyd to steer into the highway median.

Boyd got his Dodge Ram TRX back on the road and went after Spivey.

"He keeps throwing the gun in our faces, acting like he's about to shoot us. If he keeps this up, I am going to shoot him," Boyd told the 911 operator. "He's trying to run from me now."

Shortly before 6 p.m., Spivey turned onto Camp Swamp Road, stopped and got out of his truck, not far from the house where he and his sister had lived as children.

The gunfire was recorded on the 911 call, and when it ended, Boyd called his mother. "You and Dad please get here ASAP," he said. "Mama, I killed him."

Boyd got out of his truck and exhorted other drivers who had stopped. "He f—ing shot at us!" he said. "Did you see him do that to us?"

At 6:10 p.m., Horry County police officer Kerry Higgs arrived with lights and siren going. "He shot us first," Boyd told him.

Horry County Fire Rescue paramedics and firefighters arrived. Four of them examined Spivey's body and identified a major wound in his torso. The bullet appeared to have entered his back. They were surprised: *Shouldn't the entrance wound be in his chest?*

"He was turned around, facing backwards," Higgs re-

plied. He demonstrated his theory by twisting his head to one side and pretending to shoot over his shoulder.

Paramedic Jeremiah Zimmerman followed up with Higgs. "A question, because I'm nosy," Zimmerman said. If Boyd shot him in the back, he said, "is he at fault now?"

"No," Higgs said.

Early the next morning, Jones sent his initial findings to colleagues. "All indications are that the actions of Mr. Williams and Mr. Boyd are justified," he said.

Good to go

Four days after the shooting, Foley spoke by phone with Jones. The detective said Boyd and Williams were cooperating, and their lawyer was arranging for them to turn over their phones. Jones would give her an update in a few weeks.

"Let me assure you of this," Jones said in the recorded call. "I was at that scene. Everybody was a victim, everybody was a suspect. Everybody still is a victim. Everybody still is a suspect."

Jones's boss had a different story. A day after the shooting, Strickland told Boyd, "I can't officially tell you this, but I'm gonna tell you, you-all are good to go," he said in one of four recorded calls that day.

Strickland told Boyd in a recorded call that he personally picked Jones for the case and had said to the captain overseeing the investigation that Boyd was a good guy.

Strickland told Boyd that the case would be wrapped up by the end of the week. He apologized about police seizing Boyd's gun and truck, and he said they would be returned shortly.

Both men laughed when Strickland offered to buy Boyd's 2022 Dodge Ram for \$10,000, a fraction of its value.

Strickland said he was serious, though, about buying the 5-carat diamond engagement ring that Boyd's ex-fiancée had returned. The deputy chief said his wife had seen the for-sale post and liked that the diamond was bigger than her own.

"We'll work something out," Boyd said. Then they talked about arranging a guy's night out with their friends in law enforcement after the investigation was over.

Boyd told his mother that he had checked Spivey's body before police arrived. "I told Bradley, I said, 'There's no blood on his back, so we did not shoot him in the back at any point.' And Bradley said, 'Thank God, because that's bad in a self-defense case. You don't want a bullet going through their back.'"

Days after the shooting, Boyd called Jones to complain about anonymous callers to his restaurant who accused him of murder. Boyd said he wanted the police department to put out a statement clearing his name.

Jones, who was in Charleston, S.C., awaiting the conclusion of Spivey's autopsy, told Boyd he had nothing to worry about.

"Chill," the detective said, "let everything kind of work itself out, bud."

The big game

In November 2023, Boyd and Williams gave police their phones and Boyd's dash-mounted Samsung tablet. In response to the Spivey family's complaints, the South Carolina Law Enforcement Division agreed to review the police file, which included the calls between Boyd and Strickland.

The state finished its review on Jan. 26, 2024. An agent added a note to the case file saying, "no additional evidence was discovered."

Foley called lawyer Mark Tinsley about representing the family. Tinsley had never sued a defendant in a stand-your-ground case, largely because the law grants civil immunity to the killer. In fact, he found no record of any lawyer filing such a case in South Carolina.

Tinsley agreed nonetheless, saying he was offended by the idea that a person could follow a driver for 9 miles and then claim self-defense for killing him. "How do you stand your ground when you're chasing someone else's?" the personal-injury lawyer said.

Last April, a state prosecutor told the family that the case would be closed without charges. On June 3, the family filed a wrongful-death lawsuit against Boyd and Williams.

On the Friday of Super Bowl weekend, Foley picked up a copy of the voluminous police file, which was encoded in a flash drive. Foley was determined to cover as much ground as she could before Monday, when Boyd was scheduled to appear at a deposition in the lawsuit. By kickoff in the Chiefs-Eagles game that Sunday, she found the Boyd recordings.

At 6:57 a.m. Monday, Foley texted Tinsley.

"I've been up all night going through the evidence files," she wrote. "Please call me."

'They took an oath'

Three hours later, Tinsley began questioning Boyd at Moss's law office. Tinsley planned to submit Boyd's sworn testimony at a court hearing where a judge would decide if the family's lawsuit could move forward.

Tinsley sought to persuade the judge that Spivey's killing wasn't done in self-defense. He hoped to show first that Boyd wasn't truthful about his role in bringing about the confrontation and that he was angry, not afraid for his life when he shot Spivey.

The lawyer didn't bring up Boyd's recorded conversations in the police file.

Tinsley asked Boyd, "If you chased Mr. Spivey, do you think you can claim self-defense?"

"I didn't chase Mr. Spivey," Boyd said.

Yet Foley had

found a recorded call of Boyd telling his mother a different story: "I was, like, 'He just ran me off the road and aimed a gun at Bradley's head? F—this guy,' and I chased him. Oh, I was on his ass, and his truck couldn't outrun my truck, and he knew it. So, yeah, he was terrified."

"You're not gonna apologize for anything, right?" Tinsley asked later in the deposition.

"No, sir," Boyd said. "Because you loved it?" Tinsley asked.

"I wouldn't say that. I might have—I mean, I might have made a comment that I'd like to see if you'd show it to me," Boyd replied, "but this was a pretty bad situation."

In the police file, there is a recording of Boyd telling Williams on the day after the fatal shooting: "I know it's f— up to say, but I had a f—ing blast," he said. "I had a good time."

Tinsley asked Boyd whether he used the dash-mounted tablet to make a video during his encounter with Spivey. "Did you use that at all?" he said.

"No, sir," Boyd replied.

State investigators in February 2024 received a screenshot of a text Boyd sent to his ex-fiancée that said, "I have video of" Spivey aiming his gun at cars, according to a spokeswoman for the state law enforcement division. The state sent that and other material to Jones, the spokeswoman said. The text about the video doesn't appear in the police file.

When Boyd's tablet was returned to police in November 2023, it contained no videos of the chase or the gunfight. The tablet indicated that videos had been saved on an external memory card, which isn't in the police file. "As far as I know, the tablet was not utilized at all and there were no images or recordings from the tablet," said Moss, Boyd's lawyer.

Two weeks later, Tinsley spoke with Horry County Attorney David Jordan about the case. "Have you listened to all these calls?" Tinsley asked. "Because if you haven't, you need to."

Jordan called Tinsley days later. "How the f— weren't these guys charged," Tinsley recalled him saying. Jordan briefed the Horry County police chief and county administrators about the recordings. He didn't respond to requests for comment.

Strickland was forced to resign from the police department on March 11. Horry County Police Chief Kris Leonhardt said in a news conference the next day that he asked state and federal authorities to review the case.

Bert von Herrmann, Strickland's lawyer, said his client did nothing to influence the investigation.

Strickland's recorded assurances to Boyd were "all bluster," von Herrmann said. "That is him trying to comfort Weldon, and probably quite frankly, a little self-aggrandizing."

Horry County police detectives are assigned on-call duty in advance, meaning that Jones responded to the Spivey homicide because it was Jones's turn, according to an Horry County spokeswoman. Horry County police didn't make Jones available for comment.

Tinsley alerted state officials to the audio recordings in early March. The South Carolina Law Enforcement Division announced on March 13 an investigation into Strickland's conduct, but not the Spivey killing.

A prosecutor from the state attorney general's office recently listened to the recorded conversations, a spokesman said. The recordings didn't change last year's decision to close the case without charges, the spokesman said.

Foley said she and her family want authorities to reopen the homicide case, in light of all they have learned. Her brother, she said, deserved better.

"Horry County did not do the investigation they needed to do," Foley said. "They took an oath to do the job, and they didn't do it."

Part 1 of the Series



Scan this code to read how two drivers crossed paths on Highway 9 and one ended up dead.



Jennifer Foley arranges a memorial to her brother, Scott Spivey, on the mantel of her parents' home in Tabor City, N.C.





tive Healthy Eating Index,” a plant-rich diet plus proteins like dairy and fish, had an 86% higher chance of healthy aging than those whose diets scored lowest for that diet. Sticking to a plant-based diet with limited animal products also appeared to make a difference, but less so. Diets high in trans fats, salt, sugary drinks, and red or processed meats, like steak, ham and sausages, were linked to worse healthy-aging results.

**Balance is key** Other studies have looked at whether specific diets help people live longer. One published in 2023 found that sticking to similarly healthy diets was linked to a lower risk of premature death. Still, robust nutrition research is notoriously difficult to conduct. Many studies rely on participants recalling what they ate over weeks, months, or years, which can be imprecise at best and inaccurate at worst. This makes it impossible to know how much sugar or saturated fat people were actually eating, says

Eric Brown, a research scientist in the National Institutes of Health's Pediatric Translational Research Branch, who wasn't involved with the study.

The questionnaires also don't account for *when* participants eat. Some research suggests timing, such as whether people eat earlier or later in the day, can affect health markers like blood pressure, as well as how the body processes sugar. Other factors can skew results, too.

“If you eat a lot of vegetables, that could also be a marker for that you prepare a lot of home-cooked meals and that you eat fresh foods,” says Inge Lindseth, a registered dietitian and nutrition researcher in Oslo, Norway, who wasn't involved with the study.

The researchers tried to account for such issues by statistically controlling for factors such as participants' body-mass index, how much they exercised, and their financial standing.

Repeating the research in more diverse populations will help determine whether the results apply more broadly, says Tessier. Meanwhile, she adds: It's still important to remember diet isn't a one-size-fits-all approach.

“This doesn't mean ignoring personal preferences or cultural traditions,” she says. “Balance and enjoyment are definitely key.”

# Your Midlife Eating Habits Can Help You Live Healthier

A plant-rich diet with some fish and dairy may make the biggest difference, new research says

By ALEX JANIN

If you reach 70 years old healthy and disease-free, you may have your diet to thank, new research suggests. The study, which analyzed the health data of more than 100,000 people over 30 years, is the first of its kind to examine how dietary patterns affect not only lifespan, but how well people age. The upshot: A diet emphasizing vegetables, fruits, healthy fats and some animal products like fish and dairy over the long term was the most likely to make a difference in aging in good health. Sticking to a balanced diet with a moderate amount of healthy animal proteins, in fact, outperformed following a more plant-based diet, in terms of healthy aging. The researchers also found that eating more ultra-processed foods was

linked to worse physical *and* cognitive health in older age. The study, published in Nature Medicine and led by researchers at Harvard University, the University of Copenhagen, and the University of Montreal, adds to the growing body of research on people's healthspans—the number of years we live in good health—and the factors that can extend those years. Though Americans are living longer, they spend less of their later years in good health. One reason is that medical advances have turned once-fatal conditions into chronic ones. Another is the rising prevalence of conditions such as obesity, diabetes and substance-use disorders. “Staying active and independent is a priority from a public health standpoint, and for many of us,” said Anne-Julie Tessier, an assistant professor at the University

of Montreal who led the study. The dietary habits most closely linked to healthy aging, she added, likely work by reducing inflammation, improving metabolism, and supporting gut health. **What is healthy aging?** The researchers defined healthy aging as reaching 70 in good cognitive, physical and mental health and without developing a major chronic disease such as cancer, diabetes or heart disease. Just under 10% of the people tracked over the 30-year period hit that milestone, they found. To examine the connection to diet, the study's authors looked at three decades' worth of questionnaires that asked health professionals in midlife to recall and record, on average, what foods they ate and how often. They filled these out every four years. For example,

how many shakes of salt did they add to food each day, and did they eat fresh or canned peaches? The data come from two well-known, long-term research projects in the health world—the Nurses' Health Study and the Health Professionals Follow-up Study, which were started at Harvard in the 1970s and 1980s and have been used to investigate a range of topics, from the effects of oral contraceptives to cigarette-smoking risks. Researchers assigned scores to how closely people's diets adhered to the key qualities of eight dietary patterns. They include various forms of the Mediterranean diet, which has been praised by doctors and scientists for everything from its brain-boosting to its anti-inflammatory benefits. People who scored high in adhering to the so-called “Alterna-

# ‘Andor’ Could Become Star Wars’ Only Hope

By JOHN JURGENSEN

**TONY GILROY** is a Jedi of screenwriting. He was a longtime Hollywood script fixer, wrote most of the Jason Bourne movies and was Oscar-nominated for “Michael Clayton,” a George Clooney legal thriller that film buffs and fellow writers speak of in reverential tones. If only Gilroy, 68, could sell everyone on the show he spent the last six years crafting. “I can't get people who have loved all my work to watch it because, you know, it's Star Wars,” he said. “It just drives me crazy.” “Andor,” the series he created, has a rarefied place in a galaxy far, far away. Hits like “The Mandalorian” have been harder to come by in recent years for the franchise, which has dealt with canceled movie projects and panned TV shows. That has raised fears of stagnancy among a fan community that's been more pissed off than passionate. “Andor,” which debuted in 2022 and earned an Emmy nomination for best drama series, became a bright spot for fans and the franchise's reputation, if not a ratings hit. It's a show for grown-ups featuring spycraft, heists and sophisticated political themes. Nobody brandished a lightsaber or mentioned the Force in the first season. The plot follows the sketchy beginnings of a rebel movement that exists as a staple of the Star Wars movies. “Andor” devotes equal time and human detail to the bad guys, an array of career-obsessed intelligence agents trying

to ferret out an insurgency. The story expands in the second and final season, launching April 22. The rebels (including the title character, a mercenary-turned-leader played by Diego Luna) fight among themselves and reckon with the bloody costs of escalating a revolution. Imperial bureaucrats use this insurrection as cover for a fascist power grab, planting news in the galactic media and implementing draconian laws. “It's the best thing I've ever done. It's the biggest thing I'll ever do,” Gilroy said, reciting his sales pitch to friends who are hesitant or oblivious about the show. “Get the freaking Disney+ app. If you don't like the show, write me, I'll send you a check.” The head writer and showrunner of “Andor” controlled his corner of the Star Wars universe from a cozy home office on the Upper West Side of Manhattan in New York. The townhouse where Gilroy has lived for 25 years became his headquarters when the pandemic halted preparations for “Andor,” then he kept it that way. At a desk on the ground floor, with a trio of prints by artist Robert Mangold at his back, he wrote scripts and honed those by other writers, including “House of Cards” creator Beau Willimon and Dan Gilroy, Tony's brother. (His other brother, John Gilroy, was an executive producer and the editor of “Andor.”) Using a proprietary Lucasfilm portal, Gilroy had all the raw materials for “Andor,” from audition videos to daily footage from the set.



The second and final season of ‘Andor,’ starring Diego Luna, launches April 22.

That allowed him to oversee a cast and crew of more than 2,000 people, most of them operating in a studio complex outside London. “The entire show was run from that cockpit,” he said during a recent interview there. “From 6 o'clock in the morning until the second vodka at the end of the day.” It's easy to see the political turmoil of 2025 in the plotlines of “Andor.” A senator who funnels secret funding to the rebellion (Genevieve O'Reilly) gives a speech invoking the gulf between truth and the government's version of it. Protesters who face off with imperial forces chant “the galaxy is watching.” Gilroy plays down the show's parallels to real-world news cycles, saying the writers were drawing

from history's repeated patterns of uprising and clampdown. He was into all this before “Andor.” “I never really spent much time in school, so you learn how to teach yourself as life continues,” Gilroy said. The son of a screenwriter and Pulitzer-winning playwright, Frank D. Gilroy, he dropped out of college to play music and tend bar while establishing his own writing career. Gilroy remembers going to the original “Star Wars” in 1977 and the excitement of being part of a mass happening. But he wasn't much interested in the narrative saga that carried on in later sequels and prequels. That detachment made Gilroy a mercenary when he was hired to revise a Star Wars script that would become the 2016 movie “Rogue One.” It was set in a time leading up to the original Luke Skywalker trilogy, and introduced Diego Luna's character, Cassian Andor. Gilroy's overhaul of

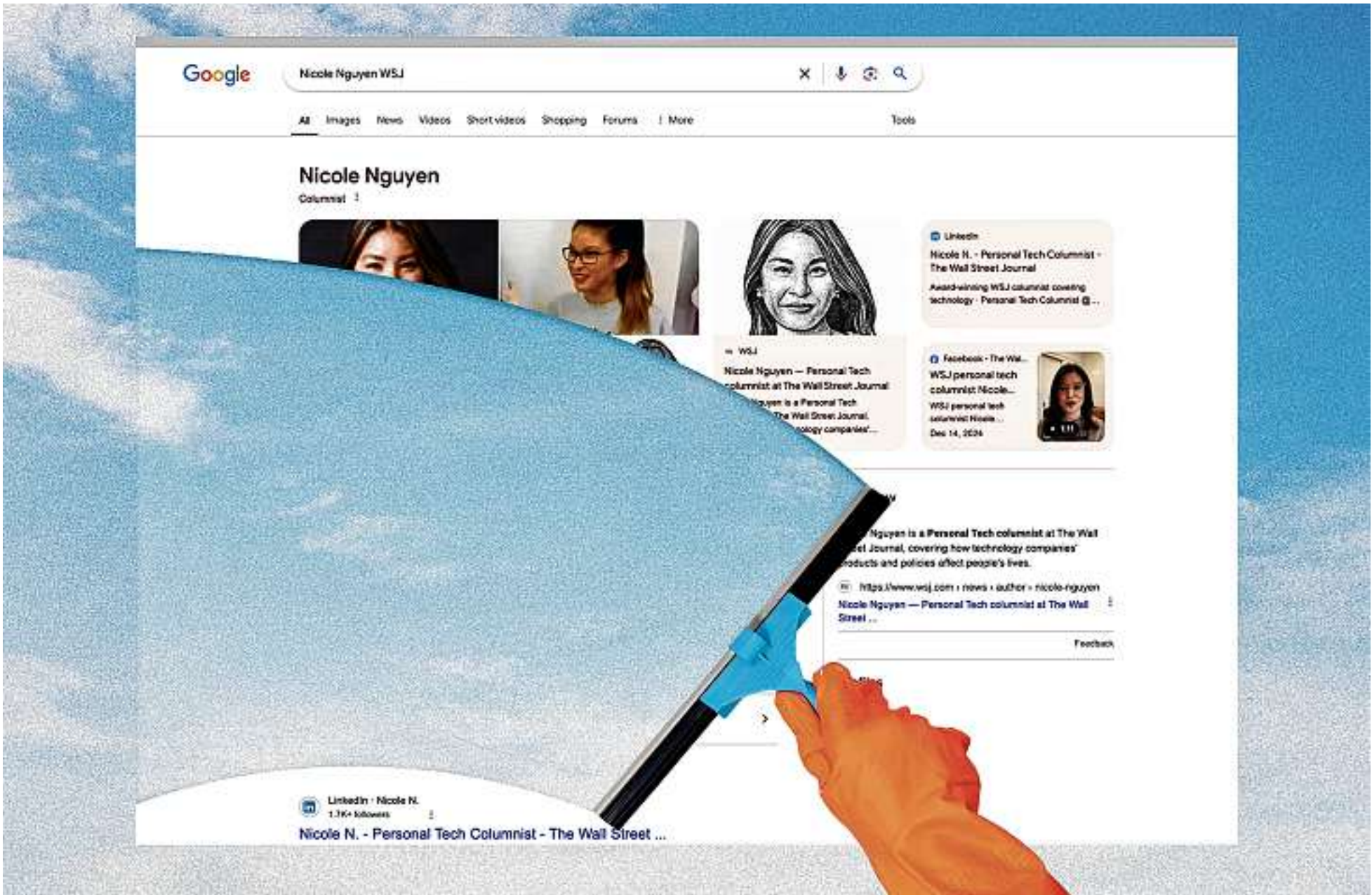
the movie resulted in the franchise's most hard-bitten version of the hero's quest to date. In the main characters' mission to steal the blueprint for the infamous Death Star, they sacrifice their lives. As a long-form prequel to “Rogue One,” “Andor” is about the radicalization of Luna's character. In season 2 he clashes over strategy with the architect of the rebellion, played by Stellan Skarsgård. The character, who leads a double life as a smiling antiquities dealer, pushes for explosive action, no matter the cost in lives, to trigger full-scale revolt. Gilroy compares Skarsgård's character to a tech founder: “He built a startup company in his garage and it has to go public, but secrecy doesn't scale up well.” Before he signed on as showrunner, Gilroy said, he had months of discussions with executives at Disney and Lucasfilm about his intention to “take the Latin mass out of the Church.” There'd be no Jedi knights; there would be a brothel. The “Andor” team wanted their actors grounded: “Don't play Star Wars for us,” Gilroy said. “There's an altitude sickness that people get when they come on these shows. I think the reverence is beautiful, but it's not helpful.” For season 2, episodes will drop in four weekly batches of three episodes each, which correspond to time jumps in the story that spans four years total. As “Andor” concludes, Gilroy is exiting the Star Wars universe, he said, then added, “I mean, never say never.”

WES LUTCH (4) GETTY IMAGES

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PERSONAL JOURNAL.



# You Can Delete Yourself From The Internet—And You Should

Enlist a service that deletes your online personal data and reduce exposure to identity theft, scams



**PERSONAL TECHNOLOGY**

BY NICOLE NGUYEN

**GOOGLE UPDATED ITS** Results About You tool, and using it has been an eye-opening experience. It uncovered my home address, phone number and email on so-called people-search websites, along with my birth date and grandma's name—even though I requested removal from some databases years ago. It regularly sends emails alerting me to more exposed data. And yes, your info is out there, too. Plug your data into Google's free tool—the company promises not to use it for any other purposes—and wait a few hours for the alerts to roll in. Why care? Easily available personal data can expose you to identity theft, impersonation scams or junk mail. For executives, government officials and public figures, the stakes are even higher. Attackers use people-search databases to increasingly target them for harassment or violence, according to a 2024 report by cybersecurity firm Recorded Future. Even private individuals have become victims of doxxing, the malicious sharing of personal info. If you've ever gotten a speeding ticket, bought a home, answered a census survey or registered to vote, those details are now a part of public record. Credit card sign-ups, magazine subscriptions and warranty cards are



It takes DeleteMe a week to comb through thousands of sites and compile a privacy report

other reliable sources of personal information. Data brokers scrape these records, plus social-media profiles, then package the data as dossiers on millions of Americans. There are hundreds of people-search services. And while most offer an opt-out process, the data can reappear when companies refresh their databases. That's why you might want to enlist another service, the kind that deletes your data from the internet. While they can't completely scrub your info from the web, they can make it harder to pin you down. Here's how it all works. **Find: See what's out there** The profiles data brokers create are only getting more detailed. They now include lists of peo-

ple's friends, family and children, as well as location and vehicle information such as license plate number and car purchase history. "In 2023, we would find about 300 pieces of someone's personally identifiable information out there. Now there are over 600 pieces," says Rob Shavell, chief executive of DeleteMe, a personal-information removal service. To see what's out there, google your name and location. You might find a profile—or two—with your personal data on people-search websites such as Spokeo or Whitepages. You can also use Google's free scanner: Go to myactivity.google.com and under Other activity, click Results About You. Here, you can review search results where your personal information appears, and request that Google re-

move the result. The Google tool only shows search results. To cover your bases, you also need to go to the data brokers directly. Two info removers, DeleteMe and Optery, offer free exposure reports. If you pay for a subscription, they will submit opt-out requests to the services for you. **Request: Opt out or automate** If you want to go it alone, set aside a weekend. People-search sites have different removal processes, and there are hundreds of such services. Some may request *more* personal data to remove your listing. Only provide information they already have about you. If you must upload a photo ID, mask the license number and your photo. Use a burner phone number and temporary email address with Maildrop. You can also pay for a service to do the leg work. After submitting your personal information, these services comb the web and use bots or humans to fulfill the removal requests. Optery, which launched in 2021, has a budget-friendly \$4-a-month plan covering over 300 sites. You can pay more (up to \$25 a month) for removals from more websites, along with extra human assistance. It works quickly: I saw a report of exposed info after a day. After two, my information had already been removed from over 100 sites. DeleteMe is the seasoned old-timer, with 14 years in the business. The service offers a yearly plan for

\$129. This makes sense—information can reappear after months. After a week, I received my first report, which included the data broker with the most info on me, and a list of about 80 people-search sites with a profile on me. The estimated removal time on some sites was as quick as 24 hours, and on others, up to six weeks. The service will automatically do another scan in three months. The DeleteMe plan also offers disposable phone numbers and email addresses, so you can stay hidden. Aura is a more complete solution, starting at \$120 a year. In addition to removing you from people-search sites, it offers anti-virus software and identity theft insurance. It's a better option for a family that needs a wide range of coverage.

**Repeat...and minimize your footprint** When you opt-out of a people-search site, your data might not disappear right away. For California residents, removal must happen within 45 days. About a dozen states have similar privacy laws. If you live in other states, "all bets are off," says Optery founder Lawrence Gentilello. Removals could take longer—or in rare cases, not happen at all. These same companies might collect the same data again from public sources or third parties. That's why you're never completely erased. You should scan the web a few times a year and repeat the process. Or, again, just pay an info-removing service to keep running. Unless you live off the grid, avoid online shopping and other aspects of modern life, it's difficult to prevent your personal information from getting out there. But you can lay low: Turn on Do Not Track on your phone, avoid posting publicly on social media and mask your identity by using burner email addresses and phone numbers. You might even want to use a fake birthday—but you still have to remember it.

ELENA SCOTT/WSJ

## Airlines Feel The Pinch As Tariffs Hit Fliers

By Dawn Gilbertson

The following items first appeared in Dawn Gilbertson's weekly Travel newsletter. Have you cut back on travel or put off upcoming trips due to the tariff tumult and related economic jitters? Airlines are already feeling the pinch, especially in the economy cabin, Delta executives said. United added to the chorus of companies worried about travel demand. Airlines plan to cut unprofitable flying in response. That often

means higher fares on remaining routes, but only time will tell as the economic situation unfolds. Here's the upshot for travelers so far. **Cutbacks in off-peak flying**, which usually have the cheapest fares. Think Tuesday and Wednesday flights. **Fewer flights to Canada and Mexico.** Delta's president said the airline has seen a "significant" drop off in Canada bookings as some Canadians boycott U.S. vacations. Mexico vacation bookings have also been hit. **Potential Europe flight deals.** The key word here is potential, at least on the major airlines. Transatlantic travel is holding up, especially in premium cabins, Delta said. So don't expect any bargains for this summer and early fall unless things deteriorate rapidly. Delta said bookings to Europe remain strong and that it has not yet seen a "crack" in bookings from Europe to the U.S. due to tariff or border policy backlash.



**Lounge away** In travel news, Delta Air Lines opened a new Delta Sky Club in Atlanta and confirmed it will be bringing a swank Delta One Lounge to its home airport in the future. There are currently three

of them: New York JFK, Boston and LAX. **Airlines merge perk** Alaska Airlines launched a quick way to earn elite status in its frequent-flier program: Book a roundtrip flight

**Delta's president said the airline has seen a drop-off in Canada bookings.** between the continental U.S. and Sydney, Australia, or Auckland, New Zealand on Alaska and Hawaiian Airlines by the end of the year. The two airlines merged last year. **Vegas bus** Visiting Las Vegas? Save your feet or taxi money by hopping on The Deuce, especially when it's hot out. The bus traverses the Las Vegas Strip and also goes downtown. You can get a two-hour pass for \$6, a 24-hour pass for \$8 and a three-day pass for \$20. Discounts are available for students, seniors and active-duty military and veterans. To subscribe to Dawn's Friday newsletter go to [wsj.com/newsletters](https://www.wsj.com/newsletters).

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# ARTS IN REVIEW

ART REVIEW

## Modern Painter, Master Thief

An exhibition highlights the work of Wayne Thiebaud, who playfully transformed ideas taken from art history



'Betty Jean Thiebaud and Book' (1965-69), by Wayne Thiebaud, left; 'Comtesse d'Haussonville' (1845), by Jean-Auguste-Dominique Ingres, above; Thiebaud's 'Supine Woman' (1963), below left.

By JUDITH H. DOBRZYNSKI

**W**hen you look at a painting of parfaits and sundaes by Wayne Thiebaud, do you think of the bottles and vessels lined up in the still lifes of Giorgio Morandi? Or view a row of his cupcakes and see Monet's grain stacks? His "Display Cakes" (1963) and acknowledge a strange similarity between the placement of those cakes and the hats in Degas's "The Millinery Shop" (1874-86)?

You may now. It might come as a surprise even to ardent fans of Thiebaud's sunny, brightly colored, impasto depictions of everyday life to learn that behind many of his paintings, there's another painting. Yet as Thiebaud (1920-2021) himself once said: "My world is one of crime. I steal from every artist around the world."

Plenty of evidence is on view in "Wayne Thiebaud: Art Comes From Art" at the San Francisco Legion of Honor through Aug. 17. Curated by the museum's Timothy Anglin Burgard, the exhibition presses the case that Thiebaud was a painter's painter, mining art history, luxuriating in the physicality of paint, intent on imparting and eliciting emotion—and, implicitly, not a practitioner of affectless, consumerist Pop art, with which he is frequently associated. Seeking both inspiration and insight into artistic practice, Thiebaud copied works by artists he admired not just as a neo-



phyte, but at least until he was 80 years old. He collected them, too: 37 works, ranging from an 1817 drawing by Jean-Auguste-Dominique Ingres to a 1993 painting by Frank Auerbach, a German-born figurative painter, are on view in the introductory gallery, along with many of his copies—a prelude to the 65 original Thiebauds that follow.

Visitors first meet "Betty Jean Thiebaud and Book" (1965-69), a portrait of the artist's wife that embodies the show's thesis. Leaning on her elbow, hand on cheek, her mane of salt-and-pepper hair meeting her charcoal sweater, she strikes a pose that, as the wall label indicates, recalls at least four art-historical masterpieces: Dürer's "Melancholia I" (1514), Ingres's "Comtesse d'Haussonville" (1845), Degas's "Portrait of Edmond Duranty" (1879) and Picasso's "Woman With a Book" (1932).

Exhibition-goers are now primed to see clues in Thiebaud's paintings that they probably missed before. The next

gallery, the first of four hung loosely chronologically, presents his "Guitar" (1962/2002). This straightforward image on a white background, with a slight shadow, clearly mimics Picasso's sheet-

metal "Guitar" (1914), shown (as all inspirational works are) in an accompanying thumbnail image. It's an example of what Mr. Burgard categorizes as overt homage, just as Thiebaud's shiny "Black Shoes" (2018) honor, but contrast with, Van Gogh's depictions of battered shoes.

Thiebaud's other approaches, termed "covert theft" and "intuitive transformation," are more engrossing. His "Eating Figures (Quick Snack)" from 1963 shows a couple, sitting on barstools against a white background, eating hot dogs and balancing candy-striped milkshake cups on their thighs. Eyes half-closed, they look alienated from each other and the world. Compare it with Degas's "L'Absinthe" (1875-76), showing a similar couple

in a drab café, also disenchanted and unable to find solace in their choice of calumet. Even more provocatively, the exhibit likens Degas's "The Bellelli Family" (1858-69), a complex masterpiece of an unhappy couple and their two girls whose positioning broadcasts their estrangement, with Thiebaud's "Five Seated Figures" (1965). In it, similarly expressionless adults sit in a more balanced grouping, but also disassociated, avoiding eye contact. Thiebaud has torn up Degas's compositions but aimed to create similar moods using analogous elements.

Thiebaud greatly admired Manet. His "Supine Woman" (1963), depicting a female dressed in white, flat on her back, is the uncanny opposite of Manet's "The Dead Man" (1864-65), which portrays a toreador in black costume. Manet's great "A Bar at the Folies-Bergère" (1882) prompted at least a dozen Thiebaud paintings, with two here (along with a copy he made of the waitress): the buxom "Makeup Girl" (1981-2012), with noticeably similar long bangs and a direct, disconsolate gaze, offers cosmetics at a department store counter, not drinks at a bar. "Woman in White Hat" (2014/15) stares out too, but seems to be putting on makeup before a mirror—perhaps referring to the mysterious one in Manet's painting.

While he is known as a figurative painter, Thiebaud's artistic net

extended to abstract painters. His "Electric Chair" (1957)—a seat, in a tangle of wires and drips, rendered in black against a red background—reinterprets Franz Kline's abstract "The Bridge" (1955), whose bold, broad strips of black Thiebaud saw in New York. A semi-figurative still life, "French Fries" (1961)—a dish sitting above what might be a black tray that occupies most of the painting, riffs on Rothko's moody color-block paintings. And "Road Through" (1983)—a narrow black street passing between two steep cliffs—refers to Barnett Newman's "zip" paintings.

Although "Art Comes From Art" in no way provides a full view of Thiebaud, it does present a key to how he thought—consciously and unconsciously. When he is not appropriating, he inverts; he reverses; he playfully injects humor; he changes pivotal colors and viewpoints; he deconstructs to reconstruct, and he comes up with something original. With this in mind, "Art Comes From Art" invites audiences to look closely at Thiebaud's paintings, to remember (or learn) art history, and to find more in his works than they see on first glance.

**Wayne Thiebaud: Art Comes From Art**  
Legion of Honor, through Aug. 15

*Ms. Dobrynski writes about art for the Journal and other publications.*



'Makeup Girl' (1981-2012), above left, and 'Five Seated Figures' (1965), above right; they are among the 65 original works by Thiebaud in the show, curated by Timothy Anglin Burgard.



# Top Seeds Didn’t Break the Bank

The Oklahoma City Thunder and Cleveland Cavaliers enter the NBA playoffs as No. 1 seeds. They got there on the cheap.

By ROBERT O’CONNELL

In the modern NBA, there’s one tried and true way to compete for a championship: spend oodles of money, and then spend some more.

For proof of just how well cold hard cash can turn into a ring, just look at two of the most storied franchises in the league. The defending champion Boston Celtics have spent nearly \$247 million, including a luxury tax designed to penalize the highest spenders. The Los Angeles Lakers, meanwhile, will cut checks for almost \$242 million.

The remarkable thing about this postseason, however, is that the very best teams in basketball are turning the lessons of recent years on their head. The Oklahoma City Thunder and Cleveland Cavaliers will enter the playoffs with the twin distinctions of having the top records in their respective conferences—and being two of the most cost-conscious teams in the NBA.

And if one of them manages to go the distance, it will prove that it’s still possible to build a champion on the cheap.

“Historically, it’s been pretty clear that you’re going to end up a tax team if you want to be a title team,” said Keith Smith, an NBA salary cap expert for Spotrac.

The NBA set its luxury-tax line this season at \$170.8 million. That number often separates the league’s haves from the have-nots. If a team’s owner is willing to breeze over the line—sending at least a \$1.50 to the league for redistribution for every dollar they spend on salary past that point—they have more room to pile up star players. The last three title winners have paid out luxury taxes to the tune of \$231.4 million combined.

Which only makes what Cleveland and Oklahoma City have pulled off this year that much more impressive. With a roster earning just a hair over \$170 million, the 15th-highest payroll in



Shai Gilgeous-Alexander, left, earns less than \$36 million this season—nearly \$20 million behind Stephen Curry, this year’s highest-paid player. At \$165.6 million, the Thunder have the sixth-lowest payroll in the NBA.

the NBA, the Cavaliers put up a 64-18 record and enter the playoffs with hopes of winning their first non-LeBron James championship in franchise history.

The Thunder went one better. At \$165.6 million, they have the sixth-lowest payroll in the NBA. And not only do they have the best record in the league, but by net rating, a measure of offensive and defensive dominance, they are the single greatest team since

Michael Jordan suited up for the Chicago Bulls.

“We got here by trying to be extremely thoughtful, trying to be disciplined,” Thunder general manager Sam Presti said before last offseason. “Trying to operate with independent thinking about how to build the team from the ground up.”

What’s the secret to assembling the NBA’s best teams for bargain-basement prices? Hire

young visionaries and avoid experienced—but expensive—veterans. Both Cleveland and Oklahoma City are led by star players on their second pro contracts, before the league’s pay structure lets them reach their peak earning years. Under the NBA’s salary rules, players on their second deals can earn up to 30% of the league’s salary cap, whereas stars later in their careers can earn up to 35%.

That means that Oklahoma City’s Shai Gilgeous-Alexander, the NBA’s likely MVP, earns less than \$36 million this season—nearly \$20 million behind Stephen Curry, this year’s highest-paid player.

But there’s one deal even better than a second-contract superstar: a first-contract contributor. Players on their rookie contracts are the most affordable assets in the NBA—and, if they turn into high-level players, the greatest bang for your buck in all of basketball. The Cavaliers’ Evan Mobley and the Thunder’s Jalen Williams make \$16 million combined. Both made this year’s All-Star team.

The affordability of Cleveland’s and Oklahoma City’s roster doesn’t just keep things reasonable for their owners. It actually provided both teams a competitive edge. Consider the case of the Minnesota Timberwolves, who have this season’s second-highest salary-plus-tax bill at more than \$295 million. Last season, the Timberwolves made a surprise run to the Western Conference Finals. But over the summer, they traded Karl-Anthony Towns, a move that saved them almost \$50 million in luxury tax penalties but sent them tumbling down the standings.

Cleveland and Oklahoma City didn’t have to make those difficult decisions and were able to keep their squads together. But those front offices are well aware that players don’t stay inexpensive forever. Soon enough they will have to pay them like the stars they are. This summer, Gilgeous-Alexander will be eligible for a contract extension that could make him the first player in NBA history to earn \$80 million in a season.

In other words, the Thunder and the Cavaliers both know that the clock is ticking on their affordable rosters. But both are making the same bet: These teams are convinced that they can win a championship trophy before the bill comes due.

## The WSJ Daily Crossword | Edited by Mike Shenk

1 2 3 4 5 6 7 8 9 10 11 12 13 14

15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64

### KEY CHANGES | By Mike Shenk

Across

39 \_\_\_ Lanka

1 Nightclubs

40 Disconcert

9 Darkly gloomy

42 “The Persistence of Memory” painter

16 Speak highly of

43 Shade source

17 They may be marked with x’s

44 Witnesses to an event

19 Feel bad about

47 Word on an octagon

20 Peach center

48 Flutist’s neighbor, in an orchestra

21 Worry

49 Classroom challenges

22 Sings like Ella Fitzgerald

50 River of Spain

25 Sun Bowl setting

53 Outdo

27 Dishonorable fellows

54 Solemn promise

28 Burden for an ant

55 AND, OR or NOT, in mathematical logic

32 Paris pal

61 Having more customers, say

33 Back end

62 Oscar nominee for “Tootsie”

34 Drive forward

63 Coldly determined

35 Checkers side

36 Mag. workers

37 Rock’s \_\_\_ Leppard

Down

64 Keyboard key, and a hint to this puzzle’s theme

1 Spying org.

2 Completely

3 Marshy area

4 Turns aside

5 Picture puzzle

6 Jane of English literature

7 One \_\_\_ customer

8 More limber

9 On-air campaign expense

10 Ball

11 Shape badly

12 Support for a casket

13 Farmer Hoggett’s wife, in “Babe”

14 Take a breather

31 Kelly of morning TV

22 Sacred beetle of ancient Egypt

23 Managed to acquire

24 Maker of Samba sneakers

25 Important times

26 Genre for Ursula K. Le Guin

38 Dream location

29 Sports shockers

30 Cabernet sauvignon alternative

31 Zeppelins’ kin

33 Amy Winehouse song

37 Move swiftly

38 Golfer Ernie

41 Police informer, informally

42 Tyrants

45 Official witness

46 Songstress Celine

47 Smelly flow

49 Shaped like a doughnut

50 Recedes

51 Boxing ring event

52 Flower on a Valentine

54 Designer Wang

56 Snaky fish

57 Vitality

58 Key beside Q

59 British singer Rita

60 Basic education trio, humorously

Previous Puzzle’s Solution

T I T T E A G L E S A R I A  
A N O S W O O S H N E O N  
P A R A L L E L P A R K I N G  
I C E M A I S I B E A M  
R A R A D E B I P A N  
S N O C O M E D Y T O B E  
C A R E W B A S E D  
P A R T A N D P A R C E L  
P O L E S R U B I O  
E R D E B R O N C O T I P  
W E E C R O P O M N I  
R E C O I N A P A T I T E  
S P A R R I N G P A R T N E R  
Y E A S L I E S A T U N C  
R U N T S E D E R S S T E

The contest answer is **RARE BIRD ENCOUNTERS**. The clue for EAGLES at 4-Across suggests the golf meaning: two under par. The letters PAR appear six times in the grid, and looking two rows under those instances yields pieces that spell the contest answer.



Aaron Judge is quickly building a case to stand alongside the greatest sluggers in history.

## The Greatest Yankees Hitters: Ruth, DiMaggio...and Judge?

By JARED DIAMOND

Every major-league franchise has its roll call of legends, but no one comes close to the New York Yankees.

The list of illustrious names to have played in pinstripes reads like a roll call of baseball greatness.

Babe Ruth. Lou Gehrig. Joe DiMaggio. Mickey Mantle. Together they could fill an entire wing of the Hall of Fame.

But now, decades after the last of those immortals took their place in Cooperstown, it might be time to make some room.

Aaron Judge, the 6-foot-7 behemoth who seems to bash moonshots on a nightly basis, is quickly building a case to stand alongside not only the best Yankees in history, but as one of the greatest sluggers to ever play the game.

That became clearer this month when Judge celebrated his 1,000th MLB game by hitting his 321st home run—the exact number Ruth had through 1,000 games with the Yankees.

Judge has already put together two of the most impressive seasons ever. In

2022, he set the American League record with 62 homers, surpassing another Yankee hero, Roger Maris. Probably, he might have one-upped that performance last year, when he hit .322 with 58 homers, 144 RBIs and a ridiculous 1.159 OPS, the highest in an uninterrupted campaign since Barry Bonds two decades prior.

Now, a few weeks into the 2025 campaign, the 32-year-old Judge is back doing more of the same. He entered Sunday with a .397 batting average, 25 RBIs and an OPS of 1.236.

“He’s the best player on this planet,” Yankees outfielder Cody Bellinger said.

If anything, that is underselling it. Judge is now in his 10th season with the Yankees, enough time to take stock of what is setting up as a truly extraordinary career. Judge has a lifetime OPS of 1.015, which ranks seventh all-time among players in the American and National Leagues. The only right-handed hitters above him both retired about 80 years ago: Jimmie Foxx and Hank Greenberg.

Using weighted runs cre-

ated plus, an advanced metric widely seen as the most comprehensive measure of a hitters’ performance, Judge fares even better: He trails only Ruth and Ted Williams. Judge has homered every 11.3 at-bats, more frequently than anyone other than Mark McGwire.

Even without the help of complicated statistics, any casual observer can see that what Judge is doing isn’t normal. He’s attempting to join one of the game’s most exclusive clubs: 60 homers in multiple seasons. The other members are McGwire and Sammy Sosa, who have both been linked to performance-enhancing drugs.

The only thing that may keep Judge from the highest echelon of hitters is something he can’t control: his age. Most elite righty hitters—Albert Pujols, Mike Trout, Miguel Cabrera and Alex Rodriguez—were MLB regulars by 21.

Judge took a different path, playing three years in college and spending time in the minors to hone his swing. He joined the Yankees at 24... and homered in his first at-bat.

He hasn’t stopped since.



OPINION

Our Capitulation Nation



INSIDE VIEW  
By Andy Kessler

The White House claims that phones were “ringing off the hook” with countries caving to President Trump’s reciprocal tariffs. Meanwhile, the bond market backed up and stocks sold off until Mr. Trump folded like a cocktail napkin by pausing most reciprocal tariffs. Ukraine has signed a memorandum of understanding leading toward a mineral-rights deal. So much folding. We’re having an Origami moment. Does anyone have a backbone anymore?

Apparently not. After years of living off electric-vehicle tax credits and green loans, Elon Musk folded and is now a born-again free marketer. He even suggested a free-trade zone with the European Union, calling America-first architect Peter Navarro “a moron.” Mr. Musk needs to work on his insults to become truly Trumpian.

Nothing is sacred. A recent New York Times op-ed urged Democrats to embrace free trade. What’s next for the left? Free markets? Small government? Speaking of folding, California Gov. Gavin Newsom is tackling crime and transgender athletes. The capitulation will continue until morale improves.

Mr. Trump caved to Apple’s Tim Cook with talk of tariff exemptions for phones and computers coming from China, or,

as the president would say, “CHY-na.” The only real push-back to tariff mania are suits by the Liberty Justice Center and others claiming violations of the “major questions” doctrine. They’re right.

It sure looks to me as if the Trump administration conceded a hands-off approach to crypto, one that will eventually burn a lot of individual investors, for a \$TRUMP meme coin.

Everybody’s folding, from Trump and Democrats to law firms and universities.

Corporations are folding along the dotted line as well. IBM, Constellation Brands (Modelo,) Major League Baseball and Victoria’s Secret are the latest to join the parade and drop or diminish their diversity, equity and inclusion programs.

And now I’m wondering if climate hysteria is ever sustainable. Mentions of climate action and other greenie things on corporate-earnings calls are down 75% from three years ago. That’s a sign this stuff was never really important, just a reversal of their former folding to the progressive fringe—a weather vane of principles.

Mr. Trump’s executive orders ending low-flow shower heads and electric-vehicle policies were practically met with crickets. There were more pro-

tests against the Department of Government Efficiency, including signs saying, “Musk is stealing our money.” So protesters are fighting for waste and inefficiency and have given up on most everything else? I don’t get it.

Earlier this month we did see crowds at “Hands Off” protests, but it was such a mishmash of grievances, as is typical of progressive rallies. Protesting everything is protesting nothing. There were even accusations of astroturfing, paying for protestors. Bernie Sanders and Alexandria Ocasio-Cortez are on a banal “Fighting Oligarchy” tour, with Bernie even showing up at the Coachella music festival (tickets \$539 and up). Yes, social justice for the affluenza-infected Lady Gaga crowd.

Also, remind me not to hire the law firms Kirkland & Ellis, Latham & Watkins and Simpson Thacher & Bartlett to defend me. Yes, Big Law’s got a big flaw. They arguably know law better than anyone and are primo negotiators. Yet they refused to fight Mr. Trump’s threat of punitive action for targeting him, folding like a collapsible chair when faced with an existential threat.

Same for the Ivy League. Columbia University’s insta-caving to the Trump administration was startling. I agree administrators blew the Gaza protests. But with the threat of losing federal money, they folded like a queen-high poker hand. All universities should protect their future integrity by weaning themselves off the federal payroll. And sign institutional-

neutrality pacts based on the University of Chicago’s Kalven Report.

Harvard is fighting back. For now, anyway. They have \$53 billion reasons to. But a threatened loss of their tax-exempt status means they’ll inevitably fold as quickly as they abandoned their principles, which include “seeking truth through open inquiry.”

In every negotiation, both sides should feel they’ve won some and lost some, but come out ahead. Kind of like trade. Are negotiations dead?

There are so many ways to describe the act of folding: Acquiesce. Surrender. Submission. It’s the final one that seems to give Mr. Trump his jollies. It’s all a power play to him. Total submission is heaven for someone in real estate who has been beholden his whole life to the whims of bankers, who added caveats and clauses that made Mr. Trump feel boxed in. Now he tells everyone else what to do—until they fold.

I agree with some of his actions (please legalize toilets that really flush). I disagree with others. These pages have proudly protested tariffs longer than anyone. No folding here.

In trade, China hasn’t yet folded. Forget clothing, TVs or even semiconductors, unless a deal means 1.4 billion Chinese can use YouTube and Facebook and I can buy a \$20,000 BYD Chinese-made electric car, I’ll consider the entire tariff tantrum a failure. Unless one side folds first.

Write to [kessler@wsj.com](mailto:kessler@wsj.com).

BOOKSHELF | By Mark Yost

North By Northwest

Into the Ice

By Mark Synnott  
Dutton, 432 pages, \$33

Many people reassessed their lives during the Covid-19 pandemic. Mark Synnott, an accomplished mountaineer and writer, decided it was a good time to refit his sailboat, the Polar Sun, and navigate the Northwest Passage—the icy arctic waterway that connects the Atlantic and Pacific oceans. “Could I sail a forty-year-old fiberglass boat from Maine to Alaska—a voyage of some seven thousand miles—and live to tell my own tale?” he wonders. “It was a question that would soon all but consume me, in the same way I knew it had done to the European explorers who had ventured into these same waters long ago when this part of the world was still a blank on their maps.”

For centuries the search for a polar trade route had indeed consumed—and bettered—more-experienced explorers, until Roald Amundsen successfully sailed it between 1903 and 1906. Today the passage remains a daunting journey fraught with danger, from the frigid waters and the shifting currents to “Jakobshavn, the fastest-moving glacier in Greenland, which surges forward” up to 130 feet a day and is responsible for about 10% “of all icebergs spawned from the Greenland Ice Cap.” Further complicating his adventure: Mr. Synnott was recently remarried and the father of a young son.

As he sat on his boat docked in its marina and contemplated the idea of traveling the Northwest Passage, Mr. Synnott, then in his 40s, asked himself: “What do you really want to do with the time that you have left?” To which he reflected: “I [want] to spend as much of it as possible with the two human beings sleeping below and with my three other children.” But he also admits that “I’m someone who has always needed more than that. I need epic adventure and exploration in my life.” He would find both on this journey, and live to write about it in “Into the Ice: The Northwest Passage, the Polar Sun, and a 175-Year-Old Mystery.”

The mystery of the subtitle is what happened to John Franklin. On May 19, 1845, the Royal Navy expedition commander set off from Greenhithe, England, in search of the Northwest Passage. In September 1846, Franklin’s two ships, HMS Erebus and HMS Terror, became trapped in the ice near King William Island in northern Canada. “Not a single one of them made it out,” Mr. Synnott tells us of Franklin’s crew, “and no detailed written account of their ordeal has ever been found.” Solving that mystery became Mr. Synnott’s mission and justification for leaving his family behind for much of this six-month lark. Lucky for him, his second wife, Hampton, herself somewhat of a free spirit, supported him.

Lucky for him, too, that he surrounded himself with the right people. Mr. Synnott was smart enough to know that he didn’t have the sailing chops to pull this off alone. So he enlisted an old climbing buddy, Ben Zartman, who had far more experience on the high seas. He also convinced National Geographic to foot the bill for the expedition, in exchange for bringing along a camera crew to produce a podcast and an hourlong television special.

In pursuit of postpandemic adventure, the author sailed the icy arctic waterway that connects the Atlantic and Pacific oceans.

Once under way, serendipity intervenes in Greenland, where Mr. Synnott meets Jens Erik Kjeldsen, another peripatetic soul who had sailed the Northwest Passage in 2018 with his wife, Dorthie. Mr. Kjeldsen introduces Mr. Synnott to Povl Linnet, who promptly produces, Mr. Synnott tells us, a “rusted steel cable as thick as my wrist.” It was from the Fox, the boat that Francis Leopold McClintock sailed in the 1850s in search of Franklin and his expedition. The two Greenlanders give Mr. Synnott navigation advice and charts of safe anchorages, including a whaling station. “If you look around,” Mr. Linnet tells him, “you might find the iron bollards that Franklin used” to secure his ships nearly two centuries before.

Once inside the passage, Mr. Synnott and the rest of his small crew of five encounter Alan Cresswell and his partner, who are also sailing the passage, but in their much-sturdier 50-foot aluminum sailboat. Together, the two boats buddy-sail through the most treacherous parts of the voyage. Mr. Cresswell leads the Polar Sun out of a crushing ice floe to open water and safety.

Mr. Synnott’s narrative often brings readers onto the boat with him, sluicing through the icy waters. “As a weak sun struggled to burn through the haze overhead, we all stripped down and jumped off the stern, hooting and hollering as we briefly plunged into the 42-degree water,” he writes of their celebration upon crossing the Arctic Circle.

There’s also a good deal of environmental preaching. Mr. Synnott frets that the very success of this trip depends on man-made global warming to keep the Northwest Passage free from ice too thick for his boat to break through. When his boat is in peril of being crushed by a growing ice field, he’s more worried about spilling diesel fuel than about surviving the arctic. Yet after describing all the horrors of 19th-century commercial whaling, Mr. Synnott is able to enjoy the whale meat left for him by some Greenland fishermen. “The meat was tender, with a delicate flavor I would describe as much closer to beef or venison than fish. I savored every morsel.”

Mr. Synnott does a lot of soul searching along the way. He recalls an essay by John Harries, a “high-latitude sailor” who in 2014 wrote about people attempting the Northwest Passage, comparing them to the tourists who scale Mount Everest. “For these crews it’s not a voyage with the associated appreciation and learning about the surrounding lands and seas,” Mr. Harries writes. “It’s a mad dash, starting way too late in the season, just to say they did it.”

“Was I really all that different from the hordes he was describing?” Mr. Synnott wonders. “I had, in fact, climbed Everest, and now here I was apparently going for the sailing equivalent.”

In the end, he completes the passage, arriving in coastal Alaska on Sept. 20, 2022, having spent 112 days and logged 6,736 miles aboard the Polar Sun. He doesn’t solve the Franklin mystery, but finds a storybook ending all the same. “Hampton and I put our house in New Hampshire up for rent, sold our cars, and pulled Tommy out of school,” he writes. “And with any luck, by the time you’re reading this, we’ll be on our way to the South Pacific.”

Mr. Yost writes about history for the Journal.



AMERICAS  
By Mary Anastasia O’Grady

Cuba’s communist dictatorship is broke and seems to have run out of suckers who might lend it more. This month we learned that it’s turned to confiscating dollars and euros from foreign businesses on the island. It may get a few million. But going after corporate profits is like hanging a “closed” sign on the moribund economy.

The regime’s desperation is no mystery. Its 1959 pact with the people says it will provide the essentials for living in exchange for the nation’s freedom. That was never a good deal. Today it’s a joke. The legendary repression continues while medicine, housing and fuel are in short supply. Inflation is galloping. Parents find it hard to feed their children. In September the government cut back bread rations to 60 grams a day from 80 grams. In December, after more than six decades, it finally said it will eliminate the ration book, admitting that it cannot provide even a skimpy list of staples.

Reuters reported last month that “transportation by ground, sea and air in Cuba” fell by 19% in 2024, reaching a level not seen in 20 years, according to American University economist Ricardo Torres. The infrastructure, from roads to electricity, has collapsed.

One demographer estimates 18% of the population emigrated between 2022 and 2023. Those left behind stare into an abyss of hopelessness. Beneath the surface, there’s hunger for change. The island is a powder keg.

Add to this grim economic outlook a U.K. court ruling in December that the National Bank of Cuba and the Cuban state are liable for two 1984 loan agreements, governed by U.K. law, in the amount of about \$75 million. Cayman Islands investment fund CRF bought the debt in the secondary market and took the bank and Havana to court for non-payment. Earlier this month the U.K. Supreme Court refused to hear Cuba’s appeal.

The unfavorable decision for Cuba scratches the surface of its liabilities. Over the lifetime of the Cuban revolution the regime has borrowed an estimated \$50 billion to \$100 billion. It’s hard to calculate the total because Havana’s reckless sovereign lending has been forgiven so many times. The Soviet Union and later Venezuela propped up the regime, but countries like China, Japan, France, Canada, Italy, Brazil and Mexico also extended credit to Havana. Outstanding debt is now about \$40 billion, according to the Miami-based Havana Consulting Group.

Foreign companies, invited into the country beginning in the mid-1990s, also have helped the regime stay alive by making direct investments on the island. But capitalism

doesn’t work in an economy run by totalitarian gangsters, which is why 30 years after the “opening,” the country’s foreign direct investment remains paltry. Havana wants to blame its poverty on the U.S. embargo. But Cuba’s dismal track record with sovereign lenders and the private sector goes a lot further in explaining why capital steers clear of the island.

Havana targets foreign-company profits as it shrinks public bread rations.

Earlier this month the news broke that Cuban officials have quietly told some foreign companies operating on the island that they could no longer take their profits out of the country. It’s impossible to know how many investors are affected because there has been no official proclamation. Companies seem to be getting the news in private “interviews,” as the regime reportedly calls the struggle sessions.

Foreign investment in Cuba is heavily Spanish and there were rumors circulating last week that Madrid’s Socialist government has tried to keep a lid on the story by pressuring Spanish media not to cover it. Still the news has leaked out. An April 10 headline in the Spanish outlet EFE read, “Cuba blocks the repatri-

ation of foreign currency to foreign companies based in the country.”

Investors don’t seem to want to complain publicly. But they’re talking. “We completely disagree. It’s not the [Cuban] government’s money, but rather the companies’ money,” one unidentified businessman who claimed his assets had been “frozen” told EFE. He said he was informed his money could be used only inside Cuba. EFE also reported that “in some cases” companies “have complained to their respective governments, according to business and diplomatic sources familiar with the situation who requested anonymity.”

These arbitrary expropriations send a message that what entrepreneurs earn in Cuba, they don’t own. Still, money losers might take comfort in knowing things could be worse. In 2011-12 English architect Stephen Purvis spent 18 months in Cuban dungeons after Raul Castro decided he wanted to take over the British expat’s business on the island. Mr. Purvis told his chilling tale in a 2017 memoir titled “Close but No Cigar.”

Cuba still receives hard-currency remittances and payments from governments that engage with the regime’s human trafficking of doctors and nurses. But it isn’t enough. Draining foreign businesses won’t be either. If the dictatorship hopes to survive it needs a new ideological sugar daddy willing to burn money.

Write to [OGrady@wsj.com](mailto:OGrady@wsj.com).

Harvard’s Timing Reveals Its Priorities

By J.T. Young

It’s said that timing is everything. Harvard is a case in point.

On April 11 the American Association of University Professors—along with Harvard’s AAUP chapter—sued the Trump administration for demanding “vague yet sweeping” changes as “preconditions for Harvard to remain a responsible recipient of federal taxpayer dollars.” The demands emerged from the Trump administration’s March 31 investigation into Harvard for, the complaint says, “asserted but unspecified failures to address antisemitism.”

On April 14, Harvard President Alan Garber refused to comply with the administration’s demands, writing: “The University will not surrender its independence or relinquish its constitutional rights.” In response, the Trump administration suspended nearly \$2.3 billion in federal grants and

contracts to Harvard, stating that “the harassment of Jewish students is intolerable” and universities “must commit to meaningful change if they wish to continue receiving taxpayer support.”

Mr. Garber tried in his statement to sidestep Harvard’s many incidents of antisemitism: “We have made it abundantly clear that we do not take lightly our moral duty to fight antisemitism. . . . We plan to do much more.”

What he neglected to note is that in January Harvard settled two lawsuits accusing the school of tolerating antisemitism. As part of the settlements, the university agreed to expand its nondiscrimination policies and adopt the International Holocaust Remembrance Alliance’s definition of antisemitism. But this was more than 18 months after the problems began.

Recall a January 2024 tweet by Bill Ackman, a Harvard alumnus who had given

millions to the university: “I first became concerned about @Harvard when 34 Harvard student organizations, early on the morning of October 8th before Israel had taken any military actions in Gaza, came

It answers money threats with alacrity but took months to address antisemitism.

out publicly in support of Hamas, a globally recognized terrorist organization, holding Israel ‘solely responsible’ for Hamas’ barbaric and heinous acts.”

So, we can start Harvard’s antisemitism clock from at least Oct. 8, 2023.

Recall also that in a Dec. 5, 2023 hearing before the House Committee on Education and the Workforce, Rep. Elise Stefanik (R., N.Y.) asked

Harvard’s then-president, Claudine Gay, if “calling for the genocide of Jews” violated “Harvard’s rules of bullying and harassment.” Ms. Gay responded infamously: “It can be, depending on the context.”

It’s fair to wonder how long it would have taken Harvard to address incidents like these if the school hadn’t been sued.

By contrast, it took only days for Harvard to decide it had a moral duty to reject the Trump administration’s conditions for billions of dollars in federal funding. If only basic morality had inspired the same sense of urgency as federal money.

Mr. Young is author of “Unprecedented Assault: How Big Government Unleashed America’s Socialist Left.” He served in President George W. Bush’s Office of Management and Budget and Treasury Department and was a congressional staffer, 1987-2000.



OPINION

REVIEW & OUTLOOK

Gender Storytime at the Supreme Court

Does the First Amendment give parents a right to opt out of gender ideology at the local public elementary school? The Supreme Court will consider that Tuesday in *Mahmoud v. Taylor*, and what a sign of the times. Consider two of the storybooks that parents in Montgomery County, Md., say the district added to its reading curriculum:

“My Rainbow” is about an autistic child named Trinity who says, “I *need* long hair,” because, “I’m a transgender girl.” It ends with a surprise gift from Trinity’s mother: a teal, pink, and purple wig.

“Born Ready” is about Penelope, who tells her mother: “I don’t *feel* like a boy. I AM a boy.” After this single discussion, the mother answers with unquestioning affirmation: “Yes. We will make a plan to tell everyone we love.”

Teachers in Montgomery County were also given guidance on replying to classroom questions and comments. One idea was to “disrupt” students from “either/or thinking” about the sexes. If a child suggested it’s “weird” to say a girl can become a boy, the proposed response was to explain how this comment is “hurtful,” and that when each of us is born, “people make a guess about our gender.”

Many families objected, including the Muslim, Catholic, and Ukrainian Orthodox plaintiffs. In their telling of the facts, they initially were offered a chance to opt out of the storybooks. But the school district quickly reversed and then said it wouldn’t inform parents when the books were going to be read.

Montgomery County replies that it sought to supplement its reading program with LGBTQ characters to “reflect the diversity” of its families. At first it “tried to accommodate parent requests to opt their children out of class,” until that “became unworkably disruptive,” because of “unsustainably high numbers of absent stu-

dents.” For the record, “My Rainbow” has since been removed, though the county doesn’t explain exactly why.

The parents rely on the precedent *Wisconsin v. Yoder* (1972), in which the Court said Amish families couldn’t be made to send their children to school past eighth grade. If the First Amendment protects dropping out, how can it not cover a “much narrower request” by parents wanting merely to skip “discrete instruction that deliberately seeks to confound their religious values”?

Montgomery County argues that if families choose to attend public schools, they “are not cognizably coerced by virtue of their children’s exposure there to religiously objectionable ideas.” If the First Amendment gives parents a right to pick and choose from the curriculum, the county says there’s “no discernible limit,” and it would work the same in science or history classes. Public schools “simply cannot accommodate” such exceptions.

They can’t? Public schools accommodate a lot, including students who are disabled or learning English as a second language. Part of living in a pluralistic society is that some families might reject, say, ideas on evolution. Their children can still benefit from learning algebra and “Romeo and Juliet.” How hard would it be for Montgomery County schools to announce that “Born Ready” will be read in three weeks, and students can be excused?

This gets to the parents’ second argument, which is that the district’s policy isn’t neutral, and the ending of opt outs was targeted. Their brief cites a member of the school board saying it’d be an impossible disruption if teachers had to “send out notices so white supremacists could opt out of civil-rights content.” The implicit comparison is between religious families and racists. Who needs a tolerance lesson?

If elementary schools read children’s books with LGBTQ themes, can parents opt out?

Olympia designs a tax targeting Elon Musk and his EV credits.

Washington State’s Tax on Tesla

Washington state progressives haven’t found a tax they don’t want to impose, and behold the bell-ringer: a tax specifically designed to siphon Elon Musk’s profit from their own environmental regulations. This is a tale of policy contortions only today’s climate left could create.

In 2020 Washington joined California regulations to reduce emissions from gas-powered cars and phase out gas vehicles by 2035. The non-gas guzzlers of the future must be a maximum of 20% plug-in hybrid and at least 80% electric by the same date.

Those rules ramp up starting next year, and car companies will need to meet the 2026 requirement that 35% of new cars be electric or plug-in hybrid. By 2028 those percentages rise to 51% and 68% in 2030. Laggards will need to offset their gas cars with credits purchased from other more saintly manufacturers. And guess which company has more emissions credits in the bank than anyone?

Unlike Honda or Ford, all Tesla cars are electric, so the company is sitting on a mountain of emissions credits it will be able to sell to traditional auto manufacturers that can’t meet the benchmarks. Tesla sold less than 10% of all vehicles in Washington state last year, but the company currently has about 54% of all emissions credits in the state, according to the Washington Policy Center.

The only thing progressives hate more than global warming these days is the prospect of enriching Elon Musk. So after years of trying

to force the country to buy electric vehicles, they’re now planning to punish the company producing them. Two companion bills filed by

Olympia Democrats would tax the EV credits that Mr. Musk is poised to cash in on at 10% of the credit price.

One of the bills explains that while reducing emissions is good, “the creation of these

tradeable and bankable credits creates the opportunity for a financial windfall accruing to firms that are not burdened by the legacy production of internal combustion engine vehicle.” That would be Tesla.

As such, “it is the intent of the legislature to address this unintended outcome by taxing the windfall profits.” Small hitch for Olympia’s mathematicians: If Tesla controls the market for EV credits, what’s to stop Mr. Musk from raising the price he charges for credits to make up for the money he’s losing on the new tax?

The EV credit scheme is a terrible idea, but employing punitive taxation at a single company is abusive lawmaking that progressives would decry if the Trump Administration tried it. Olympia Republicans filed a bill last week that would prohibit the Tesla tax “or any other tax that applies to only one individual, business, or entity.” That bill that won’t go anywhere, but it makes the right point.

Democrats are seeking every avenue to hurt Mr. Musk short of setting Teslas on fire (and some progressives are doing that). But Democrats don’t mind incinerating their environmental principles in the process.

Gov. Youngkin will soon sign the measure he proposed last year.

No More Cellphones in Virginia Classrooms

Those wondering how to get smartphones out of schools now have an example in Virginia. This month the state’s Democratic-led General Assembly passed legislation mandating cellphone-free education in K-12 public schools. Republican Gov. Glenn Youngkin is expected to sign it.

Mr. Youngkin started this initiative last July with an executive order directing his Department of Education to issue guidance on eliminating phones in public schools. Within months 102 of Virginia’s 131 school districts had implemented some form of cellphone ban.

The inspiration for Mr. Youngkin’s order was social psychologist Jonathan Haidt’s 2024 book, “The Anxious Generation.” The book argues with compelling evidence that mental-health issues are highly concentrated among screen-addicted children. Adolescents who spend more than three hours a day on social media face double the risk of mental-health problems—and the average American teenager spends nearly five. Rates of anxiety, depression, suicide and self-harm among youth have soared since 2010.

As Mr. Haidt wrote last April, studies show “mental health improvements, increases in physical activity, and reductions in bullying when schools go phone-free.”

Cellphones also impede academic performance: 72% of U.S. high-school teachers say phones are a major distraction in class, while 53% of public-school leaders believe they have a negative effect on student grades. Is it a coincidence that National Assessment of Educa-

tional Progress scores have plummeted?

As Virginia legislators sought to codify Mr. Youngkin’s order, they wanted to exempt lunch hours from the phone ban. But phones are also distracting during lunch. Mr. Youngkin sent copies of “The Anxious Generation” to all 140 lawmakers, along with a letter urging them to reconsider the

exemption.

On April 2 the Legislature unanimously approved his amended version, which directs school boards to “restrict student cell phone and smart device possession and use on school property from bell to bell”—meaning the entire school day.

Comprehensive data on the effect of the phone bans won’t be available until after the 2024-25 school year, but case studies are promising. Wakefield High School in Arlington, Va., participated in a pilot cellphone ban beginning in September. Students locked their phones in pouches upon arrival and retrieved them after the final bell. By April students and teachers reported higher rates of concentration, less violence and more robust face-to-face communication in school, according to CNN.

At least eight other states have imposed bans on classroom cellphone use. Dozens more are considering the same—including New York, where one high school that banned cellphones in 2023 saw student grades increase to pre-pandemic levels within a year.

The evidence is strong that cellphone addiction undermines student learning, while promoting other ills. Kudos to Virginia for doing something about it.

LETTERS TO THE EDITOR

The Folly of Treating Ankara as an Adversary

Elliott Abrams’s letter “Put Not Your Trust in Recep Tayyip Erdogan” (April 16) urges treating Turkey as an adversary but never asks the next question: Then what? Turkey is invaluable to the central task of American statecraft: countering the challenge from the revisionist bloc of China, Russia, Iran and North Korea. The Turks operate across multiple theaters—Eastern Europe, the Balkans, the Middle East and the South Caucasus—support Ukrainian sovereignty and field the North Atlantic Treaty Organization’s second-largest army. Without them, the West can neither contain Russia nor sustain a viable order in Eurasia. Ask a Polish or Romanian official how secure they’d feel if Washington considered Ankara an adversary.

Turkey doubtless poses serious challenges, including growing hostility toward Israel. But labeling it an enemy risks deepening that hostility. Before making a bad situation worse, shouldn’t the Trump administration at least attempt to strike a balance between Ankara and Jerusalem? Doing so would prove fruitful, not

least because the Turks play a central role in containing Iran. Turkish-backed forces helped topple Bashar al-Assad and expelled Iranian militias from Syria—outcomes that directly served American and Israeli interests. Equally important, Turkey guarantees the independence of Azerbaijan, a crucial Israeli ally. Azerbaijan anchors the corridor linking Europe to Central Asia, which blocks Russian and Iranian influence, and gives Kazakhstan and Uzbekistan a path to economic, if not political, autonomy from Moscow and Beijing.

Mr. Abrams calls for continued U.S. support for the “Kurds”—a euphemism for the YPG, the Syrian branch of the PKK, a U.S.-designated terrorist organization. Backing a terrorist militia over a NATO ally isn’t strategy; it’s provocation masquerading as principle.

American statecraft must work with what exists. Military alliances are built on shared interests, not values. Strategy begins with that recognition.

MICHAEL DORAN  
Hudson Institute  
Washington

Neither Harvard Nor Trump Gets a Good Grade

Your editorial “Donald Trump Tries to Run Harvard” (April 16) offers an excellent analysis of the administration’s overreach in bringing Harvard to heel. Federal funding comes with civil-rights strings attached, not viewpoint-diversity strings. Yet no one in Cambridge, Mass., or Washington is handling this affair especially well. Both parties are lumping discrete issues together.

The government has the better of at least one argument: It can and should use funding as leverage to force Harvard to confront its national-origin-discrimination problem. Harvard admits that it has had severe issues with anti-Jewish and anti-Israeli harassment, and though it settled with one group of Jewish plaintiffs, it remains locked in litigation with others. It is likewise under Education Department investigation for failing to handle discrimination adequately. The school can’t simply claim it is trying its best and expect the feds to go away. It certainly can’t do so without attempting to identify and expel students who ha-

assed Jews and Israelis, or those who led groups that did the same. It is absurd to suggest that federal civil-rights enforcement is a violation of academic freedom. That’s an argument against civil-rights law, not heroic defiance of an overweening government.

But the Trump administration does itself a disservice when it lumps in related but not fully congruent gripes about how Harvard conducts its academic affairs. The lack of viewpoint diversity may contribute to discrimination—e.g., biasing students against Zionism—but combined with tangential demands about DEI and administrative bloat, it looks as if it’s bullying for the sake of a larger anti-Harvard campaign. If the administration is going to freeze funding due to discrimination, it needs to ensure that it connects its means and ends logically and then follows the proper procedures so it doesn’t lose its progress in court.

TAL FORTGANG  
Manhattan Institute  
New York

Will Students Trade the Screen for the Novel?

I applaud Joseph Bottum for designing a course to learn whether the current generation of college students will read canonical novels if required to do so (“College Students Don’t Read. But Can They?,” op-ed, April 11). I would, however, stop short of calling this venture an “experiment.” Mr. Bottum’s course will attract a self-selected group of students who likely will have already cultivated the desire to read from cover to cover. The large number who don’t have this desire will likely continue to enroll in courses—even in English departments, I suspect—that require them to read only short passages from a smattering of works.

I hope, then, that the University of Colorado’s English department will eventually cross-list Mr. Bottum’s course. This would create a better test of whether his approach can prompt students, who heretofore have been apprehensive about reading entire novels, to crave the call for self-reflection, empathy and understanding of the human that these sublime works deliver.

PROF. RONALD P. SEYB  
Skidmore College  
Saratoga Springs, N.Y.

Ali Khamenei: A Moral Man?

John Kerry and Thomas Kaplan evidently understand something about Ayatollah Khamenei that has evaded Iranian dissidents like me: that because the ayatollah says he objects to nuclear weapons on moral grounds, he isn’t committing to acquiring nuclear weapons (“Trump Has a Chance to Strike a Deal With Iran,” op-ed, April 14). How could we have believed he isn’t a deeply moral man? It must be our bias from witnessing the Islamic regime gun down protesters during the 2009 Green Revolution. We got the same impression after watching family and friends get arrested, denied due process and tortured for wanting a better future. Kicked out of our homeland under threat of death for seeking freedom of religion, rendered refugee and homeless, how could we see the ayatollah otherwise? Perhaps Messrs. Kerry and Kaplan have seen a side of the regime we haven’t. If so, they should supply the evidence.

WILLIAM MEHRVARZ  
Washington

Circle Back on the Deep Dive

The letters in response to Joseph Epstein’s op-ed “The Incredibly Massive Number of Hackneyed Words Is Surreal” (April 4) have been excellent but not exhaustive. Allow me to add the rules of language that seem to dominate every committee meeting, board room and professional setting today:

First, every sentence must begin with the word “so.” Assuming the subject is within one’s wheelhouse, he must then look at the metrics of whatever idea is under discussion and be sure it is both “scalable” and “sustainable.” That is a conversation that will take up a lot of real estate and inevitably require a deep dive. But fear not: One can always circle back.

SUSAN M. KAHN  
Coral Gables, Fla.

Pepper ... And Salt

THE WALL STREET JOURNAL



“A hectic day in the fourth grade does not cause PTSD.”

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OPINION

# The Economics of Polarization

By Roland Fryer

Nothing throws America's divisions into stark relief quite like having Donald Trump in the White House. But Mr. Trump is an effect of polarization as much as a cause. We've been growing apart politically for decades.

Data from Gallup show wide and growing divergence across numerous issues between 2003 and 2023. Nowadays, 85% of Democrats but only 30% of Republicans think the government should ensure that everyone has healthcare, a gap that grew 24 points during those two

People tend to interpret ambiguous information as confirming whatever they believed to begin with.

decades. The notion that the federal government has too much power now gains support from 73% of Republicans and 31% of Democrats, a 51-point shift—yes, the partisan split was slightly in the other direction during the George W. Bush administration. The split on whether abortion should always be legal has widened 30 points, on whether human activity is the main cause of global warming by 33 points.

Lawmakers have become more polarized over the past half-century too, and there's polarization over why this happened. Some data suggest that Republican politicians pulled to the right, but conservatives note that the government has moved sharply to the left. In a 2021 Quinnipiac University poll, 52% of Americans said the Democrats had moved too far left, while 35% said the Republicans had moved too far right.

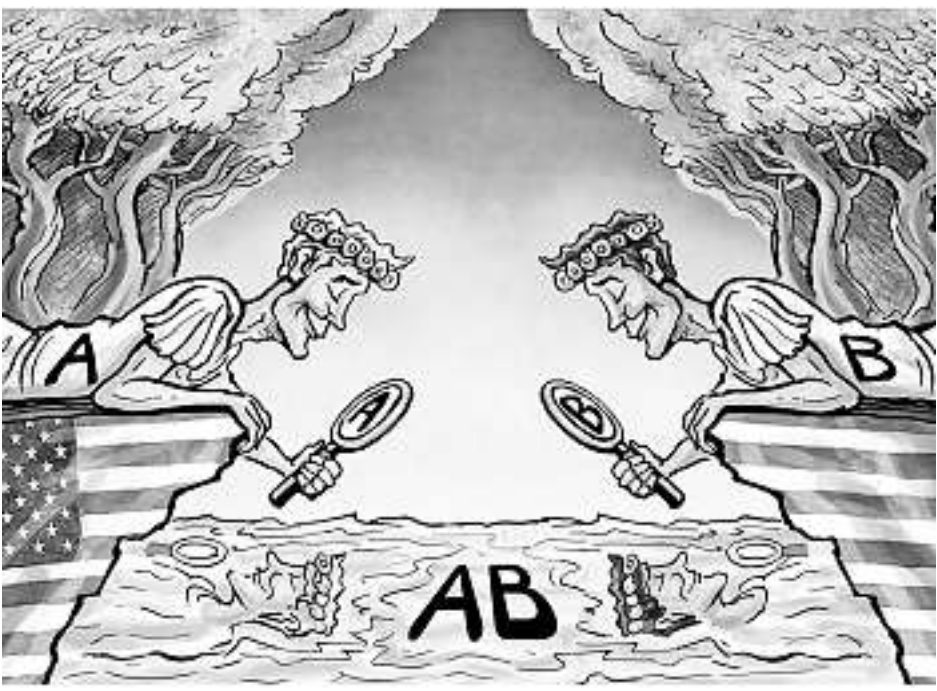
Even seemingly nonpartisan measures such as the consumer sentiment index reveal partisan polarization: Republicans had more-positive views than Democrats about their economic situation during the first Trump term, and this abruptly flipped in early 2021.

How can two people who observe the same information come away with starkly different conclusions? And why do views on factual questions, such as the cause of global warming or the strength of the economy, break down so neatly on ideological lines?

A key insight came to me in a roundabout way. My wife is a great driver, but she blows the horn too much for my taste. Any slight—perceived or real—and you get a loud honk if she is behind the wheel. One morning as we were commuting, a car pulled past her on the highway and veered just slightly our way, so that its tires drifted into our lane. She honked. I tried to reason with her—his driving was within the usual margin of error. Her response: “I have kept myself from many, many accidents by being a proactive honker.”

We observed the same incident, drew opposite conclusions, and each became more convinced we'd been right all along. Is this consistent with rational thought? And could it explain why Americans have become so polarized?

As soon as she dropped me off on campus, I ran to my office to tell a fellow economist this anomaly I had observed. Was my wife irrational? Was I? Or did we need to think about inference and decision-making a bit differently? I confided first in Matthew Jackson, who special-



agents who have differing prior beliefs and who see exactly the same sequence of evidence can often end up polarized, with one person being absolutely sure of A and the other of B.

We explored the model's implications in an online experiment with more than 600 subjects, modeled on a pioneering 1979 paper by Charles G. Lord and colleagues. First, participants were presented with questions about their beliefs on climate change and the death penalty. Then they read a series of summaries of research about each topic. After each summary, we asked participants if they thought the summary provided evidence for or against the topic on a 16-point scale. After all of the summaries were presented, we repeated the initial questions about their beliefs about the topic.

There was a very significant correlation between a subject's prior belief and his interpretation of the evidence. More than half of our sample exited our experiment with more extreme beliefs than at the start, even though the evidence presented to them was neutral.

The discouraging implication is that in a world where information is plentiful, people will become more divided, not less. That's true even if they all see the same information—which they don't when they can choose between Fox News and MSNBC. And it's true even if our widening divisions prove deeply unhealthy for our country.

*Mr. Fryer, a Journal contributor, is a professor of economics at Harvard, a founder of Equal Opportunity Ventures and a senior fellow at the Manhattan Institute.*



**LIFE SCIENCE**  
By Allysia Finley

American waistlines have ballooned in tandem with government welfare. Is there a connection? Studies show people on food stamps eat less healthily than other low-income Americans.

Now governors in Indiana, Arkansas and Idaho want to conduct real-world experiments by asking the U.S. Department of Agriculture for waivers to exclude soda, candy and other sweets from the Supplemental Nutrition Assistance Program—i.e., food stamps.

Cue the protests from so-called antihunger groups on the left, which have made common cause with the candy and beverage industries. In their telling, it's unfair to “punish” lower-income folks by barring them from using government handouts to purchase sugary treats.

But nobody is being punished. Recipients could still buy a Kit Kat or a Coke with cash. The states' experiment will test whether the restrictions might nudge recipients to eat better. Would candy and soda manufacturers be screaming from the rooftops if they didn't think this

might plausibly be a result?

Soda and sweets aren't the only causes of obesity, but they are major sources of empty calories. The stated purpose of the food stamp program is to support “nutrition.” Are Skittles nutritious?

Alcoholic drinks, which you can't buy with food stamps, at least have some demonstrated cardiovascular benefits. Soft drinks and candy have no health upside. They create pleasure by triggering a release of dopamine in the brain, but this can lead to overindulgence and cravings when the sugar high ebbs. They can thus lead people to consume more calories than they otherwise would.

Call it a diet-multiplier effect. A soda once in awhile won't give you diabetes or pile on the pounds, but having one every day can add up. While such ultraprocessed foods aren't literally “poison,” as Robert F. Kennedy Jr. calls them, they aren't good for you. Why should government pay for them?

Democrats deceptively labeled food stamps the “Supplemental Nutrition Assistance Program” in 2008 to increase public support. But for far too many Americans, food stamps aren't supplementing earned income. They are supplementing other government programs.

The U.S. boasts more than a dozen “nutrition,” programs including free school lunches and the WIC program for low-income pregnant women, infants and children. The Biden administration even allowed states to spend federal Medicaid dollars on “nutritious” meal deliveries. This seems a tacit admission that food stamps aren't paying for nutritious foods.

The government shouldn't subsidize soda and candy, which have no nutritional benefit whatever.

Studies bear that out. More food-stamp money is spent on soda and sweets than fruits, vegetables, eggs, pasta, beans and rice combined, according to a report by the Foundation for Government Accountability. Sweetened beverages and candy alone account for 11% of food-stamp spending. A 2018 study in the Journal of the American Medical Association examined diet quality of food-stamp beneficiaries and people of similar incomes who didn't receive the handouts from

2003 to 2014. Nonbeneficiaries ate better than beneficiaries, and their diets improved more over time.

USDA studies show that working-age adults on food stamps consume more added sugars and refined grains, and less fruit and vegetables, than nonrecipients with similar demographics. They are also more likely to have severe hypertension and low levels of “good” cholesterol. Women on food stamps in particular have much higher obesity and diabetes rates.

Does this prove that food stamps make people fat? Well, no. People who subsist on government payments may be less health-conscious. On the other hand, it's plausible that when government gives people more money to spend, they buy more junk. (Food stamps may also free up more money in household budgets to buy tobacco, alcohol and other unhealthy products.)

America doesn't have a “hunger” problem. About three-quarters of adult food-stamp beneficiaries are overweight or obese. Only 3% are underweight. More Americans of all income levels die from illnesses caused by overindulgence than by hunger. Yet the government keeps subsidizing more people to eat more junk.

Over the past 15 years, food-stamp enrollment has swelled as eligibility standards eased through waivers of work requirements while Congress and the USDA boosted payments. There are now 50% more people on food stamps than in 2008, and payments have nearly tripled.

In 2021 the Biden administration boosted allotments by 27% on average based on the rationale that—get this—Americans need to consume more calories because they are fatter. The costs of poor diet are socialized through government and private insurance. Taxpayers pick up the tab via Medicaid, and healthy people have to pay higher insurance premiums.

Zooming out, the bigger problem is that government encourages indulgent and indolent behavior. You don't have to work to qualify for most welfare programs, and if you work too much, you lose eligibility. Members of both parties lack the appetite to shrink bloated entitlements. Democrats are protesting even modest trims to Medicaid spending on able-bodied adults.

No single program is to blame for the government's \$36.7 trillion debt. Spending adds up like calories and pounds, and America would be healthier and more prosperous if Washington went on a diet.

## Trump Follows but Rebrands Biden's Semiconductor Policy

By Jim Secreto

As President Trump tries to revitalize domestic manufacturing, he's been clear about his love of tariffs—and his disdain for the Chips Act. Calling the 2022 law a “horrible, horrible thing” and urging Congress to repeal it, he cast doubt on the future of a once-in-a-generation \$39 billion bet to bring semiconductor manufacturing back to the U.S.

Despite the rhetoric, the Trump administration isn't walking away. It's rebranding. Enter the Investment Accelerator, an initiative entrusted to Commerce Secretary Howard Lutnick. It doesn't repeal or replace the Chips Act. Instead, it

places a layer on top of it: an executive director, yet to be named, who will report directly to Mr. Lutnick and coordinate implementation with a broader mandate to reduce regulatory burdens and attract domestic and foreign investment.

This allows the administration to continue semiconductor incentives under a new name. Politically, it lets Mr. Trump pursue a crucial industrial strategy—reshoring semiconductor fabrication—without crediting President Biden or former Commerce Secretary Gina Raimondo, who launched the effort and lined up \$500 billion in planned private-sector investment over the next decade.

The Investment Accelerator also

provides political cover for America-first conservatives to support semiconductor subsidies. The executive order establishing the office says the new structure is about getting the “benefit of the bargain” and securing “much better deals than those of the previous administration.” While it doesn't abandon strategic incentives, it recognizes that regulatory relief can be a tool to drive investment, and that subsidies can distort markets with unintended consequences.

The stakes couldn't be higher. China has directed at least \$150 billion to its semiconductor industry. U.S. production shrank, while Taiwan now produces more than 90% of the world's most advanced chips—and faces growing pressure from Beijing. Meanwhile, America remains dangerously reliant on foreign supply chains for the semiconductors that power the artificial-intelligence systems essential to U.S. defense and intelligence operations.

While Mr. Trump promotes tariffs as the solution, few serious investors or policymakers believe tariffs alone can close the cost gap between building a chip factory in Phoenix and one in Taipei. Strategic incentives, combined with regulatory relief, are still essential to correct the market failure to protect long-term strategic interests. The Investment Accelerator confirms that.

So what should Messrs. Trump and Lutnick do? Three things. First, finish the job. Of the 34

Chips awards announced, 20 are finalized—including grants for fabrication plants by Intel, Micron, Samsung and Taiwan Semiconductor Manufacturing Co. But not all the funding has been disbursed, with portions tied to future project timelines and milestones. A second tier of preliminary awards—targeting sup-

Whether you call such an effort the Chips Act or the Investment Accelerator, the goal remains the same.

ply-chain firms, specialty fabs and defense applications—remains on the table. These are the deals for which a “better bargain” could be struck, especially with the help of experienced technical staff at the National Institute of Standards and Technology, where the Chips program still resides.

Second, cut the red tape. Big projects can get stuck in permits, environmental reviews and endless interagency sign-offs. Despite receiving \$6.6 billion in direct funding and \$5 billion in loans, TSMC faces regulatory hurdles that have delayed construction at its Arizona site, where it plans to invest up to \$100 billion. A central office inside the Commerce Department could clear the path, meaning fewer delays, tighter timelines and better accountability from Washington.

Third, go beyond semiconductors. The Chips Act focused on a single foundational technology, funding fabs and R&D. But the Accelerator's broader scope can drive growth in related industries—including the estimated \$1 trillion in U.S. capital investment projected over the next five years in data centers and infrastructure powering the artificial-intelligence boom. By coordinating across federal agencies and working with states, the Accelerator can turn megaprojects into durable regional economies and position the U.S. to lead the next generation of technological innovation.

MAGA critics of the Chips Act may cast it as a Biden-era initiative, ignore its bipartisan support, and argue that tariffs are the better way to restore U.S. manufacturing. But China's tech ambitions, persistent supply-chain risks, and the capital intensity of fab construction tell a different story.

President Harry Truman once said, “It is amazing what you can accomplish if you do not care who gets the credit.” The Trump administration's move to rebrand rather than reject the Chips program gives it a chance to get the credit for reshoring semiconductor factories. For the sake of national security, let's hope Mr. Trump finishes what the Chips Act began—and makes semiconductor manufacturing great again.

*Mr. Secreto served as deputy chief of staff to Commerce Secretary Gina Raimondo, 2024-25.*

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Lachlan Murdoch

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For more than 50 years, glyphosate has been an essential farming tool that helps raise yields and lowers costs.

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# BUSINESS & FINANCE

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THE WALL STREET JOURNAL.

Monday, April 21, 2025 | B1

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## Nvidia Finds Ways to Sell In China Despite Curbs

Company makes AI chips specifically for that market, in compliance with U.S.

By Drew An-Pham and Bradley Olson

The Trump administration's new restrictions on sales of artificial-intelligence chips to China come with a familiar feeling for Nvidia.

The chip maker has spent years finding new ways to continue doing business with a huge market while staying on the correct side of U.S. regulations.

The regulations generally limit the speed and capabilities of advanced AI processors that can be sold to China.

Each time the U.S. has announced new restrictions, Nvidia launched updated versions of its processors that fell below the performance limitations.

On Thursday, Chinese state media reported that Nvidia Chief Executive Jensen Huang met with a Chinese official in Beijing.

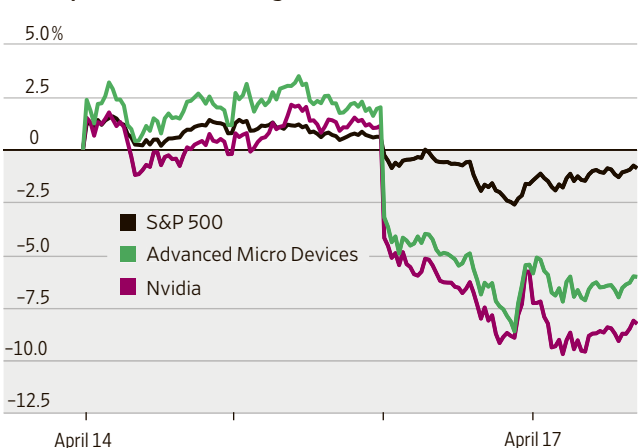
"The China market is very important to us," Huang said in a video posted by state media, wearing a suit and tie instead of his usual leather jacket. "The technical excellence of Chinese industry is quite impressive."

While regulations have made doing business with China tougher, demand for Nvidia's cutting-edge AI chips globally has been a boon to its revenue.

The whack-a-mole history of Nvidia's chips



Share-price and index change



\*Nvidia's fiscal year ends the last Sunday of January.

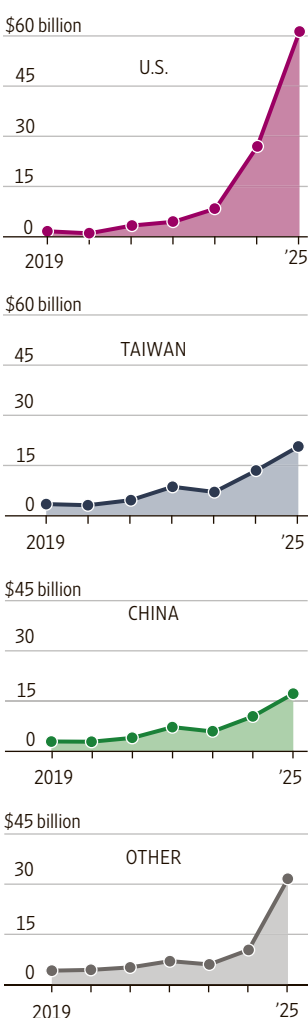
Sources: Nvidia (chronology, revenue); analyst reports and news stories (chronology); FactSet (share-price and index change)

The company has said that it follows U.S. regulations in chip sales "to the letter" and that U.S. interests are served when the tech industry ex-

ports its products around the world.

The Trump administration's new restrictions require export licenses for some arti-

Nvidia revenue by source, yearly\*



cial-intelligence chips, including Nvidia's H20 chips and **Advanced Micro Devices'** MI308 product. The move rattled the companies' shares.

## Delisting Chinese Stocks Would Carry Dangers

By Hannah Erin Lang

Trade tensions between the U.S. and China are escalating rapidly. Wall Street could be the next battleground.

In his first term, President Trump explored the idea of delisting Chinese stocks from U.S. exchanges, eventually signing an executive order that prohibited Americans from investing in some Chinese companies the U.S. said had aided China's military. In a recent interview, Treasury Secretary Scott Bessent declined to rule out a broader ban. "I think everything's on the table," Bessent said on Fox Business when asked if the government would go as far as barring all Chinese stocks from trading on U.S. exchanges. "That'll be President Trump's decision."

A broad delisting of Chinese companies could have serious consequences for financial markets in both countries.

U.S.-based investors can invest in foreign companies by purchasing American depositary receipts, or ADRs. These are certificates that represent shares in a foreign company.

A bank or broker holds underlying stock in a company and issues ADRs that track the price of those shares. American investors also can gain exposure to Chinese and other in-

ternational stocks through exchange-traded and mutual funds.

Delisting means shares are removed from a stock exchange and investors can no longer buy or sell them there. A delisting can be voluntary or involuntary.

Under the Holding Foreign Companies Accountable Act, enacted in 2020, the Securities and Exchange Commission can ban foreign companies from trading if American regulators are unable to inspect their au-

ditors for three consecutive years. Some Chinese companies have voluntarily delisted in the years after the law was passed. "The tools with which to do it don't have to be invented," said Steven Schoenfeld, CEO at MarketVector Indexes, a market research and data provider.

The Trump administration also could issue a broader investment ban by executive order, Schoenfeld said. Trump and former President Joe Biden have issued directives in the past targeting specific industries and types of Chinese companies but not a blanket ban on investing in China.

"It's much more rare, if not unprecedented, to tell Americans where they can invest," said Schoenfeld, who has worked in global market analysis. *Please turn to page B2*

Many investors would opt to sell their stakes in Chinese firms if they are delisted.

## Trump Tariffs Tilt the Scales in the Coke vs. Pepsi Battle

By Laura Cooper

Pepsi was already losing the cola wars. The trade war isn't going much better.

It all comes down to where **PepsiCo** and **Coca-Cola** make the secret-recipe concentrates that are the essence of their sodas. Concentrates are made in special facilities, then shipped to bottling plants. There, they are combined with water, bubbles and sweetener to make soda.

PepsiCo more than 50 years ago began making concentrate in Ireland because of the country's low corporate tax rate. Now, PepsiCo's tax-saving move has backfired: The concentrate for nearly all U.S. sales of Pepsi and Mountain Dew is subject to a 10% tariff.

Coca-Cola for decades also has produced concentrate in Ireland, shipping it to markets



Both Coca-Cola and PepsiCo make soda concentrates in Ireland, which are subject to new tariffs, but PepsiCo is more exposed.

around the world. But Coca-Cola makes most of the concentrate for its American sodas in Atlanta and Puerto Rico, a U.S. territory. That means

drinks like Coke and Sprite are less exposed to tariffs.

"Ireland has long had the tax advantage—until the tariffs hit," said Carlos Laboy, an ana-



lyst at HSBC. It is unclear how long they will last, but Pepsi clearly is at a disadvantage now, he said.

Coca-Cola and PepsiCo both

could be hurt by a 25% tariff on aluminum imports imposed by the U.S. in March. Coca-Cola imports some aluminum from Canada, and soda prices could

rise as a result of the levy, Coke Chief Executive James Quincey said in February. He said that his company could ease the impact by packaging more drinks in plastic bottles or sourcing aluminum from the U.S.

The tariff war comes at a particularly bad time for PepsiCo. Pepsi's U.S. market share has plummeted over the past two decades, and last year hit a new low when Dr Pepper unseated Pepsi-Cola as America's No. 2 soda. After years of focusing on food and energy drinks, PepsiCo is now trying to revive its U.S. soda sales. The new tariffs could make that more difficult.

PepsiCo also makes concentrate in Texas, Uruguay and Singapore. The company declined to comment on what steps it might take to mitigate *Please turn to page B2*

## EU Delayed Punishing Meta Before U.S. Talks

By Kim Mackrael and Sam Schechner

The European Union recently delayed penalizing **Apple** and **Meta Platforms**, temporarily avoiding a conflict with the Trump administration during a week that saw the bloc ramp up its push for a trade deal with the U.S.

The European Commission, the EU's executive body, had initially planned to announce cease-and-desist orders targeting the tech giants last Tuesday and had informed at least one of the companies of that timing, people familiar with the matter said. Both companies also could have been slapped with fines.

The decision to postpone the announcement was made shortly before EU Trade Commissioner Maroš Šefčovič met with U.S. officials in Washington last Monday, for his first in-person talks since President Trump announced a 90-day pause on some tariffs. In addition, last week Italian Prime

Minister Giorgia Meloni met with Trump, who said he would have "very little problem" making a trade deal with the EU.

The rulings are still expected to go ahead, and it isn't immediately clear how long the delay might last.

Asked Tuesday about the cases, a commission spokesman told reporters no dates had been announced but technical work had been completed. "We're currently working on the adoption of final decisions in the short term," the spokesman said. The commission said Friday that it doesn't comment on its internal planning.

The delay represents a brief reprieve for Meta Chief Executive Mark Zuckerberg. The Wall Street Journal reported last month that Meta executives had pressed U.S. trade officials to fight the expected order. Trump has complained about EU tech regulations and threatened earlier this year to respond with tariffs. *Please turn to page B4*

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Prologis sees trade war lifting demand for warehouse space. **B3**



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Who will bear the cost of Trump's economic goals? **B10**

## United Airlines Finds Refuge In Serving Wealthier Travelers

By Katherine Hamilton

High-spending U.S. travelers are keeping **United Airlines** level, but the air-travel industry isn't clear of turbulence yet.

The airline last week said it is leaning into its business catered toward wealthier consumers, including premium seating and travel to international destinations, as those spenders appear relatively unfazed by economic storm clouds when compared with other customers. United has more breathing room to lower prices for cost-conscious travelers without falling into the red, analysts say, potentially making it hard for budget airlines to compete through an economic downturn.

"We have a tale of two business models here," Boston Consulting Group managing director and partner Adam Gordon said.

As demand for premium-cabin seating and visiting international destinations re-



The carrier is leaning more on premium fares as it says more-affluent customers are less rattled by economic fears.

mains strong, airlines with business models that favor those markets will likely have an advantage, Gordon said.

United has been investing in more premium capacity and brand loyalty, and that helped offset some demand decline in other segments, United Chief Commercial Officer Andrew Nocella told investors last Wednesday. Premium-cabin

revenue increased 9% and international revenue rose 5%, while domestic passenger revenue per available seat mile fell nearly 4% in the latest quarter.

The decline in domestic main-cabin travel was the biggest drag on earnings during the period and is expected to be a challenge this quarter, as *Please turn to page B4*



BUSINESS & FINANCE

Delisting  
China  
Stocks

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sis for decades.

As of March 7, there were 286 Chinese companies listed on major U.S. stock exchanges. Many are smaller, lesser-known firms. Some of the biggest names include: Tech giant **Alibaba Group**; **Baidu**, a search engine and internet company; **PDD Holdings**, the parent company of e-commerce site Temu; **Tencent Music Entertainment Group**; and **JD.com**, an e-commerce company.

If Chinese stocks are delisted, investors could convert their ADRs into shares listed on the Stock Exchange of Hong Kong. But it is likely that many investors would opt to simply sell their stakes in Chinese firms, Schoenfeld said.

That would send a serious ripple through global markets. Goldman Sachs analysts estimate that U.S. institutional investors currently own about \$830 billion worth of Chinese stocks.

A quick selloff would drag down stock valuations and could create liquidity problems. For companies like Alibaba, JD.com or Baidu, average daily trade volumes in the U.S. vastly exceed those in Hong Kong, analysts at Morgan Stanley found.

Investors could try to be patient—after previous delistings, investors had nearly a year to shed their holdings. But waiting might not produce better results. The value of those securities would likely drop on news that delisted companies were losing access to U.S. capital markets.

If Chinese investors unload their U.S. securities, it would send shock waves through U.S. markets. They hold an estimated \$370 billion of U.S. equities and \$1.3 trillion in bonds, Goldman Sachs analysts say.



A broad delisting of Chinese companies could have serious consequences for financial markets in both the U.S. and China.

Tariffs  
Shift Coke,  
Pepsi Battle

Continued from page B1  
tariffs. It also declined to say whether the levies could drive up prices for sodas like Pepsi or Mountain Dew.

The cola wars aren't the only rivalry whose scales have been tipped by tariffs. In the world of jeans, **Levi Strauss** sources from many countries whose goods are now subject to 10% tariffs. Wrangler's parent company, on the other hand, makes 40% of its jeans and other pants in the Western Hemisphere, including seven facilities in Mexico. The products Wrangler makes in Mexico are compliant with the U.S.-Mexico-Canada Agreement trade pact, and so are currently exempt from the 25% tariff on Mexican imports President Trump introduced earlier this year.

The escalating tariff war

also could embroil toothpaste makers. Most of the Crest toothpaste consumed in the U.S. is made domestically, while **Colgate-Palmolive** produces some of its U.S.-bound toothpaste in Mexico.

For PepsiCo's independent bottlers—many of them family-owned businesses whose sales have been hurt by Pepsi's market-share decline—the trade war is a fresh blow.

Bottlers receive small orders of concentrate in 55-gallon containers; large orders arrive in tanker trucks. Some independent bottlers say they expect tariffs on concentrate to push up their costs, and worry that the trade war could make it difficult for PepsiCo to compete with its rivals.

PepsiCo opened its first concentrate plant in Ireland in 1974. In the Irish city of Cork, PepsiCo now operates its global concentrate head office, two concentrate plants and a research and development center. And the soda giant recently doubled down on Ireland: In 2022, it invested €166 million, or about \$189 million, in its manufacturing facilities in Cork.

The tariff war comes at a particularly bad time for PepsiCo.



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Podcast: Why More Americans Are Sitting On Cash Piles Instead of 'Buying the Dip'



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BUSINESS NEWS

# Castlelake To Invest Via Invictus

Asset manager sees attractive prices in the residential mortgage market

By ISAAC TAYLOR

Asset manager Castlelake has agreed to invest up to \$2 billion in residential mortgages through Invictus Capital Partners, a firm that specializes in residential credit.

The arrangement will let Minneapolis-based Castlelake invest in high-quality mortgage loans acquired by Invictus, providing Castlelake fund investors with a chance to gain from “attractive pricing” of assets in the growing residential-loan market. In turn, the firm’s capital commitment will help Invictus expand its loan investment operations.

Washington, D.C.-based Invictus sources mortgage loans through its Verus Mortgage Capital subsidiary and a network of over 300 correspondent originators. Invictus-sponsored vehicles have acquired more than \$38 billion in residential mortgage loans, according to the firm.

Castlelake cited the credit underwriting approach used by Invictus, saying it aligns with its own “granular data-driven investment philosophy and focus on downside protection.” The firms are looking to deploy capital from Castlelake over the next 12 to 18 months in mortgages with an average loan-to-value ratio of 65% to 75%.

The \$2 billion target isn’t a hard cap, leaving the door open to further capital injections.

“Should the opportunity set develop in the way that we’re

hoping, I could see us growing it materially from there,” said Isaiah Toback, a Castlelake partner and deputy co-chief investment officer.

The agreement is part of Castlelake’s asset-based investment strategy, which includes backing residential and commercial mortgages, among other types of assets. Residential mortgage makes up 20% to 25% of Castlelake’s asset-based private-credit strategy, Toback said.

In the past year or so, Castlelake has agreed to acquire or finance roughly \$7 billion of residential and commercial real-estate loans through partnership arrangements, according to the firm.

The residential real-estate market isn’t immune from recent events rocking other sectors. Tariffs on imports from Mexico and Canada are set to increase building costs, once fully implemented. Tougher immigration enforcement might accelerate rises in labor costs for home builders.

As the potential for disruption in the market increases, credit investors could stand to benefit as liquidity becomes more of a concern, Toback said.

Other large asset managers have recently bolstered their credit-focused activity in the residential property sector.

Private-equity giant Carlyle Group last year purchased a minority stake in Unison, a provider of cash to homeowners in exchange for a share of the equity in their property.

Castlelake, founded in 2005, manages assets of roughly \$25 billion. Invictus manages more than \$20 billion in gross assets.



The industrial real-estate company also said it expects to benefit from a slowdown in warehouse construction.

## Prologis: Trade War Helps Demand

By LIZ YOUNG AND DENNY JACOB

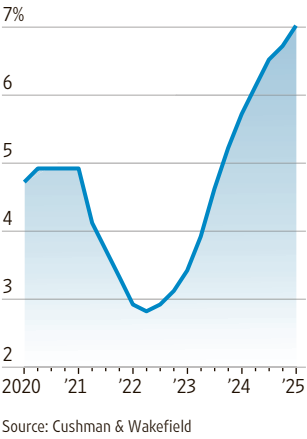
The world’s biggest industrial real-estate giant said the escalating global trade war will spur demand for U.S. storage space as companies stockpile inventory close to consumers.

Prologis said its customers, which include Amazon.com, Home Depot and FedEx, have been rushing in merchandise, rerouting shipments and taking on overflow storage space as they seek to get ahead of President Trump’s new tariffs.

Chief Executive Hamid Moghadam, who plans to retire Jan. 1, told The Wall Street Journal he believes Prologis is well-positioned during a period of uncertainty in the broader market because the company operates warehouses across the U.S. and around the world near population centers.

“The logistics chain has two parts. It has the production part and it has the consump-

U.S. industrial real-estate vacancy rate, quarterly



tion part,” Moghadam said last week. “We don’t play in the production part for a very simple reason: We’re not good at predicting tariffs; we’re not good at predicting labor costs; we’re not good at predicting government policy; but we are

very good at predicting consumption.”

Companies stockpiling inventory close to U.S. shoppers could help boost demand in a warehousing sector that has struggled with slow leasing activity over the past two years following a frenzied period of expansion during the pandemic.

The average warehouse vacancy rate across the U.S. increased to 7% in the first quarter, up from 5.7% a year earlier and more than twice the 3% vacancy rate in late 2022, according to real-estate services firm Cushman & Wakefield.

Moghadam spoke with the Journal after San Francisco-based Prologis reported higher earnings and revenue for the quarter ended March 31 as leasing activity accelerated. The company logged net earnings attributable to common stockholders of \$592 million, or 63 cents a share, compared

with \$584 million, or 63 cents a share, in the same period in 2024.

Stripping out certain one-time items, core funds from operations came in at \$1.42 a share. Analysts polled by FactSet had forecast \$1.38 a share. Revenue grew 9.2% to \$2.14 billion in the period, ahead of the \$1.96 billion expected by analysts. The company left its guidance unchanged for the rest of the year, forecasting net earnings per share attributable to common shareholders of between \$3.45 and \$3.70.

Prologis said it expects to benefit from a slowdown in warehouse construction, which has constrained supply and kept upward pressure on industrial real-estate rents. About 73 million square feet of new warehouse space was completed in the first quarter of this year, down 41% from the same period last year, according to Cushman.

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# Johnson & Johnson Moves To Focus Its AI Strategy

Pivot focuses on highest-value uses for GenAI; some projects will close

By ISABELLE BOUSQUETTE

**Johnson & Johnson** has shifted its generative AI strategy away from broad experimentation across the health-care conglomerate to a more focused approach. Chief Information Officer Jim Swanson said the move ensures that the company allocates resources only to the highest-value generative AI use cases, while it cuts projects that are redundant or not working, or where a technology other than GenAI works better. “That was a pivot we made after about a year of learning,” Swanson said. “Now we’ve moved from the thou-

sand flowers to a really prioritized focus on GenAI.” The “thousand flowers” approach involved a number of use case ideas from across the company, which made their way through a centralized governance board. At one point, employees were pursuing nearly 900 individual use cases, many that were redundant or simply didn’t work, he said. As the company tracked the broad value of AI, including generative AI, data science and intelligent automation, it found that only 10% to 15% of use cases were driving about 80% of the value, he added. Now J&J is drilling down into high-value GenAI use cases around drug discovery and supply chains, as well as an internal chatbot to answer questions on company policy. “We’re prioritizing, we’re scaling, we’re looking at the things that make the most

sense,” he said. “That was part of the maturation process we went through.” Since ChatGPT’s 2022 debut, companies have been grappling with where and how to start applying generative AI. Many, including J&J, encouraged experimentation, a strategy that set employees up to learn and test the technology and build applications that could see wider adoption across the enterprise. In some cases, they are finding that too much experimentation can be at odds with finding business value, and as a result they are looking to forge a new strategy. J&J began its pivot last year, removing a centralized governance board responsible for vetting employee GenAI ideas. It then distributed governance responsibilities to various corporate functions, including commercial, supply

chain and research, that had a better handle on whether the use cases were actually driving value in their area. Those groups were able to shut down or consolidate redundant use cases and focus resources into the ones that were working. One example that is working is a “Rep Copilot,” which helps coach sales representatives on how to engage with healthcare professionals about new treatments. The company is piloting this in its Innovative Medicine business segment, which develops new treatments for oncology and other areas, and is working to expand that pilot to its MedTech segment, which sells robotics and hardware like hip replacements and lenses. GenAI also is being used for an internal chatbot that ingests information about company policies and benefits to



J&J’s chief information officer, Jim Swanson.

help reduce the some 10 million interactions employees have every year with the services team. In drug discovery, the company is looking at whether GenAI can help researchers find the optimal moment to add a solvent to turn a liquid molecule into a solid. J&J is testing how AI can help identify and mitigate supply-chain risks, including the impact of a shortage of a given raw material, Swanson said. The company is tracking

progress in three buckets: the ability to successfully deploy and implement use cases; how widely they are adopted; and the extent to which they deliver on business outcomes. Swanson said the broad experimentation phase was necessary to learn about what the technology was good at. “We had the right plan three years ago, but we matured our plan based on three years of understanding,” he said. “This is the better way to run now.”

# Marketers Hand Brands Over to AI

By PATRICK COFFEE

Marketers are growing more comfortable with letting generative artificial intelligence assume the voices of their brands. The work often involves AI agents, which are a series of interconnected AI entities, that perform different tasks in the marketing chain, as the industry inches closer to a day when proponents say AI will develop, produce and even approve entire campaigns with little or no human intervention. “You’re going to relinquish control. It makes too much sense,” said Noah Brier, co-founder of AI-focused consulting firm Alephic. Building better AI tools to evaluate the work of other AI tools will be key, since the amount of marketing content being generated is already too great for proper human review, Brier said. But brands aren’t waiting for perfection. **Prudential Financial** recently addressed the challenge of producing large volumes of personalized content by working with **Adobe** and AI startup **Gradiant** to cre-

ate a virtual employee. “We are, for the first time, onboarding a digital co-worker in the content authoring space,” said Hema Widhani, chief digital and marketing officer at the financial services and insurance company. The AI “employee” has been tasked with using behavioral, geolocation and real-time data to generate webpages tailored to the interests of each of the millions of customers and advisers who visit Prudential’s web properties, Widhani said. At last year’s Aspen Ideas Festival, Prudential debuted an AI photo booth that outlines each user’s potential retirement, complete with a simulated image of the person 20 years into the future. The project is designed to help identify new customers, said Widhani. Since Prudential works in a highly regulated industry, it has begun testing evaluative AI agents that review all AI-generated content to determine whether it complies with relevant legal regulations and adheres to the company’s brand voice, Widhani said. AI-generated content that is published under the names of Pru-



Prudential Financial debuted an AI photo booth last year.

dential executives requires an extra layer of scrutiny, she said. Every AI process still requires some form of final human review, but the personalized webpages may become an exception to that rule, said Widhani. The nature of the relationship between AI and its human reviewers is changing. Consumer health products company Opella operates an AI “factory” that produces advance care planning materials

for medical professionals alongside the hundreds of webpages, images and Instagram posts that it generates every day, said Chief Growth Officer Alberto Hernandez. The company, which sells products from the antihistamine Allegra to the laxative Dulcolax, embeds at least one human regulatory adviser within its AI marketing team to catch and undo hallucinations, or instances when the algorithms spit out false information, Hernandez said. To ensure that its AI-powered marketing materials are medically accurate, Opella also trains its AI agents using the expertise of healthcare professionals. Opella is confident enough in the AI agents that have been trained on the voices of its many brands that it doesn’t alter the work they produce beyond human fact-checking, Hernandez said. “We don’t allow for interactions and changes, because [in the time it takes to] say, ‘No, can you make the logo bigger,’” the AI could have produced another 100 pieces of content, Hernandez said.

# EU Delays Apple, Meta Penalties

Continued from page B1

Zuckerberg defended the company last week in a U.S. Federal Trade Commission trial that could end up forcing Meta to sell valuable pieces of its business empire. European officials say they won’t water down their tech regulations in response to U.S. pressure. But some lawmakers in the European Parliament have questioned whether the cases have become political as the EU seeks to negotiate a deal with the U.S. on trade. The Meta and Apple cases both relate to alleged breaches of the EU’s Digital Markets Act, a law that seeks to make it easier for smaller companies to compete with their big tech rivals. The commission opened investigations in March 2024 and issued preliminary findings in both cases last summer. The cases carry a potential fine of up to 10% of the companies’ global annual revenue, though people familiar with the matter have said they expect fines would be much lower. The cease-and-desist orders, which target business practices, are expected to have a bigger impact on the two companies than any potential fines. A spokesman for Meta referred to earlier comments that

its concern isn’t only about fines. “It’s about the commission seeking to handicap successful American businesses simply because they’re American, while letting Chinese and European rivals off the hook,” the statement said. The commission says it enforces the bloc’s laws equally for all companies that operate in the EU. The Meta case relates to whether the company should be forced to allow users to use Facebook and Instagram for free without seeing personalized ads. The EU said last year that Meta’s policy of requiring users to choose between buying a subscription or allowing Meta to use their data for targeted advertising didn’t comply with the Digital Markets Act. In an effort to appease regulators, Meta last fall introduced an alternative that allows users to see “less personalized ads” without buying a subscription. A separate case against Apple deals with the iPhone maker’s App Store rules, which the commission has previously said prevent app developers from freely directing users to alternative ways to make purchases. The commission has said Apple restricts how developers can communicate with users and charges fees for facilitating transactions outside the App Store in a way that goes beyond what is necessary. Apple referred on Friday to its previous comments on the case. It has said that developers who use the App Store benefit from Apple’s proprietary technology and other tools.

## The cease-and-desist orders are still expected to go ahead.

# BUSINESS & FINANCE

# United Has A Refuge In Wealthy

Continued from page B1

well, Nocella said. But the airline was still able to post adjusted earnings per share of 91 cents, ahead of the 74 cents expected by Wall Street. “To still have positive numbers because of international, I think a lot of that is mix,” Jefferies analyst Greg Konrad said. United has retired aircraft early and is cutting capacity during off-peak hours to mitigate costs where it is seeing declines. So far, bookings have stabilized over the past six weeks, Nocella said. The airline also will lower prices, making it harder for budget carriers to offer something more attractive to customers that might spill over from United, Chief Executive Scott Kirby said. “The only way a spill airline gets passengers is through lower prices,” Kirby said. “But when times get tougher, the brand-loyal airlines like United have more seats to sell, which we do sell at lower prices, but that disproportionately im-

pacts all the spill carriers that we compete with.” Corporate travel and trips originating from outside the U.S. also declined in the first quarter, but United has been reducing how much it relies on those markets for revenue. Business revenue now makes up 8% of passenger revenue, a smaller share than before the pandemic, Kirby said. Travel from Canada to the U.S. was down 9% while European origin travel dropped 6%, but trips from outside the U.S. make up only 20% of United’s point of sale, Nocella said. United shared two outlooks for the full year: one that was the same as the guidance it had provided in January and one of \$7 to \$9 a share based on what it expects would happen if the U.S. enters a recession. While it didn’t provide total clarity or confidence, United offered more insight than peers after **Delta Air Lines** withdrew full-year guidance, giving investors some hope that earnings wouldn’t hit zero in an economic downturn, analysts said. “They’re telling you that there’s a number they can produce in earnings, even in a recessionary scenario, that doesn’t have a zero in the dollar position,” Bernstein analyst David Vernon said. “Investors should take some comfort in that.”

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## NOTICE OF SALE

**IN THE UNITED STATES BANKRUPTCY COURT FOR THE DISTRICT OF DELAWARE**  
In re: DANIMER SCIENTIFIC, Chapter 11  
INC., et al., Case No. 25-10518 (MFV)  
Debtors.<sup>1</sup> (Jointly Administered)

**NOTICE OF SALE, BIDDING PROCEDURES, AUCTION, AND SALE HEARING.** The above-captioned debtors and debtors in possession (collectively, the “Debtors”) filed the Motion of Debtors for Entry of Orders (A) Approving (i) Bidding Procedures, (ii) Certain Bid Protections in Connection With A Stalking Horse Agreement, if Any, (iii) Form and Manner of Notice of Sale, Auction, and Sale Hearing, and (iv) Assumption and Assignment Procedures; (B) Scheduling Auction, Sale Hearing, and Related Deadlines; (C) Approving (i) Sale of Substantially All of Debtors’ Assets Free and Clear of Liens, Claims, Interests, and Encumbrances, and (ii) Assumption and Assignment of Executory Contracts and Unexpired Leases; and (D) Granting Related Relief (Docket No. 93) (the “Bidding Procedures Motion”).

**PLEASE TAKE FURTHER NOTICE** that, on April 17, 2025, the United States Bankruptcy Court for the District of Delaware (the “Court”) entered that certain order (Docket No. 196) (the “Bidding Procedures Order”) approving, among other things, the Bidding Procedures, which establishes the key dates and times related to the sale of the Debtors’ Kentucky Assets, Bainbridge Assets, Novomer Assets, and Greenfield Assets (collectively, the “363 Sale Assets”), including an Auction (if necessary) and a Sale Hearing. All interested bidders should carefully read the Bidding Procedures Order and the Bidding Procedures in their entirety.<sup>2</sup>

Copies of the Bidding Procedures Order, the Bidding Procedures, the Stalking Horse Agreement, and any other related documents are available upon visiting the Debtors’ restructuring website at <https://cases.stretto.com/danimer-scscientific>.

**Important Dates and Deadlines**

- **Bid Deadline.** Any person or entity interested in bidding on any of the 363 Sale Assets must submit a Qualified Bid on or before **May 2, 2025 at 5:00 p.m. (prevailing Eastern Time)** (the “Bid Deadline”).
- **Auction.** If the Debtors receive more than one Qualified Bid for any of the 363 Sale Assets, the Debtors will conduct the Auction, which has been scheduled for **May 5, 2025 at 9:00 a.m. (prevailing Eastern Time)** either (i) at the New York offices of Vinson & Elkins, LLP, 1114 Avenue of the Americas, New York, NY 10036, or (ii) virtually, pursuant to procedures to be timely filed on the Court’s docket in advance of the commencement of the Auction, or at such later date and time and such other location as selected by the Debtors in accordance with the Bidding Procedures.
- **Sale Object Deadline.** Objections to the sale of any of the 363 Sale Assets free and clear of all liens, claims, interests, and encumbrances pursuant to section 363(f) of the Bankruptcy Code (a “Sale Object”), must be (i) filed with the Court in accordance with the Bidding Procedures and (ii) served on the Auction Notice Parties (as defined herein) on or before **May 7, 2025 at 5:00 p.m. (prevailing Eastern Time)** (the “Sale Object Deadline”).
- **Sale Hearing.** A hearing to approve and authorize the sale of any of the 363 Sale Assets to one or more Winning Bidders will be held before the Court on or before **May 12, 2025 at 3:00 p.m. (prevailing Eastern Time)** or such other date as determined by the Court.

**Filing Objections.** Sale Objectors, if any, must (a) be in writing, (b) state, with specificity, the legal and factual basis and thereof, (c) comply with the Bankruptcy Code, Bankruptcy Rules, and Local Rules, (d) be filed with the Court by no later than the Sale Object Deadline, and (e) be served on: (i) proposed co-counsel to the Debtors: (A) Vinson & Elkins LLP, 1114 Avenue of the Americas, 32nd Floor, New York, New York 10036, Attn: George R. Howard (ghoward@velaw.com) and Emily Jungwirth (ejungwirth@velaw.com); (B) Vinson & Elkins LLP, 2001 Ross Avenue, Suite 3900, Dallas, Texas 75201, Attn: Trevor G. Spears (tspears@velaw.com) and Matthew D. Struble (mstruble@velaw.com); and (C) Richards, Layton & Finger, P.A., One Rodney Square, 920 North King Street, Wilmington, Delaware 19801, Attn: Daniel J. DeFranceschi (dfranceschi@rlf.com); Zachary L. Shapiro (shapiro@rlf.com); Matthew P. Milana (milana@rlf.com); and Emily R. Mathews (mathews@rlf.com); (ii) the U.S. Trustee, Attn: Megan Selbier (Megan.Selbier@usdoj.gov); (iii) the official committee of unsecured creditors appointed in these Chapter 11 Cases and their counsel, Brown Rudnick LLP, 7 Times Square, New York, NY 10036, Attn: Benoit Silverberg (bsilverberg@brownrudnick.com) and One Financial Center, Boston, MA 02111, Attn: Shari Dvoskin (sdvoskin@brownrudnick.com); (iv) counsel

to the DIP Lenders: (A) Proskauer Rose LLP, Eleven Times Square, New York, New York 10036, Attn: David M. Hillman (dhillman@proskauer.com) and Dylan J. Marker (dmarker@proskauer.com); and (B) Landis Rath & Cobb LLP, 919 Market Street, Suite 1800, Wilmington, Delaware 19801, Attn: Adam G. Landis (landis@rlcaw.com) and Matthew R. Pierce (mpierce@rlcaw.com); and (C) those persons who have formally appeared in these Chapter 11 Cases and requested service pursuant to Bankruptcy Rule 2002.

**Additional Information.** The Bidding Procedures set forth the requirements for becoming a Qualified Bidder and submitting a Qualified Bid, and any party interested in making an offer to purchase the 363 Sale Assets must comply with the Bidding Procedures. Only Qualified Bids will be considered by the Debtors, in accordance with the Bidding Procedures.

Any party interested in submitting a Qualified Bid should contact the Debtors’ advisors: (A) Vinson & Elkins LLP, Attn: George R. Howard (ghoward@velaw.com), Emily Jungwirth (ejungwirth@velaw.com), Trevor G. Spears (tspears@velaw.com) and Matthew D. Struble (mstruble@velaw.com); or (B) AllPartners LLP, Attn: Frank Pometti (fpometti@allpartners.com); Nick Lobdello (nlobdello@allpartners.com); and John Creighton (jcreighton@allpartners.com).

**Reservation of Rights.** Except as otherwise set forth herein and in the Bidding Procedures, the Debtors reserve the right, in their reasonable business judgment, in a manner consistent with their fiduciary duties and applicable law, to modify the Bidding Procedures; waive terms and conditions set forth therein with respect to all Potential Bidders; extend the deadlines set forth therein; announce at the Auction modified or additional procedures for conducting the Auction; and alter the assumptions set forth therein, provided that the Debtors will not be authorized to make material modifications to the Bidding Procedures without further order of the Court. The Debtors may provide reasonable accommodations to any Potential Bidder(s) with respect to such terms, conditions, and deadlines of the bidding and Auction process to promote further bids on the 363 Sale Assets, in each case, to the extent not materially inconsistent with the Bidding Procedures and the Bidding Procedures Order. All parties reserve their rights to seek Court relief with regard to the Auction, the Bidding Procedures, and any related items (including, if necessary, to seek an extension of the Bid Deadline).

**FAILURE TO ABIDE BY THE BIDDING PROCEDURES, THE BIDDING PROCEDURES ORDER, OR ANY OTHER ORDER OF THE COURT IN THESE CHAPTER 11 CASES MAY RESULT IN THE REJECTION OF YOUR BID.**

**FAILURE OF ANY PERSON OR ENTITY TO FILE AND SERVE AN OBJECTION IN ACCORDANCE WITH THE BIDDING PROCEDURES ORDER BY THE SALE OBJECT DEADLINE WILL FOREVER BAR SUCH PERSON OR ENTITY FROM ASSERTING ANY OBJECTION TO THE MOTION, THE ORDER APPROVING THE SALE TRANSACTION, THE PROPOSED SALE TRANSACTION, OR ANY OTHER AGREEMENT EXECUTED BY THE DEBTORS AND A SUCCESSFUL BIDDER AT THE AUCTION.**

Dated: April 17, 2025, Wilmington, Delaware, (s/ Matthew P. Milana, Richard, Layton & Finger, P.A., Daniel J. DeFranceschi (No. 2732), Zachary L. Shapiro (No. 5103), Matthew P. Milana (No. 6681), Emily R. Mathews (No. 6866), One Rodney Square, 920 North King Street, Wilmington, Delaware 19801, Tel: (302) 651-7700, Fax: (302) 651-7701, Email: dfranceschi@rlf.com, shapiro@rlf.com, milana@rlf.com, mathews@rlf.com) -and- VINSON & ELKINS LLP, George R. Howard (admitted pro hac vice), David S. Meyer (admitted pro hac vice), The Grace Building, 1114 Avenue of the Americas, 32nd Floor, New York, New York 10036-7708, Tel: (212) 237-0000, Fax: (212) 237-0100, Email: ghoward@velaw.com, dmeyer@velaw.com -and- Matthew D. Struble (admitted pro hac vice), Trevor G. Spears (admitted pro hac vice), 2001 Ross Avenue, Suite 3900, Dallas, Texas 75201, Tel: (214) 220-7700, Fax: (214) 220-7716, Email: mstruble@velaw.com, tspears@velaw.com, Proposed Attorneys for the Debtors and Debtors in Possession

## BANKRUPTCIES

**IN THE UNITED STATES BANKRUPTCY COURT FOR THE DISTRICT OF DELAWARE**  
In re: CREATIVEMASS HOLDINGS, INC. et al., Chapter 11  
Debtors.<sup>1</sup> Jointly Administered (Subchapter V)

**NOTICE OF (I) COMMENCEMENT OF JOINT PREPACKAGED SUBCHAPTER V DEBTORS’ PLAN OF LIQUIDATION, (II) CONFIRMATION HEARING, AND (III) RELATED DEADLINES TO ALL HOLDERS OF CLAIMS, HOLDERS OF EQUITY INTEREST AND PARTIES IN INTEREST IN THE ABOVE CAPTIONED CHAPTER 11 CASES, PLEASE TAKE NOTICE THAT:**

On April 14, 2025 (the “Petition Date”), the above-captioned debtors and debtors in possession (collectively, the “Debtors”) filed with the United States Bankruptcy Court for the District of Delaware (the “Bankruptcy Court”) the Joint Prepackaged Subchapter V Debtors’ Plan of Liquidation (D.I. 9) (as amended, supplemented, or otherwise modified from time to time, the “Plan”) pursuant to sections 1125 and 1126(b) of title 11 of the United States Code (the “Bankruptcy Code”). Copies of the Plan may be obtained, upon request, from the Debtors’ proposed counsel, Pashman Stein Walder Hayden, P.C., at the addresses specified below. The Plan is also available for inspection free of charge on the Debtors’ case website at <https://cases.stretto.com/Creativemass> (the “Case Website”). The Plan is also available for inspection for a fee on the Bankruptcy Court’s website at [www.dbs.uscourts.gov/the/CourtWebsite/](http://www.dbs.uscourts.gov/the/CourtWebsite/), a provision to be filed with the Clerk of the Bankruptcy Court, 824 North Market Street, 3rd Floor, Wilmington, Delaware 19801, where they will be available for review between the hours of 8:00 a.m. to 4:00 p.m., prevailing Eastern Time.

**PLEASE TAKE FURTHER NOTICE** that a hearing to consider confirmation of the Plan, any objections to the foregoing, and any other matter that may properly come before the Bankruptcy Court (the “Confirmation Hearing”), will be held before the Bankruptcy Court, 824 North Market Street, Wilmington, Delaware 19801, on **May 29, 2025, 9:00 a.m. (ET)**. The Confirmation Hearing may be rescheduled from time to time without further notice other than an announcement in open court or in the filing of a notice or a hearing agenda in the Chapter 11 Cases. Notice of such rescheduled date(s) will be made available on the Case Website.

**PLEASE TAKE FURTHER NOTICE** that the deadline for filing objections (each, an “Confirmation Objection”) to the Plan is **May 15, 2025, 4:00 p.m. (ET)** (the “Confirmation Objection Deadline”). Any such Confirmation Objections must: (a) be in writing; (b) comply with the Bankruptcy Rules and the Local Rules; (c) state the name and address of the objecting party and the amount and nature of the Claim or Interest owned by such entity; (d) state with particularity the legal and factual basis for such objections, and, if practicable, a proposed modification to the Plan that would resolve such objections; and (e) be filed with the Court with proof of service thereof no later than the Confirmation Deadline and be served upon the following parties

(the “Notice Parties”): (i) the Debtors, (ii) proposed counsel to the Debtors, Pashman Stein Walder Hayden, P.C., 824 N. Market Street, Suite 800, Wilmington, Delaware 19801, Attn: Joseph C. Barsalona II (jbarsalona@pashmanstein.com), Michael Custer (mcluster@pashmanstein.com), and Alexis R. Gambale (agambale@pashmanstein.com); (iii) the Office of the United States Trustee for the District of Delaware, J. Caleb Boggs Building, 844 King Street, Suite 2207, Lockbox 35, Wilmington, Delaware, 19801, Attn: Benjamin A. Hackman (bhackman@usdoj.gov); and (iv) the Subchapter V Trustee, William A. Honorny (bhonorny@mtclp.com), 1628 John F. Kennedy Boulevard, Suite 950, Philadelphia, PA 19103.

**PLEASE TAKE FURTHER NOTICE** that because the Debtors have limited contact information Holder of Equity Interests, if you wish to receive any form of recovery on account of your Equity Interests, please contact proposed counsel to the Debtors, Pashman Stein Walder Hayden, P.C., 824 N. Market Street, Suite 800, Wilmington, Delaware 19801, Attn: Joseph C. Barsalona II (jbarsalona@pashmanstein.com), Michael Custer (mcluster@pashmanstein.com), and Alexis R. Gambale (agambale@pashmanstein.com), and relay the following information: (i) the full name of the individual or entity that is a Holder of Equity Interests of the Debtors; and (ii) the physical mailing address where the foregoing individual or entity would like to receive legal documents and any distribution materials. Should the information match the Debtors’ books and records, you will receive a Distribution. If the Debtors do not have this information, it may forever bar you from receiving any recovery in these Chapter 11 Cases.

**PLEASE TAKE FURTHER NOTICE** that any party, not listed in the Plan, believing it (i) holds a Claim against the Debtors(s), or (ii) Interest in the Debtors(s), must file a Proof of Claim and/or Proof of Equity Interest on or before the Confirmation Date, provided however, the government bar date shall be 180 days from the Petition Date per section 502(b)(9) of the Bankruptcy Code.

**UNLESS AN OBJECTION IS TIMELY SERVED AND FILED IN ACCORDANCE WITH THIS NOTICE IT MAY NOT BE CONSIDERED BY THE BANKRUPTCY COURT.**

**YOU ARE ADVISED TO CAREFULLY REVIEW AND CONSIDER THE PLAN, INCLUDING THE RELEASE, EXCULPATION, DISCHARGE, AND INJUNCTION PROVISIONS, AS YOUR RIGHTS MIGHT BE AFFECTED.**

Dated: April 16, 2025, Wilmington, Delaware  
**PASHMAN STEIN WALDER HAYDEN, P.C.**, s/ Joseph C. Barsalona II, Joseph C. Barsalona II (No. 6102), Michael Custer (No. 4843), Alexis R. Gambale (No. 7150), 824 North Market Street, Suite 800, Wilmington, DE 19801, Telephone: (302) 592-6496, E-mail: jbarsalona@pashmanstein.com, mcluster@pashmanstein.com, agambale@pashmanstein.com, Proposed Counsel to the Debtor and Debtor in Possession

<sup>1</sup> The Debtors in these Chapter 11 Cases, along with the last four digits of each Debtor’s federal tax identification number are as follows: Creativemass Holdings, Inc. (2172) and Creativemass Enterprises US LLC (9046). The Debtors’ mailing address is 1209 N Orange Street, Wilmington, DE 19801.



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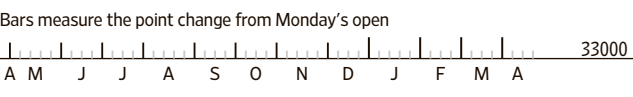
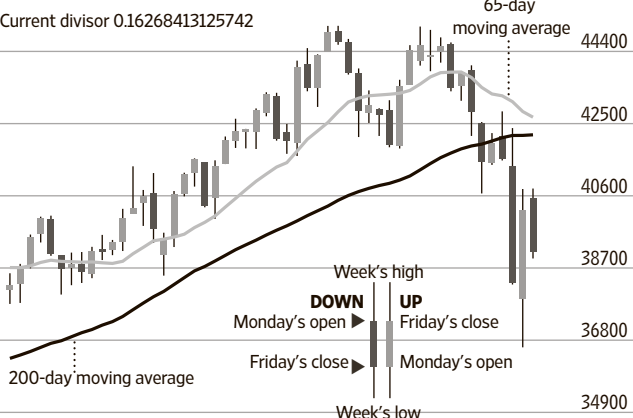
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# MARKETS DIGEST

## Dow Jones Industrial Average

**39142.23** ▼1070.48, or 2.66% last week  
High, low, open and close for each of the past 52 weeks  
Last Trailing P/E ratio 23.19  
Year ago P/E estimate \* 18.59  
Dividend yield 1.82  
All-time high 45014.04, 12/04/24



**NYSE weekly volume**, in billions of shares  
Primary market Composite  
\*Weekly P/E data based on as-reported earnings from Biriiny Associates Inc.; \*Based on Nasdaq-100 Index

## Major U.S. Stock-Market Indexes

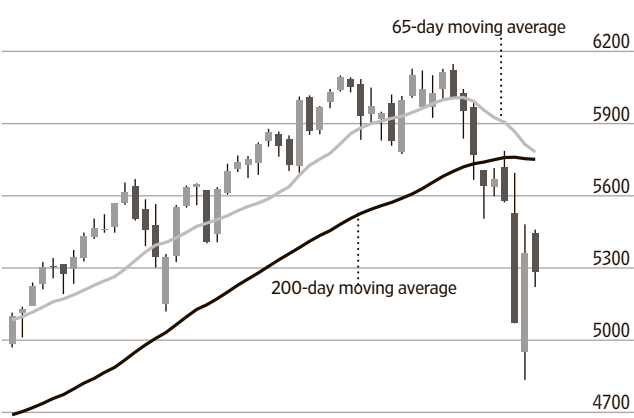
|                                 | High     | Low      | Latest Week Close | Net chg  | % chg         | Low      | 52-Week Close (●) | High     | % chg        | YTD   | 3-yr. ann.   |
|---------------------------------|----------|----------|-------------------|----------|---------------|----------|-------------------|----------|--------------|-------|--------------|
| <b>Dow Jones</b>                |          |          |                   |          |               |          |                   |          |              |       |              |
| Industrial Average              | 40791.18 | 38950.31 | <b>39142.23</b>   | -1070.48 | <b>-2.66</b>  | 37645.59 | ●                 | 45014.04 | <b>-3.6</b>  | -8.0  | <b>4.3</b>   |
| Transportation Avg              | 13675.36 | 12967.74 | <b>13438.72</b>   | 29.23    | ●             | 12637.04 | ●                 | 17754.38 | <b>-10.1</b> | -15.5 | <b>-3.3</b>  |
| Utility Average                 | 1032.21  | 997.01   | <b>1020.55</b>    | 24.07    | ●             | 860.30   | ●                 | 1079.88  | <b>18.6</b>  | 3.8   | <b>-1.3</b>  |
| Total Stock Market              | 53711.79 | 51453.81 | <b>52114.42</b>   | -649.16  | <b>-1.23</b>  | 49067.76 | ●                 | 61024.05 | <b>4.7</b>   | -10.8 | <b>5.5</b>   |
| Barron's 400                    | 1145.58  | 1112.64  | <b>1131.14</b>    | 9.47     | ●             | 1058.38  | ●                 | 1356.99  | <b>3.6</b>   | -9.6  | <b>4.4</b>   |
| <b>Nasdaq Stock Market</b>      |          |          |                   |          |               |          |                   |          |              |       |              |
| Nasdaq Composite                | 17136.56 | 16066.46 | <b>16286.45</b>   | -438.01  | <b>-2.62</b>  | 15267.91 | ●                 | 20173.89 | <b>4.4</b>   | -15.7 | <b>6.8</b>   |
| Nasdaq-100                      | 19115.15 | 17995.96 | <b>18258.09</b>   | -431.96  | <b>-2.31</b>  | 17037.65 | ●                 | 22175.6  | <b>5.0</b>   | -13.1 | <b>9.5</b>   |
| <b>S&amp;P</b>                  |          |          |                   |          |               |          |                   |          |              |       |              |
| 500 Index                       | 5459.46  | 5220.79  | <b>5282.70</b>    | -80.66   | <b>-1.50</b>  | 4967.23  | ●                 | 6144.15  | <b>5.4</b>   | -10.2 | <b>6.3</b>   |
| MidCap 400                      | 2782.64  | 2692.35  | <b>2744.39</b>    | 21.84    | ●             | 2560.93  | ●                 | 3390.26  | <b>-2.9</b>  | -12.1 | <b>1.4</b>   |
| SmallCap 600                    | 1189.64  | 1149.96  | <b>1174.88</b>    | 10.14    | ●             | 1106.12  | ●                 | 1544.66  | <b>-5.5</b>  | -16.6 | <b>-2.9</b>  |
| <b>Other Indexes</b>            |          |          |                   |          |               |          |                   |          |              |       |              |
| Russell 2000                    | 1902.32  | 1842.15  | <b>1880.62</b>    | 20.42    | ●             | 1760.71  | ●                 | 2442.03  | <b>-3.2</b>  | -15.7 | <b>-2.1</b>  |
| NYSE Composite                  | 18604.06 | 18124.48 | <b>18367.12</b>   | 147.47   | ●             | 17188.46 | ●                 | 20272.04 | <b>5.6</b>   | -3.8  | <b>3.6</b>   |
| Value Line                      | 536.21   | 518.55   | <b>528.61</b>     | 2.04     | ●             | 495.50   | ●                 | 656.04   | <b>-7.6</b>  | -13.5 | <b>-5.3</b>  |
| NYSE Arca Biotech               | 5434.00  | 5226.43  | <b>5275.52</b>    | 2.81     | ●             | 4861.76  | ●                 | 6318.63  | <b>8.5</b>   | -8.2  | <b>0.3</b>   |
| NYSE Arca Pharma                | 921.47   | 873.37   | <b>911.02</b>     | 31.21    | ●             | 860.88   | ●                 | 1140.17  | <b>-5.8</b>  | -2.5  | <b>1.8</b>   |
| KBW Bank                        | 113.17   | 108.31   | <b>110.60</b>     | 2.05     | ●             | 98.18    | ●                 | 140.59   | <b>12.7</b>  | -13.2 | <b>-1.8</b>  |
| PHLX <sup>®</sup> Gold/Silver   | 199.41   | 183.84   | <b>191.49</b>     | 3.48     | ●             | 131.01   | ●                 | 195.33   | <b>41.5</b>  | 39.6  | <b>4.5</b>   |
| PHLX <sup>®</sup> Oil Service   | 57.62    | 53.88    | <b>57.10</b>      | 2.510    | ●             | 50.70    | ●                 | 92.31    | <b>-34.5</b> | -21.4 | <b>-12.6</b> |
| PHLX <sup>®</sup> Semiconductor | 4098.44  | 3733.99  | <b>3832.47</b>    | -158.43  | <b>-3.97</b>  | 3562.94  | ●                 | 5904.54  | <b>-14.7</b> | -23.0 | <b>8.2</b>   |
| Cboe Volatility                 | 35.17    | 28.29    | <b>29.65</b>      | -7.91    | <b>-21.06</b> | 11.86    | ●                 | 52.33    | <b>64.7</b>  | 70.9  | <b>9.3</b>   |

§Nasdaq PHLX

Sources: FactSet; Dow Jones Market Data

## S&P 500 Index

**5282.70** ▼80.66, or 1.50% last week  
High, low, open and close for each of the past 52 weeks  
Last Trailing P/E ratio \* 21.37  
Year ago P/E estimate \* 19.78  
Dividend yield \* 1.47  
All-time high 6144.15, 02/19/25



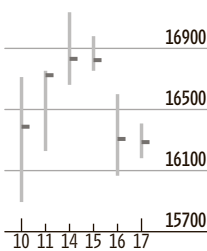
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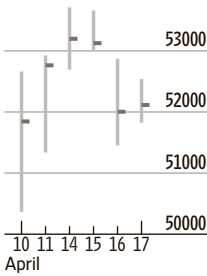
## Nasdaq Composite

**▼ 438.01, or -2.62% last week**



## DJ US TSM

**▼ 649.16, or -1.23% last week**



## New to the Market

## Public Offerings of Stock

## IPOs in the U.S. Market

None expected this week

## Lockup Expirations

Below, companies whose officers and other insiders will become eligible to sell shares in their newly public companies for the first time. Such sales can move the stock's price.

| Lockup expiration | Issue date    |                                | Symbol      | Offer price(\$) | Offer amt (\$ mil.) | Through Friday (%) | Lockup provision |
|-------------------|---------------|--------------------------------|-------------|-----------------|---------------------|--------------------|------------------|
| April 20          | Oct. 23, '24  | High Roller Technologies       | <b>ROLR</b> | 8.00            | 10.0                | <b>-64.38</b>      | 180              |
|                   | Oct. 29, '24  | Gelteq                         | <b>GELS</b> | 4.00            | 5.2                 | <b>-76.75</b>      | 180              |
|                   | Oct. 23, '24  | Founder Group                  | <b>FGL</b>  | 4.00            | 4.9                 | <b>-62.25</b>      | 180              |
|                   | Oct. 23, '24  | Li Bang International          | <b>LBGI</b> | 4.00            | 7.0                 | <b>-68.75</b>      | 180              |
|                   | Oct. 23, '24  | SAG Holdings                   | <b>SAG</b>  | 8.00            | 7.0                 | <b>-88.38</b>      | 180              |
|                   | Oct. 24, '24  | Ingram Micro Holding           | <b>INGM</b> | 22.00           | 470.6               | <b>-26.50</b>      | 180              |
|                   | Oct. 25, '24  | Septerna                       | <b>SEPN</b> | 18.00           | 331.2               | <b>-67.22</b>      | 180              |
|                   | Oct. 25, '24  | WeRide                         | <b>WRD</b>  | 15.50           | 440.7               | <b>-46.90</b>      | 180              |
|                   | Oct. 29, '24  | Polyrizon                      | <b>PLRZ</b> | 4.38            | 4.2                 | <b>-91.78</b>      | 180              |
| April 21          | Jan. 22, '25  | Diginex                        | <b>DGNX</b> | 4.10            | 10.6                | <b>2707.32</b>     | 90               |
|                   | Jan. 22, '25  | CTRL Group                     | <b>MCTR</b> | 4.00            | 9.2                 | <b>87.75</b>       | 90               |
| April 23          | Jan. 23, '25  | Skyline Builders Group Holding | <b>SKBL</b> | 4.00            | 6.9                 | <b>183.50</b>      | 90               |
| April 25          | April 26, '24 | CleanCore Solutions            | <b>ZONE</b> | 4.00            | 5.0                 | <b>-70.25</b>      | 365              |

Sources: LSEG Data and Analytics; Dow Jones Market Data

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## IPO Scorecard

Performance of IPOs, most-recent listed first

| Company                    | SYMBOL       | Thurs.'s close (\$) | % Chg From Offer price | 1st-day close | Company                     | SYMBOL       | Thurs.'s close (\$) | % Chg From Offer price | 1st-day close |
|----------------------------|--------------|---------------------|------------------------|---------------|-----------------------------|--------------|---------------------|------------------------|---------------|
| Chagee Hldgs               | <b>CHA</b>   | 32.44               | <b>15.9</b>            | ...           | Ruanyn Edai Technology      | <b>RYET</b>  | 7.72                | <b>93.0</b>            | 71.2          |
| Everbright Digital Holding | <b>EDHL</b>  | 4.20                | <b>5.0</b>             | ...           | BeLive Hldgs                | <b>BLIV</b>  | 3.04                | <b>-24.0</b>           | -14.1         |
| Haoxin Hldgs               | <b>HXHX</b>  | 2.10                | <b>-47.5</b>           | -32.5         | Lianhe Sowell Intl          | <b>LHSW</b>  | 3.65                | <b>-8.8</b>            | -14.9         |
| Aether Hldgs               | <b>ATHR</b>  | 5.26                | <b>22.3</b>            | -8.5          | Sizzle Acquisition II       | <b>SZZLU</b> | 9.99                | <b>-0.1</b>            | -0.2          |
| Cuprina Hldgs              | <b>CUPR</b>  | 4.95                | <b>23.8</b>            | 19.3          | SmartStop Self Storage REIT | <b>SMA</b>   | 33.94               | <b>13.1</b>            | 3.6           |
| iOTthree                   | <b>IOTR</b>  | 1.15                | <b>-71.3</b>           | -67.6         | Top Win Intl                | <b>TOPW</b>  | 2.55                | <b>-36.2</b>           | -33.4         |
| MasterBeef Grp             | <b>MB</b>    | 3.08                | <b>-23.0</b>           | -24.9         | Energys Grp                 | <b>ENGS</b>  | 5.11                | <b>13.4</b>            | 3.1           |
| Tianci Intl                | <b>CIIT</b>  | 2.78                | <b>-30.5</b>           | 6.9           | Siddhi Acquisition          | <b>SDHIU</b> | 10.05               | <b>0.5</b>             | 0.3           |
| Titan Acquisition          | <b>TACHU</b> | 10.00               | ...                    | 0.1           | Watson Financial            | <b>WTF</b>   | 6.48                | <b>62.0</b>            | -67.4         |
| FatPipe                    | <b>FATN</b>  | 12.43               | <b>116.2</b>           | -4.4          | UY Scuti Acquisition        | <b>UYSCU</b> | 10.07               | <b>0.7</b>             | 0.1           |

Sources: Dow Jones Market Data; FactSet

## Public and Private Borrowing

### Treasuries

|                                                                           | Tuesday, April 22 | Monday, April 21                                                            |
|---------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------|
| Auction of 6 week bill; announced on April 17; settles on April 24        |                   | Auction of 13 and 26 week bills; announced on April 17; settles on April 24 |
| Auction of 2 year note; announced on April 17; settles on April 30        |                   | Wednesday, April 23                                                         |
| Thursday, April 24                                                        |                   | Auction of 2 year FRN; announced on April 17; settles on April 30           |
| Auction of 7 year note; announced on April 17; settles on April 30        |                   | Auction of 5 year note; announced on April 17; settles on April 30          |
| Auction of 4 and 8 week bills; announced on April 22; settles on April 29 |                   | Auction of 17 week bill; announced on April 22; settles on April 29         |

## A Week in the Life of the DJIA

A look at how the Dow Jones Industrial Average component stocks did in the past week and how much each moved the index. The DJIA lost 1,070.48 points, or 2.66%, on the week. A \$1 change in the price of any DJIA stock = 6.15-point change in the average. To date, a \$1,000 investment on Dec. 31 in each current DJIA stock component would have returned \$28,135, or a loss of 6.22%, on the \$30,000 investment, including reinvested dividends.

| The Week's Action |             |                      | Company            | Symbol | Close    | \$1,000 Invested (Year-end '24) |
|-------------------|-------------|----------------------|--------------------|--------|----------|---------------------------------|
| Pct chg (%)       | Stock price | Point chg in average |                    |        |          |                                 |
| 4.38              | 10.73       | 65.96                | Travelers          | TRV    | \$255.86 | \$1,066                         |
| 3.78              | 5.74        | 35.28                | Johnson & Johnson  | JNJ    | 157.47   | 1,098                           |
| 3.23              | 5.06        | 31.10                | Boeing             | BA     | 161.90   | 915                             |
| 3.04              | 15.05       | 92.51                | Goldman Sachs      | GS     | 509.49   | 894                             |
| 2.52              | 1.37        | 8.42                 | NIKE               | NKE    | 55.76    | 741                             |
| 2.23              | 3.72        | 22.87                | Procter & Gamble   | PG     | 170.63   | 1,024                           |
| 2.20              | 1.57        | 9.65                 | Coca-Cola          | KO     | 73.00    | 1,181                           |
| 1.65              | 2.24        | 13.77                | Chevron            | CVX    | 137.87   | 962                             |
| 1.41              | 3.33        | 20.47                | IBM                | IBM    | 238.81   | 1,094                           |
| 0.71              | 0.31        | 1.91                 | Verizon            | VZ     | 44.04    | 1,139                           |
| 0.45              | 0.42        | 2.58                 | Walmart            | WMT    | 93.22    | 1,035                           |
| 0.45              | 1.40        | 8.61                 | McDonald's         | MCD    | 311.30   | 1,080                           |
| 0.34              | 1.20        | 7.38                 | Home Depot         | HD     | 355.06   | 919                             |
| 0.27              | 0.80        | 4.92                 | Caterpillar        | CAT    | 294.25   | 814                             |
| 0.07              | 0.18        | 1.11                 | American Express   | AXP    | 251.31   | 852                             |
| -0.09             | -0.08       | -0.49                | Walt Disney        | DIS    | 84.81    | 762                             |
| -0.59             | -1.17       | -7.19                | Apple              | AAPL   | 196.98   | 787                             |
| -0.98             | -1.95       | -11.99               | Honeywell          | HON    | 196.60   | 875                             |
| -1.01             | -3.39       | -20.84               | Sherwin-Williams   | SHW    | 333.38   | 983                             |
| -1.14             | -3.79       | -23.30               | Visa               | V      | 329.61   | 1,045                           |
| -1.49             | -1.18       | -7.25                | Merck & Co         | MRK    | 78.00    | 791                             |
| -1.80             | -4.24       | -26.06               | JPMorgan Chase     | JPM    | 231.96   | 979                             |
| -2.74             | -1.57       | -9.65                | Cisco              | CSCO   | 55.76    | 955                             |
| -3.04             | -7.74       | -47.58               | Salesforce         | CRM    | 247.26   | 741                             |
| -3.04             | -8.69       | -53.42               | Amgen              | AMGN   | 277.29   | 1,073                           |
| -4.22             | -5.74       | -35.28               | 3M                 | MMM    | 130.21   | 1,014                           |
| -5.32             | -20.67      | -127.06              | Microsoft          | MSFT   | 367.78   | 874                             |
| -6.63             | -12.26      | -75.36               | Amazon.com         | AMZN   | 172.61   | 787                             |
| -8.51             | -9.44       | -58.03               | NVIDIA             | NVDA   | 101.49   | 756                             |
| -24.25            | -145.36     | -893.51              | UnitedHealth Group | UNH    | 454.11   | 902                             |

\*Based on Composite price. DJIA is calculated on primary-market price. Source: Dow Jones Market Data; FactSet.

## International Stock Indexes

| Region/Country | Index                    | Close     | Latest Week % chg | Low       | 52-Week Range Close                                                | High      | YTD % chg |
|----------------|--------------------------|-----------|-------------------|-----------|--------------------------------------------------------------------|-----------|-----------|
| World          | MSCI ACWI                | 793.42    | 0.35              | 742.96    | <div><div></div><div></div><div></div><div></div><div></div></div> | 887.72    | -5.7      |
|                | MSCI ACWI ex-USA         | 337.64    | 3.56              | 306.87    | <div><div></div><div></div><div></div><div></div><div></div></div> | 357.36    | 3.61      |
|                | MSCI World               | 3476.06   | 0.14              | 3254.24   | <div><div></div><div></div><div></div><div></div><div></div></div> | 3910.71   | -6.3      |
|                | MSCI Emerging Markets    | 1068.59   | 2.24              | 993.45    | <div><div></div><div></div><div></div><div></div><div></div></div> | 1187.97   | -0.6      |
| Americas       | MSCI AC Americas         | 1997.36   | -1.19             | 1877.61   | <div><div></div><div></div><div></div><div></div><div></div></div> | 2316.90   | -9.7      |
| Canada         | S&P/TSX Comp             | 24192.81  | 2.56              | 21516.90  | <div><div></div><div></div><div></div><div></div><div></div></div> | 25808.25  | -2.2      |
| Latin Amer.    | MSCI EM Latin America    | 2047.78   | 3.48              | 1842.59   | <div><div></div><div></div><div></div><div></div><div></div></div> | 2511.08   | 10.5      |
| Brazil         | Bovespa                  | 129650.03 | 1.54              | 118532.68 | <div><div></div><div></div><div></div><div></div><div></div></div> | 137343.96 | 7.8       |
| Chile          | S&P IPSA                 | 4136.73   | 4.98              | 3306.43   | <div><div></div><div></div><div></div><div></div><div></div></div> | 4136.73   | 14.4      |
| Mexico         | S&P/BMV IPC              | 53018.57  | 2.95              | 48837.72  | <div><div></div><div></div><div></div><div></div><div></div></div> | 57846.08  | 7.1       |
| EMEA           | STOXX Europe 600         | 506.42    | 4.03              | 469.89    | <div><div></div><div></div><div></div><div></div><div></div></div> | 563.13    | -0.2      |
|                | STOXX Europe 50          | 4233.13   | 3.31              | 3968.17   | <div><div></div><div></div><div></div><div></div><div></div></div> | 4809.98   | -1.8      |
| Eurozone       | Euro STOXX               | 517.79    | 3.70              | 472.81    | <div><div></div><div></div><div></div><div></div><div></div></div> | 569.12    | 2.5       |
|                | Euro STOXX 50            | 4935.34   | 3.09              | 4571.60   | <div><div></div><div></div><div></div><div></div><div></div></div> | 5540.69   | 0.8       |
| Austria        | ATX                      | 3916.09   | 5.38              | 3443.70   | <div><div></div><div></div><div></div><div></div><div></div></div> | 4350.76   | 6.9       |
| Belgium        | Bel-20                   | 4197.65   | 4.19              | 3826.58   | <div><div></div><div></div><div></div><div></div><div></div></div> | 4500.10   | -1.6      |
| France         | CAC 40                   | 7285.86   | 2.55              | 6863.02   | <div><div></div><div></div><div></div><div></div><div></div></div> | 8239.99   | -1.3      |
| Germany        | DAX                      | 21205.86  | 4.08              | 17339.00  | <div><div></div><div></div><div></div><div></div><div></div></div> | 23419.48  | 6.5       |
| Greece         | Athex Composite          | 1642.17   | 3.76              | 1341.23   | <div><div></div><div></div><div></div><div></div><div></div></div> | 1746.67   | 11.7      |
| Israel         | Tel Aviv                 | 2490.51   | 3.01              | 1912.45   | <div><div></div><div></div><div></div><div></div><div></div></div> | 2552.30   | 4.0       |
| Italy          | FTSE MIB                 | 35980.43  | 5.74              | 31107     | <div><div></div><div></div><div></div><div></div><div></div></div> | 39713     | 5.2       |
| Netherlands    | AEX                      | 852.20    | 3.98              | 796.45    | <div><div></div><div></div><div></div><div></div><div></div></div> | 948.54    | -3.0      |
| Norway         | Oslo Bors All-Share      | 1660.79   | 2.71              | 1561.41   | <div><div></div><div></div><div></div><div></div><div></div></div> | 1776.32   | 1.0       |
| Portugal       | PSI 20                   | 6735.84   | 3.30              | 6253.96   | <div><div></div><div></div><div></div><div></div><div></div></div> | 6988.33   | 5.6       |
| South Africa   | FTSE/JSE All-Share       | 89486.79  | 3.56              | 73271.43  | <div><div></div><div></div><div></div><div></div><div></div></div> | 90149.69  | 6.4       |
| Spain          | IBEX 35                  | 12918.00  | 5.14              | 10390.5   | <div><div></div><div></div><div></div><div></div><div></div></div> | 13484.3   | 11.4      |
| Sweden         | OMX Stockholm            | 894.01    | 3.39              | 824.22    | <div><div></div><div></div><div></div><div></div><div></div></div> | 1057.57   | -6.3      |
| Switzerland    | Swiss Market             | 11660.96  | 3.75              | 10887.73  | <div><div></div><div></div><div></div><div></div><div></div></div> | 13166.68  | 1.5       |
| U.K.           | FTSE 100                 | 8275.66   | 3.91              | 7679.48   | <div><div></div><div></div><div></div><div></div><div></div></div> | 8871.31   | 0.3       |
| Asia-Pacific   | MSCI AC Asia Pacific     | 179.09    | 3.35              | 162.82    | <div><div></div><div></div><div></div><div></div><div></div></div> | 197.3     | -1.4      |
| Australia      | S&P/ASX 200              | 7819.10   | 2.26              | 7343.3    | <div><div></div><div></div><div></div><div></div><div></div></div> | 8555.8    | -4.2      |
| China          | Shanghai Composite       | 3276.73   | 1.19              | 2704.09   | <div><div></div><div></div><div></div><div></div><div></div></div> | 3489.78   | -2.2      |
| Hong Kong      | Hang Seng                | 21395.14  | 2.30              | 16224.14  | <div><div></div><div></div><div></div><div></div><div></div></div> | 24771.14  | 6.7       |
| India          | BSE Sensex               | 78553.20  | 4.52              | 72079.05  | <div><div></div><div></div><div></div><div></div><div></div></div> | 85836.12  | 0.5       |
| Japan          | NIKKEI 225               | 34730.28  | 3.41              | 31136.58  | <div><div></div><div></div><div></div><div></div><div></div></div> | 42224.02  | -12.9     |
| Malaysia       | FTSE Bursa Malaysia KLCI | 1499.40   | 3.07              | 1400.59   | <div><div></div><div></div><div></div><div></div><div></div></div> | 1678.80   | -8.7      |
| Singapore      | Straits Times            | 3720.33   | 5.92              | 3176.51   | <div><div></div><div></div><div></div><div></div><div></div></div> | 3981.57   | -1.8      |
| South Korea    | KOSPI                    | 2483.42   | 2.08              | 2293.70   | <div><div></div><div></div><div></div><div></div><div></div></div> | 2891.35   | 3.5       |
| Taiwan         | TAIEX                    | 19395.03  | -0.68             | 17391.76  | <div><div></div><div></div><div></div><div></div><div></div></div> | 24390.03  | -15.8     |



## CLOSED-END FUNDS

Listed are the 300 largest closed-end funds as measured by assets. Closed-end funds sell a limited number of shares and invest the proceeds in securities. Unlike open-end funds, closed-ends generally do not buy their shares back from investors who wish to cash in their holdings. Instead, fund shares trade on a stock exchange. **NA** signifies that the information is not available or not applicable. **NS** signifies funds not in existence for the entire period. 12 month yield is computed by dividing income dividends paid (during the previous 12 months for periods ending at month-end or during the previous 52 weeks for periods ending at any time other than month-end) by the latest month-end market price adjusted for capital gains distributions. Depending on the fund category, either 12-month yield or total return is listed.

Source: LSEG

Thursday, April 17, 2025

12/11/2023, April 27, 2024

| Fund (SYM)                           | NAV   | Close/Disc | Prem   | TL    | Ret | 52wk |
|--------------------------------------|-------|------------|--------|-------|-----|------|
| General Equity Funds                 |       |            |        |       |     |      |
| Adams Diversified Inc <b>ADX</b>     | 20.10 | 18.18      | -9.6   | 13.5  |     |      |
| BlackRock Sc Eq Cap Cr <b>CI</b>     | 18.66 | 17.27      | -7.4   | 13    |     |      |
| Central Secs <b>CEY</b>              | 37.19 | 42.47      | -18.0  | 14.5  |     |      |
| CoHensStAccDvPndFm <b>CVF</b>        | 11.32 | 11.35      | +0.3   | 11.6  |     |      |
| EVNTrsAdvEqIncl <b>EVT</b>           | NA    | 21.29      | NA     | 4.4   |     |      |
| BlackRock DivsIncl Tr <b>GDV</b>     | 25.85 | 22.50      | -12.0  | 13.5  |     |      |
| Gabelli Equity Tr <b>GAB</b>         | 4.86  | 5.27       | +8.4   | 13.1  |     |      |
| General Amer <b>GAM</b>              | 56.17 | 48.96      | -12.8  | 20.3  |     |      |
| JHancockTaxAdv <b>HTD</b>            | 24.35 | 22.03      | -9.5   | 25.1  |     |      |
| Liberty All-Star Growth <b>USA</b>   | NA    | 6.10       | NA     | 0.5   |     |      |
| Liberty All-Star Growth <b>USG</b>   | NA    | 6.10       | NA     | -2.0  |     |      |
| Royce Micro Cap Tr <b>RMT</b>        | 8.92  | 7.90       | -11.4  | 5.8   |     |      |
| Royce Value Trust <b>RTV</b>         | 14.67 | 13.40      | -8.7   | 5.4   |     |      |
| Source Capital <b>SOR</b>            | 42.65 | 41.10      | -3.6   | 11.2  |     |      |
| Special Opportunities <b>SPE</b>     | 15.74 | 14.14      | -10.2  | 24.0  |     |      |
| Sprott Focus Fund <b>SF</b>          | 7.88  | 7.00       | -11.2  | -0.8  |     |      |
| SPRI Total Return <b>STEW</b>        | 20.69 | 16.52      | -20.2  | 19.3  |     |      |
| Tri-Continental <b>TV</b>            | 32.35 | 28.48      | -12.0  | 8.0   |     |      |
| Specialized Equity Funds             |       |            |        |       |     |      |
| abrdn Gbl Prem <b>ABP</b>            | 3.74  | 3.78       | +0.9   | 18.2  |     |      |
| abrdn Gbl Infra Inc <b>ASGI</b>      | 20.31 | 18.64      | -8.3   | 25.0  |     |      |
| Adams Natural Resources <b>PE</b>    | 22.88 | 20.32      | -11.2  | -4.5  |     |      |
| ASA Gold & Prec Met Ltd <b>ASA</b>   | 34.86 | 30.74      | -11.8  | 76.0  |     |      |
| BlackRock Energy & Env <b>BJR</b>    | 13.07 | 12.25      | -6.3   | 0.0   |     |      |
| BlackRock Energy Res Div <b>BDJ</b>  | 8.60  | 7.77       | -9.3   | 9.8   |     |      |
| BlackRock Enbl Gbl Div <b>BEG</b>    | 11.21 | 10.19      | -9.1   | 11.6  |     |      |
| BlackRock Enbl Intl Div <b>BOE</b>   | 5.80  | 5.26       | -9.3   | 11.2  |     |      |
| BlackRock ESGC Cap Tr <b>BCAT</b>    | 15.42 | 14.56      | -5.8   | 11.0  |     |      |
| BlackRock Hth & Sol Tm <b>BMAT</b>   | 14.42 | 13.62      | -5.5   | 7.9   |     |      |
| BlackRock Hlth Sciences <b>BME</b>   | 38.86 | 36.00      | -7.4   | -0.4  |     |      |
| BlackRock Res & Comm <b>BRC</b>      | 35.95 | 8.77       | -8.2   | 4.2   |     |      |
| BlackRock Sci&Tech Trn <b>BSTZ</b>   | 18.21 | 16.73      | -8.1   | 11.5  |     |      |
| BlackRock Sci&Tech Trust <b>BST</b>  | 32.76 | 31.49      | -3.9   | 3.7   |     |      |
| BlackRock Invs&Grwth Trn <b>BTX</b>  | 6.25  | NA         | NA     | NA    |     |      |
| BlackRock Utilty & Pwr <b>BU</b>     | 44.40 | 22.39      | 0.0    | 15.1  |     |      |
| CLASSENERGOE&Env M&P <b>BMO</b>      | NA    | 22.93      | 0.0    | 20.9  |     |      |
| ChrStnRnfrt <b>UTL</b>               | NA    | 24.84      | NA     | 20.7  |     |      |
| Cohen&SteinQualf <b>ROI</b>          | 12.52 | 11.73      | -6.3   | 17.7  |     |      |
| Cohen&SteinReal Est <b>RLTY</b>      | NA    | 14.55      | NA     | 20.6  |     |      |
| Cohen&SteinToteRt <b>RTT</b>         | 11.39 | 11.66      | +2.4   | 15.9  |     |      |
| Cohen&SteinPrf Tech <b>RNP</b>       | 20.43 | 20.55      | +1.7   | 14.9  |     |      |
| Columbia Sel Pm Tech <b>SCP</b>      | 25.22 | 26.08      | +3.4   | 26.8  |     |      |
| Duff&Pb Dvnt Inc <b>DNP</b>          | NA    | 9.16       | NA     | 17.5  |     |      |
| Duff&Pb S&Infra Inc <b>DPG</b>       | NA    | 12.40      | NA     | 41.4  |     |      |
| EnVnTrsEqIncl <b>EOI</b>             | NA    | 17.40      | NA     | 7.5   |     |      |
| EnVnTrsEqIncl <b>EOS</b>             | NA    | 19.75      | NA     | 11.6  |     |      |
| EVRS&Mn&DevEqIncl <b>ETJ</b>         | NA    | 8.17       | NA     | 9.3   |     |      |
| EnVnTrsMgbyWtrInv <b>ETB</b>         | NA    | 12.75      | NA     | 7.4   |     |      |
| EnVnTrsMgbyWtrOp <b>ETV</b>          | NA    | 12.18      | NA     | 7.7   |     |      |
| EnVnTrsMn&DevEqIncl <b>ETV</b>       | NA    | 13.25      | NA     | 12.0  |     |      |
| EnVnTrsMgblbBb <b>ETV</b>            | NA    | 7.75       | NA     | 10.2  |     |      |
| EnVnTrsMn&DevEqIncl <b>EXG</b>       | NA    | 7.65       | NA     | 8.8   |     |      |
| First Tr Enhanced Eq <b>FFA</b>      | 10.97 | 18.11      | -5.0   | 8.5   |     |      |
| Gabelli Healthcare <b>GRX</b>        | 10.71 | 9.40       | -12.2  | 6.7   |     |      |
| Galt Utility <b>GU</b>               | 3.08  | 5.24       | -70.1  | 14.9  |     |      |
| GAMCO Global NatRes <b>GGN</b>       | 4.36  | 4.30       | -1.4   | 14.9  |     |      |
| Jan Fin Flnt Opns <b>FOF</b>         | 29.86 | 30.44      | +1.9   | 21.5  |     |      |
| KayneErnstGryfnalnc <b>KYN</b>       | 13.65 | 11.74      | -14.0  | 29.5  |     |      |
| Neuberger Brmn Eqty <b>BL</b>        | 9.31  | 8.36       | -10.2  | 23.9  |     |      |
| Neuberger Brmn Gen Con <b>NBLG</b>   | 13.04 | 11.44      | -12.3  | 10.5  |     |      |
| NubrgRtIESTSec <b>NRO</b>            | 3.25  | 3.05       | -6.2   | 12.3  |     |      |
| Nw Dow 30rd Qvnt <b>DIA</b>          | 15.33 | 13.50      | -11.9  | 6.6   |     |      |
| Nw Dow 30rd Alpha <b>QQQX</b>        | 24.91 | 22.38      | -10.2  | 7.8   |     |      |
| Nw Real Est <b>JRS</b>               | 8.21  | 7.65       | -6.8   | 15.9  |     |      |
| Nw SSP500DvntDf <b>BSX</b>           | NA    | 15.75      | NA     | 13.7  |     |      |
| Nw SP500BuyVid <b>PMXX</b>           | 13.83 | 12.53      | -10.7  | 6.7   |     |      |
| NGX Gushing Midstream <b>NG</b>      | 38.29 | 38.15      | -0.4   | 7.0   |     |      |
| NGX NextGen Infra Inc <b>NXG</b>     | 40.49 | 41.90      | +2.5   | 27.7  |     |      |
| Convertible Sec's Funds              |       |            |        |       |     |      |
| AdvtCnvtbl&BdPrfSec <b>AVK</b>       | 11.06 | 10.78      | -2.5   | 6.7   |     |      |
| CalamosConvHi <b>CH</b>              | 3.99  | 3.87       | -5.1   | -0.5  |     |      |
| CalamosConvLo <b>CHI</b>             | 8.89  | 9.62       | +8.2   | -0.1  |     |      |
| V Conv & Inc <b>NCV</b>              | NA    | 12.47      | NA     | 15.7  |     |      |
| V Conv & Inc L <b>NCV</b>            | NA    | 11.12      | NA     | 15.9  |     |      |
| V Div & Inc <b>ACV</b>               | NA    | 19.14      | NA     | -3.3  |     |      |
| V Eqty & Conv Inc <b>NIE</b>         | NA    | 20.81      | NA     | 6.1   |     |      |
| World Equity Funds                   |       |            |        |       |     |      |
| abrdn Emk BvS-Chna Aff <b>AGD</b>    | 5.62  | 4.84       | -13.9  | 8.6   |     |      |
| abrdn Gbls Dv Trn <b>AGD</b>         | 10.22 | 9.52       | -6.8   | 16.7  |     |      |
| abrdn Trn Dvtn Dv <b>ADD</b>         | 8.84  | 7.96       | -10.0  | 14.6  |     |      |
| Allsprrng Gbl AdvPort <b>EOD</b>     | 5.30  | 4.76       | -10.2  | 18.5  |     |      |
| BlackRock Cap Trn <b>BCAT</b>        | 14.85 | 13.76      | -7.3   | 10.8  |     |      |
| Calamos GblDyNcl <b>CH</b>           | 6.79  | 6.03       | -11.2  | 7.2   |     |      |
| EV TrsAdvGblDvnt <b>ETG</b>          | NA    | 23.26      | NA     | 11.4  |     |      |
| EVNTrsAdvDvnt <b>ETG</b>             | NA    | 17.33      | NA     | 7.4   |     |      |
| Highland Global Alpha <b>HGLB</b>    | 11.86 | 7.43       | -37.4  | 19.4  |     |      |
| India Fund <b>IFIN</b>               | 16.92 | 15.32      | -9.5   | -0.5  |     |      |
| Japan Smaller Cap <b>JOE</b>         | 8.83  | 8.38       | -14.8  | 15.3  |     |      |
| LazardGblTrRetncl <b>LJG</b>         | 15.93 | 14.29      | -10.3  | 6.5   |     |      |
| Mexico <b>MX</b>                     | NA    | 15.01      | NA     | -11.2 |     |      |
| MS ChinaSHRf <b>CAF</b>              | 15.30 | 15.20      | -18.3  | 6.6   |     |      |
| MS India IF                          | 29.12 | 29.43      | +4.4   | 32.1  |     |      |
| New Germany <b>GF</b>                | 11.32 | 9.97       | -11.9  | 21.1  |     |      |
| NuveenMulti-Asset Inc <b>NMAI</b>    | 12.68 | 11.53      | -9.1   | 12.7  |     |      |
| Nuveen Rl Asst Inc & Gro <b>JNI</b>  | 13.01 | 12.42      | -4.7   | 28.1  |     |      |
| NVL GRE Gblncl <b>MEGI</b>           | 14.72 | 13.24      | -10.1  | 26.1  |     |      |
| Templeton Dragon <b>TMF</b>          | 14.00 | 8.79       | -15.5  | 15.4  |     |      |
| Templeton Emk <b>EMF</b>             | 14.06 | 12.13      | -13.4  | 15.1  |     |      |
| Prem12 Mo                            |       |            |        |       |     |      |
| Fund (SYM)                           | NAV   | Close      | Disc   | Yld   |     |      |
| U.S. Mortgage Bond Funds             |       |            |        |       |     |      |
| BlackRckRMort <b>BKT</b>             | 11.88 | 11.38      | -4.2   | 8.9   |     |      |
| Investment Grade Bond Funds          |       |            |        |       |     |      |
| Angel Aok FV Inc Trn <b>FINS</b>     | 14.24 | 12.91      | -9.3   | 9.7   |     |      |
| BlRck Core Bond <b>BHK</b>           | 10.01 | 10.03      | +0.3   | 8.2   |     |      |
| BR Credit Alloc <b>CTL</b>           | 10.91 | 10.21      | -6.4   | 9.3   |     |      |
| InvescoBond <b>VBF</b>               | 15.96 | 14.93      | -6.3   | 5.6   |     |      |
| JH Income <b>JHS</b>                 | 11.86 | 10.99      | -7.3   | 4.9   |     |      |
| MS Intmdt <b>MIN</b>                 | 2.75  | 2.65       | -3.8   | 2.7   |     |      |
| Loan Participation Funds             |       |            |        |       |     |      |
| BR&F Trn <b>DSU</b>                  | 10.13 | 9.91       | -2.2   | 11.3  |     |      |
| BRF/R Inc Str <b>FRA</b>             | 12.26 | 12.10      | -1.3   | 11.5  |     |      |
| BlackRock Floating Rtn <b>BGT</b>    | 11.96 | 11.69      | -2.3   | 11.6  |     |      |
| Blackstone Lng Str Crnc <b>BKX</b>   | NA    | 11.80      | NA     | 9.7   |     |      |
| BS SFR 2027 Tr <b>BSL</b>            | NA    | 11.88      | NA     | 9.4   |     |      |
| BS STr Cr 2027 Tr <b>BGG</b>         | NA    | 11.51      | NA     | 9.3   |     |      |
| EnVnTrsFltRetncl <b>EFT</b>          | NA    | 11.75      | NA     | 10.3  |     |      |
| EV SenTrfTr <b>EFF</b>               | NA    | 11.48      | NA     | 10.5  |     |      |
| FT/Str Fltg Rte Inc <b>FCT</b>       | 10.42 | 9.77       | -6.2   | 11.4  |     |      |
| HighlandOppt <b>HFO</b>              | 12.24 | 4.86       | -60.3  | 8.2   |     |      |
| InvDYCOPpOpp <b>ATA</b>              | 10.64 | NA         | NA     | NA    |     |      |
| InvSrnlnr Tr <b>VVR</b>              | 3.70  | 3.52       | -4.9   | 13.3  |     |      |
| Nuveen Credit Trn <b>JQC</b>         | 5.57  | 5.01       | -10.1  | 12.0  |     |      |
| Nw FloatRateIncl <b>FJR</b>          | 8.58  | 7.91       | -7.8   | 12.0  |     |      |
| High Yield Bond Funds                |       |            |        |       |     |      |
| abrdn Inc Credit Str <b>ACF</b>      | 5.92  | 5.53       | -6.6   | 18.8  |     |      |
| AllianceBnrlGblHlnc <b>AWF</b>       | 10.97 | 10.13      | -7.7   | 7.4   |     |      |
| Allsprrng OpDty Hld <b>AWD</b>       | 7.08  | 6.44       | -9.0   | 9.2   |     |      |
| Bairings Gbls Dv Hld <b>BGH</b>      | 15.19 | 14.08      | -7.3   | 10.2  |     |      |
| BR Corporate Hvy <b>BLT</b>          | 9.24  | 13.18      | -0.6   | 9.7   |     |      |
| BlackRock Trn Dyr Inc <b>BW</b>      | 3.47  | 3.40       | -0.5   | 9.4   |     |      |
| Global High Yield Bond Funds         |       |            |        |       |     |      |
| PGIM Global High Yield <b>GHY</b>    | NA    | 12.03      | NA     | 9.3   |     |      |
| PGIM High Yield Bond <b>ISD</b>      | NA    | 12.97      | NA     | 9.8   |     |      |
| PGIM Sh Dur Hld Yld <b>SDH</b>       | NA    | 15.64      | NA     | 7.8   |     |      |
| PioneerHlncm <b>PHT</b>              | 7.91  | 7.32       | -7.5   | 8.5   |     |      |
| WstAsstHfI <b>HIX</b>                | NA    | 4.00       | NA     | 14.0  |     |      |
| Western Asset Hlnc Inc <b>HIO</b>    | NA    | 3.76       | NA     | 10.7  |     |      |
| Western Asset Hl Yld D <b>HVI</b>    | NA    | 11.56      | NA     | 9.6   |     |      |
| Other Domestic Taxable Bond Funds    |       |            |        |       |     |      |
| Allsprrng Multi-Sector <b>ERC</b>    | 9.62  | 8.90       | -7.5   | 8.8   |     |      |
| Arns Dynamic Rclt Alloc <b>ARDC</b>  | NA    | 13.24      | NA     | 9.9   |     |      |
| BlackRock Mtl-Sctr Bln <b>BTL</b>    | 13.96 | 13.73      | -1.6   | 10.2  |     |      |
| BlackRock Tax Muni Bld <b>BBN</b>    | 16.96 | 15.77      | -7.0   | 6.6   |     |      |
| DoubleLineOpn Crd <b>DVL</b>         | 15.15 | 15.31      | +1.1   | 6.8   |     |      |
| EVNTrsDvntDvnt <b>EDV</b>            | NA    | 9.62       | NA     | 9.3   |     |      |
| Franklin Trl Dvr Inc <b>FVI</b>      | 6.63  | 6.14       | -7.4   | 11.4  |     |      |
| FSC Credit Opportunities <b>FSCO</b> | NA    | 6.63       | NA     | 10.4  |     |      |
| JH Income Investors <b>JHI</b>       | 14.11 | 12.87      | -8.8   | 18.2  |     |      |
| MFS Charter <b>MCR</b>               | 6.56  | 6.04       | -7.9   | 3.8   |     |      |
| NuveenCorePlusMnt <b>NPCT</b>        | 11.01 | 9.97       | -9.4   | 10.9  |     |      |
| NuveenTaxableMuni <b>NBM</b>         | 15.86 | 15.14      | -4.5   | 6.9   |     |      |
| PIMCO Corp & Cnstr <b>PTC</b>        | 11.36 | 11.37      | +15.9  | 9.8   |     |      |
| PIMCO Corp Gbl Sctr <b>PCN</b>       | 11.49 | 12.62      | +9.8   | 9.8   |     |      |
| PIMCOHlnc <b>PHK</b>                 | 4.52  | 4.57       | +1.1   | 11.7  |     |      |
| PIMCO IncnsStrd <b>PIF</b>           | 7.82  | 8.10       | +3.6   | 11.3  |     |      |
| PIMCO IncnsStrd II <b>PIFN</b>       | 6.90  | 7.06       | +2.3   | 11.4  |     |      |
| Putnam Mas Inc <b>PIM</b>            | 3.41  | 3.25       | -4.9   | 8.6   |     |      |
| Putnam Prem Int <b>PIPT</b>          | 3.75  | 3.53       | -5.9   | 8.6   |     |      |
| Western Asset Dvntl Inc <b>WDI</b>   | NA    | 13.69      | NA     | 12.2  |     |      |
| Western Asset Hl Yld <b>WIA</b>      | NA    | 8.14       | NA     | 7.2   |     |      |
| Western Asset Infl L Gbl <b>WW</b>   | NA    | 8.52       | NA     | 8.2   |     |      |
| World Income Funds                   |       |            |        |       |     |      |
| abrdn AP Inc Fd <b>FAB</b>           | 16.53 | 15.01      | -9.2   | 12.6  |     |      |
| MS EmkMntDvnt <b>DEB</b>             | NA    | 4.64       | NA     | 12.5  |     |      |
| MS EmkMntDvnt <b>DEB</b>             | NA    | 7.32       | NA     | 11.4  |     |      |
| PIMCO Access Fund <b>PASX</b>        | 14.79 | 15.02      | +1.4   | 11.1  |     |      |
| PIMCO Dynamic Inc Str <b>PDK</b>     | NA    | 21.13      | NA     | 15.3  |     |      |
| PIMCO Dynamic Income <b>PDI</b>      | NA    | 17.87      | NA     | 13.7  |     |      |
| PIMCO Dynamic Inc Op <b>PIO</b>      | 12.62 | 12.13      | -4.2   | 11.0  |     |      |
| PIMCO Strategic Inc <b>RCS</b>       | 4.34  | 5.38       | +24.0  | 9.2   |     |      |
| Templeton Gbl Inc <b>SABA</b>        | NA    | 8.24       | NA     | NE    |     |      |
| Templeton Emk <b>SEI</b>             | 5.67  | 5.13       | -9.5   | 10.5  |     |      |
| WstAsstEmergDvnt <b>ME</b>           | NA    | 9.12       | NA     | 10.4  |     |      |
| National Muni Bond Funds             |       |            |        |       |     |      |
| AllBerNatMuniBnd <b>AFB</b>          | 11.61 | 10.16      | -12.5  | 4.0   |     |      |
| BlackRock 2037 Muni Trn <b>BMF</b>   | 24.92 | 23.79      | -4.5   | 4.4   |     |      |
| BlackRck Inv 2030 Trn <b>BKN</b>     | 11.85 | 10.89      | -8.9   | 5.9   |     |      |
| BlackRock Muni QM Tr <b>BTT</b>      | 23.49 | 20.83      | -11.3  | 2.6   |     |      |
| BlackRock Muni <b>BFL</b>            | 10.60 | 9.95       | -9.9   | 6.0   |     |      |
| BlackRock Muni II <b>BFK</b>         | 11.05 | 9.84       | -11.0  | 6.1   |     |      |
| BlackRck Muni Qly <b>BYM</b>         | 11.14 | 10.42      | -9.7   | 6.0   |     |      |
| BR MuniAssets Fd <b>BMF</b>          | NA    | 10.96      | 10.04  | -8.4  |     |      |
| BRMH Qly 2 <b>MUE</b>                | 10.59 | 9.95       | -5.7   | 5.7   |     |      |
| BR MuniHoldings <b>MHD</b>           | 12.36 | 11.06      | -10.5  | 5.7   |     |      |
| BR MuniVest Fd <b>MVF</b>            | 7.29  | 6.73       | -7.7   | 5.6   |     |      |
| BR MuniVest 2 <b>MYT</b>             | 11.29 | 10.10      | -10.5  | 5.9   |     |      |
| BR MuniYield Fd <b>MYD</b>           | 11.01 | 9.92       | -9.9   | 6.0   |     |      |
| BR MuniYield Qly <b>MYQ</b>          | 11.99 | 10.92      | -8.9   | 5.9   |     |      |
| BR MuniYld Qly 3 <b>MYT</b>          | 10.57 | 10.40      | -9.1   | 6.0   |     |      |
| BR MuniYld Qly 2 <b>MYT</b>          | 11.46 | 10.43      | -9.0   | 6.0   |     |      |
| BNY Mellon Muni Blnfa <b>BNM</b>     | NA    | 9.91       | NA     | 3.4   |     |      |
| BNY Mellon Str Muni Bond <b>DSM</b>  | NA    | 5.51       | NA     | 3.7   |     |      |
| BNY Mellon Str Muni <b>LEO</b>       | NA    | 5.81       | NA     | 3.7   |     |      |
| DWS Muni Inc <b>KTF</b>              | 9.26  | 8.96       | -3.2   | 7.8   |     |      |
| EVN MuniBd <b>EMN</b>                | NA    | 9.54       | NA     | 5.9   |     |      |
| EVN MuniInc <b>EVT</b>               | NA    | 9.94       | NA     | 5.8   |     |      |
| EVN NatMuni <b>EVN</b>               | NA    | 16.07      | NA     | 4.8   |     |      |
| InvAdvMnltnTr <b>OKI</b>             | 8.90  | 8.52       | -7.6   | 7.3   |     |      |
| Invesco Muni <b>VIA</b>              | 6.02  | 5.72       | -5.0   | 5.8   |     |      |
| InvescoMuniTr <b>VMO</b>             | 9.94  | 9.26       | -8.6</ |       |     |      |

## Insider-Trading Spotlight

Trading by 'insiders' of a corporation, such as a company's CEO, vice president or director, potentially conveys new information about the prospects of a company. Insiders are required to report large trades to the SEC within two business days. Here's a look at the biggest individual trades by insiders, based on data received by LSEG Data and Analytics on April 18, and year-to-date stock performance of the company

**KEY:** B: beneficial owner of more than 10% of a security class   CB: chairman   CE: chief executive officer   CFO: chief financial officer   CH: chief operating officer   D: director   DO: director and beneficial owner   GC: general counsel   H: officer, director and beneficial owner   I: indirect transaction filed through a trust, insider spouse, minor child or other   O: officer   OD: officer and director   P: president   UT: unknown   VP: vice president   Excludes pure options transactions

## Biggest weekly individual trades

Based on reports filed with regulators this past week

| Date(s)    | Company                                | Symbol | Insider        | Title | No. of shrs in trans (000s) | Price range (\$ in transaction) | \$ Value (000s) | Close (\$) | Ytd (%) |
|------------|----------------------------------------|--------|----------------|-------|-----------------------------|---------------------------------|-----------------|------------|---------|
| Buyers     |                                        |        |                |       |                             |                                 |                 |            |         |
| Apr. 8-9   | Sindair                                | SBGI   | D. Smith       | H     | 185                         | 12.91-13.66                     | 2,481           | 14.14      | -12.4   |
| Apr. 15    | Goldman Sachs Group                    | GS     | J. Hess        | D     | 4                           | 507.96-517.45                   | 1,998           | 509.49     | -11.0   |
| Apr. 10-11 | Atlanta Braves Holdings                | BATRA  | J. Malone      | B     | 46                          | 41.80-41.99                     | 1,942           | 42.56      | 4.3     |
| Apr. 14-15 |                                        |        | J. Malone      | B     | 5                           | 41.99-42.00                     | 202             |            |         |
| Apr. 14-15 | Dollar Tree                            | DLTR   | S. Glendinning | CFO   | 17                          | 72.50-72.83                     | 1,237           | 79.14      | 5.6     |
| Apr. 11    | Asana                                  | ASAN   | D. Moskowitz   | CEO   | 75                          | 15.58                           | 1,176           | 14.84      | -26.8   |
| Apr. 15    |                                        |        | D. Moskowitz   | CEO   | 50                          | 15.84                           | 792             |            |         |
| Apr. 15    | Calavo Growers                         | CVGW   | L. Cole        | CEO   | 41                          | 24.57                           | 1,002           | 26.17      | 2.6     |
| Apr. 14    |                                        |        | L. Cole        | CEO   | 38                          | 24.60                           | 939             |            |         |
| Apr. 11    |                                        |        | L. Cole        | CEO   | 21                          | 23.73                           | 500             |            |         |
| Apr. 14    | Clear Channel Outdoor Holdings         | CCO    | A. Moreno      | B     | 500                         | 1.01                            | 505             | 0.96       | -30.1   |
| Apr. 10-11 |                                        |        | A. Moreno      | B     | 386                         | 1.01                            | 391             |            |         |
| Apr. 8     | NexPoint Diversified Real Estate Trust | NXDT   | J. Dondero     | PI    | 100                         | 4.08                            | 408             | 3.75       | -38.5   |
| Apr. 10-14 | Golden Matrix Group                    | GMGI   | A. Milovanovic | B     | 151                         | 1.90-1.94                       | 293             | 2.06       | 4.0     |
| Apr. 15    |                                        |        | A. Milovanovic | B     | 146                         | 1.95                            | 285             |            |         |
| Apr. 16    | FB Financial                           | FBK    | J. Cross       | D     | 7                           | 40.49                           | 292             | 40.61      | -21.2   |
| Apr. 10    | Rocket Pharmaceuticals                 | RCKT   | G. Shah        | CEO   | 20                          | 5.08                            | 102             | 6.33       | -49.6   |
| Apr. 8     | Nutex Health                           | NUTX   | J. Bates       | CFO   | 1                           | 81.81                           | 82              | 149.00     | 370.2   |

## Sellers

|            |                         |      |                 |     |       |               |         |        |       |
|------------|-------------------------|------|-----------------|-----|-------|---------------|---------|--------|-------|
| Apr. 17    | Cal-Maine Foods         | CALM | A. Baker        | CB  | 1288* | 90.60         | 116,658 | 91.49  | -11.1 |
| Apr. 11    | Palo Alto Networks      | PANW | N. Zuk          | CT  | 650*  | 162.12-168.56 | 107,490 | 167.69 | -7.8  |
| Apr. 14    | JPMorgan Chase          | JPM  | J. Dimon        | CEO | 134   | 235.68        | 134,96  | 231.96 | -3.2  |
| Apr. 14-16 | C3.ai                   | AI   | T. Siebel       | CEO | 626*  | 19.47-20.81   | 21,579  | 19.35  | -43.8 |
| Apr. 16    | Robinhood Markets       | HOOD | B. Bhatt        | DI  | 223   | 41.24         | 9,200   | 41.18  | 10.5  |
| Apr. 14-15 | MicroStrategy           | MSTR | W. Shao         | GC  | 20    | 314.16-315.00 | 6,292   | 317.20 | 9.5   |
| Apr. 11-14 | Copart                  | CPRT | S. Cohan        | D   | 100   | 60.01         | 6,001   | 59.68  | 4.0   |
| Apr. 14    | MP Materials            | MP   | J. Litinsky     | CEO | 199   | 27.08         | 5,394   | 26.35  | 68.9  |
| Apr. 15-17 | ASP Isotopes            | ASPI | P. Mann         | CEO | 950   | 5.45-5.75     | 5,326   | 5.84   | 28.9  |
| Apr. 14    | Bank of New York Mellon | BK   | C. Keating      | O   | 65    | 77.31         | 5,062   | 76.34  | -0.6  |
| Apr. 15    | GitLab                  | GTLB | S. Sijbrandij   | HI  | 109   | 42.87         | 4,656   | 42.90  | -23.9 |
| Apr. 15    | HubSpot                 | HUBS | B. Halligan     | OD  | 9     | 538.60        | 4,578   | 539.69 | -22.5 |
| Apr. 14    | Alignment Healthcare    | ALHC | D. Maroney      | P   | 230   | 19.10         | 4,392   | 18.24  | 62.1  |
| Apr. 14    | Zoom Communications     | ZM   | E. Yuan         | CEO | 41    | 71.96-72.76   | 2,968   | 71.49  | -12.4 |
| Apr. 14    | Coinbase Global         | COIN | B. Armstrong    | CEO | 15    | 178.24-179.96 | 2,682   | 175.03 | -29.5 |
| Apr. 15    | Duolingo                | DUOL | A. Von          | CEO | 8     | 324.97-330.90 | 2,619   | 326.50 | 0.7   |
| Apr. 16    | Salesforce              | CRM  | G. Harris       | OD  | 10    | 253.31        | 2,468   | 247.26 | -26.0 |
| Apr. 16    |                         |      | S. Tallapragada | O   | 10    | 253.31        | 2,468   |        |       |

\* Half the transactions were indirect \*\*Two day transaction  
p - Pink Sheets

## Insider Buying and selling by sector

Based on actual transaction dates in reports received this past week

| Sector                | Buying    | Selling   | Sector           | Buying    | Selling    |
|-----------------------|-----------|-----------|------------------|-----------|------------|
| Basic Industries      | 0         | 5,393,794 | Finance          | 866,603   | 58,020,013 |
| Capital Goods         | 0         | 4,284,081 | Health care      | 260,314   | 24,028,839 |
| Consumer durables     | 0         | 46,548    | Public Utilities | 12,480    | 0          |
| Consumer non-durables | 2,497,137 | 1,352,569 | Technology       | 2,594,993 | 67,701,352 |
| Consumer services     | 7,946,523 | 9,434,605 |                  |           |            |
| Energy                | 23,258    | 112,377   |                  |           |            |

Sources: LSEG Data and Analytics; Dow Jones Market Data

| Prem12 Mo                |      |       |       |       | 52 wk                 |                                   |                        |       |                           | Prem12 Mo                 |                         |       |      |      |      |
|--------------------------|------|-------|-------|-------|-----------------------|-----------------------------------|------------------------|-------|---------------------------|---------------------------|-------------------------|-------|------|------|------|
| Fund (SYM)               | NAV  | Close | Disc  | Yld   | Fund (SYM)            | NAV                               | Close                  | Disc  | Ret                       | Fund (SYM)                | NAV                     | Close | Disc | Yld  |      |
| Nuveen AMT-Fr Mtz Val    | NEW  | 14.58 | 13.43 | -6.9  | 7.9                   | Verusd Capital Real Asst          | 28.44                  | NA    | 6.1                       | AFA Private CreditFndr    | NA                      | NA    | NA   | 12.0 |      |
| Nuveen AMT-Fr Qlty Mun   | NEA  | 11.44 | 10.70 | -5.5  | 7.4                   | Wildermuth                        | 3.37                   | NA    | \$4.0                     | AFA Private CreditInst    | NA                      | NA    | NA   | 12.1 |      |
| Nuveen AMT-Fr Mtz Cl     | NVG  | 12.49 | 11.60 | -7.1  | 7.3                   | WILSHIRE PRIV ASSTJNST            | 12.68                  | NA    | NA                        | Alternative Credit IncA   | NA                      | NA    | NA   | 9.3  |      |
| Nuveen Dnm Crtf Opps     | NMDO | 10.28 | 9.57  | -7.2  | 7.2                   | ALCO & Preferred Stock Funds      |                        |       |                           | Alternative Credit IncC   | NA                      | NA    | NA   | 8.3  |      |
| Nuveen Dnm Crtf Opps     | NMDO | 10.86 | 9.94  | -8.5  | 7.3                   | ALPHA ALT ASSTA                   | 6.32                   | NA    | 5.5                       | Alternative Credit Incl   | NA                      | NA    | NA   | 9.6  |      |
| Nuveen Dnm Crtf Income   | NZF  | 12.46 | 11.60 | -6.9  | 7.4                   | Alphas Alternative Assets         | 6.39                   | NA    | 5.8                       | Alternative Credit IncL   | NA                      | NA    | NA   | 9.0  |      |
| Nuveen MunHlnCIncl       | NM   | 10.32 | 10.35 | -6.3  | 6.8                   | Galma US/Eqty and ID              | OPZ                    | 16.49 | 15.13                     | -8.2                      | Alternative Credit IncW | NA    | NA   | NA   | 9.3  |
| Nuveen MunVal            | NVU  | 8.94  | 8.55  | -4.4  | 4.2                   | CARLYLE ALPINV PVT MKTW           | NA                     | NA    | NS                        | AMGPantheon Crtf SolnW    | NA                      | NA    | NA   | NS   |      |
| Nuveen Quality Multi     | NM   | 11.88 | 11.15 | -6.1  | 7.4                   | LORD ABBETT CORP OPPS A           | 8.87                   | NA    | 4.6                       | AMGPantheon Crtf SolnM    | NA                      | NA    | NA   | NS   |      |
| NySelF TFX               | NXP  | 13.37 | 13.77 | -1.4  | 4.2                   | LORD ABBETT CORP OPPS I           | 8.87                   | NA    | 5.4                       | AMGPantheon Crtf SolnS    | NA                      | NA    | NA   | NS   |      |
| NyLlyMacKay Dtm Opps     | MMO  | 15.78 | 14.27 | -9.6  | 4.7                   | ONEASCENT CAP OPPTYS              | NA                     | NA    | NS                        | Angel Oak Str CrdtInst    | 21.04                   | NA    | NA   | 9.0  |      |
| PIMCO MunInc             | PML  | 8.47  | 8.04  | -5.1  | 5.7                   | Variant Altrnty IncInstnt         | 26.32                  | NA    | 4.0                       | Angel Oak Str CrdtInst    | 21.02                   | NA    | NA   | 9.0  |      |
| PIMCO MunIncII           | PMF  | 7.92  | 7.52  | -5.1  | 5.8                   | Variant ImpactInstnt1             | 27.81                  | NA    | 5.7                       | BR Credit StratA          | 8.25                    | NA    | NA   | 8.7  |      |
| Pimco MunIII             | PMX  | 7.28  | 6.71  | -7.8  | 5.5                   | Verusd Capital InfraInc           | 10.16                  | NA    | 9.7                       | BR Credit StratInst       | 8.22                    | NA    | NA   | 8.7  |      |
| PioneerHlncAdV           | MAV  | 8.58  | 7.86  | -8.4  | 4.6                   | Convertible Sec's Funds           |                        |       |                           | BR Credit StratJ          | 8.24                    | NA    | NA   | NS   |      |
| PionHlnCInclOpps         | MU   | 12.23 | 11.29 | -8.1  | 5.0                   | Galmos Dnm Conv and Inc           | CD                     | 16.57 | 20.37                     | +22.9                     | BR Credit StratU        | 8.23  | NA   | NA   | 8.7  |
| PionHlnCInclM            | MHI  | 9.47  | 8.70  | -8.7  | 4.5                   | World Equity Funds                |                        |       |                           | BR Credit StratU          | 8.25                    | NA    | NA   | 8.7  |      |
| Putnam MgdI Inc          | PMIM | 6.39  | 5.86  | -8.3  | 4.6                   | BEACON PONTIC MTLT-ALTO           | NA                     | NA    | NS                        | CION Ares Dvsfd CrdtA     | NA                      | NA    | NA   | 8.8  |      |
| Putnam MgdI Opp          | PMO  | 10.77 | 9.70  | -9.9  | 4.1                   | CalamosGbl TotRet                 | CGO                    | 10.26 | 9.82                      | -4.3                      | CION Ares Dvsfd CrdtC   | NA    | NA   | NA   | 8.0  |
| RiverNorth Fia Mnt IncII | RFMZ | 14.11 | 12.38 | -12.3 | 7.7                   | Cantor Fitzgerald SIA             | 12.17                  | NA    | 15.3                      | CION Ares Dvsfd CrdtL     | NA                      | NA    | NA   | 9.0  |      |
| RiverNorth Mgd Dur Mun   | RMM  | 14.82 | 13.24 | -10.7 | 7.6                   | Cantor Fitzgerald SCL             | 11.97                  | NA    | 14.5                      | CION Ares Dvsfd CrdtL     | NA                      | NA    | NA   | 9.5  |      |
| Western Asset Mgd Mun    | MNI  | NA    | 9.71  | NA    | 6.3                   | Cantor Fitzgerald SII             | 12.19                  | NA    | 15.4                      | CION Ares Dvsfd CrdtU     | NA                      | NA    | NA   | 8.3  |      |
| Single State Mun Bond    |      |       |       |       |                       |                                   | CANTOR Fitzgerald INSX | 12.37 | NA                        | NS                        | CION Ares Dvsfd CrdW    | NA    | NA   | NA   | 8.5  |
| BlackRock CA Mun         | BFZ  | 11.70 | 10.45 | -10.7 | 6.3                   | Sweater Cashmere                  | 24.10                  | NA    | 9.7                       | CNR Select Strategies     | 15.49                   | NA    | NA   | 0.0  |      |
| BR MH CA Qly Fd Inc      | MUC  | 14.17 | 10.28 | -10.4 | 5.7                   | Primark Pwr Eq Inv L              | 13.83                  | NA    | 8.0                       | Diamond Hill Sec Crl      | 10.28                   | NA    | NA   | NS   |      |
| BR MH NY Qly             | MUJ  | 12.03 | 10.81 | -10.1 | 5.5                   | Thornburg Inc Bldr Opps           | TBL                    | 19.06 | 17.50                     | -8.2                      | Diamond Hill Sec Crl    | 10.27 | NA   | NA   | NS   |
| BR MH NY Qly             | MHN  | 10.85 | 9.71  | -10.5 | 5.6                   | Verusd Capital Real Asst          | 28.44                  | NA    | 5.89                      | First Eagle Crdt Opps A   | 22.46                   | NA    | NA   | 9.99 |      |
| BR MunYld NY Qly         | MIY  | 11.99 | 11.11 | -7.3  | 5.3                   | Prem12 Mo                         |                        |       |                           | First Eagle Crdt Opps A2  | 22.44                   | NA    | NA   | 9.4  |      |
| BR MunYld NY Qly         | MYN  | 10.53 | 9.58  | -9.0  | 5.7                   | Fund (SYM)                        | NAV                    | Close | Disc                      | Yld                       | First Eagle Crdt Opps I | 22.32 | NA   | NA   | 10.3 |
| BR MunYld PA Qly         | MYA  | 11.97 | 10.74 | -10.3 | 6.7                   | U.S. Mortgage Bond Funds          |                        |       |                           | First Trust Private Car   | NA                      | NA    | NA   | 10.6 |      |
| BlackRock Mun Bond       | BNY  | 10.78 | 9.78  | -9.3  | 5.5                   | ATCA US Treasury                  | NA                     | NA    | 1.1                       | First Trust Private CrI   | NA                      | NA    | NA   | 10.4 |      |
| EVCA CA Municipl         | EVN  | NA    | 8.78  | NA    | 5.2                   | ARALYST/PERIN STRAT IN            | 10.11                  | NA    | NS                        | Franklin BSP Pvt Crd      | NA                      | NA    | NA   | 8.7  |      |
| Eaton Vene NY Mun Bd     | ENX  | NA    | 9.24  | NA    | 5.2                   | Ellington Inc OppA                | NA                     | NA    | 10.6                      | Franklin BSP Pvt CrdV     | NA                      | NA    | NA   | 9.2  |      |
| InvCaValMunCrtF          | VCV  | 10.20 | 9.80  | -3.9  | 6.7                   | Ellington Inc OppM                | NA                     | NA    | 11.0                      | FS Credit IncomeA         | 12.03                   | NA    | NA   | 8.7  |      |
| InvPAValMunCrtF          | VPV  | 10.79 | 9.70  | -10.1 | 7.1                   | Local Participation Funds         |                        |       |                           | FS Credit IncomeL         | 12.07                   | NA    | NA   | 8.9  |      |
| InvTrlnvGrMun            | VTN  | 10.88 | 9.95  | -9.7  | 7.3                   | 1WVS Credit IncomeA2              | NA                     | NA    | 8.5                       | FS Credit IncomeT         | 12.06                   | NA    | NA   | 8.4  |      |
| Nuveen CAIMT-Fr QalM     | NKX  | 11.84 | 11.91 | -1.2  | 6.8                   | 1WVS Credit IncomeInst            | NA                     | NA    | 8.3                       | FS Credit IncomeU         | 12.01                   | NA    | NA   | 8.2  |      |
| Nuveen CA Val            | NCA  | 8.93  | 8.34  | -6.6  | 4.0                   | Apollo Diversified CdcA           | NA                     | NA    | 8.2                       | FS Credit IncomeU-2       | 12.12                   | NA    | NA   | 8.1  |      |
| Nuveen CAQly Multi       | NAC  | 11.38 | 10.72 | -5.8  | 7.4                   | Apollo Diversified CdcC           | NA                     | NA    | 7.5                       | JHANCOC QS ASSETSA        | 18.98                   | NA    | NA   | NS   |      |
| NuVNY Qal Multi Inc      | NKJ  | 12.32 | 11.36 | -7.8  | 7.4                   | Apollo Diversified CdcF           | NA                     | NA    | 10.2                      | JHANCOC QS ASSETSB        | 18.99                   | NA    | NA   | NS   |      |
| Nuveen AMT-Fr Mtz Val    | NEW  | 14.58 | 13.43 | -6.9  | 7.9                   | Apollo Diversified CdcL           | NA                     | NA    | 8.0                       | JHANCOC QS ASSETSC        | 18.98                   | NA    | NA   | NS   |      |
| Nuveen NY Val            | NNY  | 8.63  | 8.01  | -7.2  | 4.1                   | Apollo Diversified CdcM           | NA                     | NA    | 7.9                       | KKR CREDIT OPTPTYD        | NA                      | NA    | NA   | 8.6  |      |
| Nuveen NY Qal Multi Inc  | NAI  | 11.26 | 11.15 | -1.0  | 7.1                   | Axononic Alternative I/A          | NA                     | NA    | 8.9                       | KKR CREDIT OPTPTYT        | NA                      | NA    | NA   | 8.8  |      |
| Nuveen PA Qal Multi Inc  | NPO  | 12.01 | 10.12 | -8.2  | 7.7                   | Axon Alternative I Ins            | NA                     | NA    | 8.9                       | KKR CREDIT OPTPTYU        | NA                      | NA    | NA   | 8.1  |      |
| Nuveen VA Qal Multi Inc  | NPV  | 11.38 | 10.94 | -2.1  | 6.7                   | Bluerock Hl Inst CrdF             | NA                     | NA    | 11.6                      | KKR CREDIT OPTPTYV        | NA                      | NA    | NA   | 8.1  |      |
| PIMCO CA PCQ             | 9.51 | 8.34  | -12.3 | 4.8   | Bluerock Hl Inst CrdF | NA                                | NA                     | 11.8  | NIAGARA INCOME OPPORTUNIT | NA                        | NA                      | NA    | 11.9 |      |      |
| PIMCOAMMUNICI            | PMF  | 6.06  | 5.35  | -11.7 | 4.5                   | Bluerock Hl Inst CrdF             | NA                     | NA    | 11.6                      | Palmer Square Opp Inc     | 17.64                   | NA    | NA   | 7.9  |      |
| Pimco CA MunII           | PZC  | 7.01  | 6.22  | -11.3 | 5.4                   | Bluerock Hl Inst CrdI             | NA                     | NA    | 11.6                      | TCW PRIVATE ASSETINCA     | 10.00                   | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | BNYM Alcntr Gbl MS CrdF           | 85.41                  | NA    | NA                        | TCW PRIVATE ASSETINCI     | 10.00                   | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | CliffwaterCIFdI                   | NA                     | NA    | 8.2                       | World Income Funds        |                         |       |      |      |      |
|                          |      |       |       |       |                       | CliffwaterCIFdE                   | NA                     | NA    | 8.2                       | BlueBay Destra I/E A      | 21.64                   | NA    | NA   | 10.7 |      |
|                          |      |       |       |       |                       | CNR Strategic Credit              | 6.21                   | NA    | 23.8                      | BlueBay Destra I/E E      | 21.66                   | NA    | NA   | 10.9 |      |
|                          |      |       |       |       |                       | Floating-Rate OppsA               | 9.82                   | NA    | 9.8                       | BlueBay Destra I/E L      | 21.62                   | NA    | NA   | 10.4 |      |
|                          |      |       |       |       |                       | Floating-Rate OppsJ               | 9.81                   | NA    | 10.1                      | BlueBay Destra I/E T      | 21.50                   | NA    | NA   | 10.2 |      |
|                          |      |       |       |       |                       | Floating-Rate OppsU               | 9.82                   | NA    | 9.3                       | BOOSTER INCOME OPTPTS     | 9.43                    | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | Equalize Community Tend           | 9.45                   | NA    | 3.3                       | Calamos-Aksia AIC A       | 10.61                   | NA    | NA   | 9.8  |      |
|                          |      |       |       |       |                       | FedProjGrT FinanceTendr           | 9.92                   | NA    | 7.4                       | Calamos-Aksia AIC C       | 10.58                   | NA    | NA   | 9.0  |      |
|                          |      |       |       |       |                       | Flat Rock Core Income             | NA                     | NA    | 10.2                      | Calamos-Aksia AIC H       | 10.61                   | NA    | NA   | 10.1 |      |
|                          |      |       |       |       |                       | Flat Rock Enhanced Inc            | NA                     | NA    | 11.2                      | Calamos-Aksia AIC M       | 10.61                   | NA    | NA   | 9.2  |      |
|                          |      |       |       |       |                       | InvDvYCrpOppA                     | 10.64                  | NA    | 11.2                      | Carlyle Tact Pvt CredA    | NA                      | NA    | NA   | 9.3  |      |
|                          |      |       |       |       |                       | InvDvYCrpOppR6                    | 10.65                  | NA    | 11.7                      | Carlyle Tact Pvt CredI    | NA                      | NA    | NA   | 9.8  |      |
|                          |      |       |       |       |                       | InvDvYCrpOppY                     | 10.66                  | NA    | 11.5                      | Carlyle Tact Pvt CredL    | NA                      | NA    | NA   | 9.2  |      |
|                          |      |       |       |       |                       | Invesco Sr Loan A                 | 5.49                   | NA    | 9.3                       | Carlyle Tact Pvt CredM    | NA                      | NA    | NA   | 9.1  |      |
|                          |      |       |       |       |                       | Invesco Sr Loan C                 | 5.50                   | NA    | 8.5                       | Carlyle Tact Pvt CredN    | NA                      | NA    | NA   | 9.8  |      |
|                          |      |       |       |       |                       | Invesco Sr Loan IB                | 5.49                   | NA    | 9.5                       | Carlyle Tact Pvt CredU    | NA                      | NA    | NA   | 9.1  |      |
|                          |      |       |       |       |                       | Invesco Sr Loan I                 | 5.49                   | NA    | 9.4                       | Carlyle Tact Pvt CredY    | NA                      | NA    | NA   | 9.4  |      |
|                          |      |       |       |       |                       | Invesco Sr Loan Y                 | 5.49                   | NA    | 9.5                       | Fid MS CrA                | NA                      | NA    | NA   | 6.6  |      |
|                          |      |       |       |       |                       | Monacih Credit IncomeL            | 10.01                  | NA    | 11.0                      | Fid MS CrC                | NA                      | NA    | NA   | 5.8  |      |
|                          |      |       |       |       |                       | Munira Alt Incl                   | 9.88                   | NA    | 10.1                      | Fid MS CrJ                | NA                      | NA    | NA   | 6.8  |      |
|                          |      |       |       |       |                       | Nuveen Enh CLO IncA1              | 18.87                  | NA    | NS                        | Fid MS CrL                | NA                      | NA    | NA   | 6.3  |      |
|                          |      |       |       |       |                       | Nuveen Enh CLO IncA2              | 18.88                  | NA    | NS                        | Jackson Credit Oppsl      | 10.14                   | NA    | NA   | 8.1  |      |
|                          |      |       |       |       |                       | Nuveen Enh CLO IncA3              | 18.88                  | NA    | NS                        | Lord Abbett Cred Opps Fd  | 8.36                    | NA    | NA   | 8.8  |      |
|                          |      |       |       |       |                       | Octagon XAI CLO IncA              | NA                     | NA    | NS                        | Lord Abbett Cred Opps G   | 8.36                    | NA    | NA   | 9.6  |      |
|                          |      |       |       |       |                       | Octagon XAI CLO IncB              | NA                     | NA    | NS                        | Lord Abbett Crd OppU      | 8.37                    | NA    | NA   | 8.8  |      |
|                          |      |       |       |       |                       | OFCS Credit Company OCCI          | NA                     | 6.44  | NA                        | Oaktree Dvsfd IncomeL     | 8.94                    | NA    | NA   | 9.3  |      |
|                          |      |       |       |       |                       | PC PCAAM AIT                      | 10.23                  | NA    | NS                        | PIMCO Flexible EM Inst    | NA                      | NA    | NA   | 8.5  |      |
|                          |      |       |       |       |                       | PRINCIPAL PRVT CR I/A             | 10.19                  | NA    | NS                        | National Multi Bond Funds |                         |       |      |      |      |
|                          |      |       |       |       |                       | PRINCIPAL PRVT CR I/MS            | 10.21                  | NA    | NS                        | BLACKROCK MUN CAPA        | 12.77                   | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | PRINCIPAL PRVT CR I/US            | 10.23                  | NA    | NS                        | BLACKROCK MUN CAPJ        | 12.77                   | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | SEI Alternative Income-F          | 10.18                  | NA    | 7.3                       | Ecofin Tax-Exempt Pn CrJ  | 7.64                    | NA    | NA   | 10.0 |      |
|                          |      |       |       |       |                       | SEI Alternative Income-F          | 10.18                  | NA    | 7.5                       | Lord Cap Pt Mun Cred Inc  | 8.48                    | NA    | NA   | 5.3  |      |
|                          |      |       |       |       |                       | Voya Enhanced Sec IncA            | NA                     | NA    | NS                        | Lord Abbett Mun OppsA     | 9.87                    | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | Voya Enhanced Sec IncC            | NA                     | NA    | NS                        | Lord Abbett Mun OppsB     | 9.87                    | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | Voya Enhanced Sec Incl            | NA                     | NA    | NS                        | Nuveen En HY Mun BdtA1    | 7.35                    | NA    | NA   | 5.0  |      |
|                          |      |       |       |       |                       | Yieldstreet Alt Inc               | NA                     | NA    | 6.6                       | Nuveen En HY Mun BdtJ     | 7.36                    | NA    | NA   | 5.3  |      |
|                          |      |       |       |       |                       | AB CarVal Credit OppAdv           | 9.90                   | NA    | 7.0                       | Nuveen En HY Mun BdtI     | 7.35                    | NA    | NA   | 5.8  |      |
|                          |      |       |       |       |                       | AB CarVal Credit OppA             | 9.89                   | NA    | NS                        | NyLI MacKay Mu IOA1       | 9.72                    | NA    | NA   | 3.4  |      |
|                          |      |       |       |       |                       | AB CarVal Credit OppC             | 9.88                   | NA    | NS                        | NyLI MacKay Mu IOA2       | 9.72                    | NA    | NA   | 3.4  |      |
|                          |      |       |       |       |                       | AB CarVal Credit OppU             | 9.88                   | NA    | NS                        | NyLI MacKay Mu IOA3       | 9.71                    | NA    | NA   | 3.2  |      |
|                          |      |       |       |       |                       | Optmptnrc Crd IntrvI              | 11.52                  | NA    | 8.7                       | NyLI MacKay Mu IOA7       | 9.72                    | NA    | NA   | 3.9  |      |
|                          |      |       |       |       |                       | PIMCO Flexible Cr I-A1            | NA                     | NA    | 10.1                      | PIMCO Flex Mun IncA3      | 9.87                    | NA    | NA   | 3.5  |      |
|                          |      |       |       |       |                       | PIMCO Flexible Cr I-A2            | NA                     | NA    | 10.1                      | PIMCO Flex Mun IncA4      | 9.87                    | NA    | NA   | 3.7  |      |
|                          |      |       |       |       |                       | PIMCO Flexible Cr I-A3            | NA                     | NA    | 9.8                       | PIMCO Flex Mun IncInst    | 9.87                    | NA    | NA   | 4.2  |      |
|                          |      |       |       |       |                       | PIMCO Flexible Cr I-A4            | NA                     | NA    | 9.8                       | ROCKEFELLER MUN OPPS      | 19.38                   | NA    | NA   | NS   |      |
|                          |      |       |       |       |                       | PIMCO Flexible Cr Inst            | NA                     | NA    | 10.6                      | Single State Mun Bond     |                         |       |      |      |      |
|                          |      |       |       |       |                       | Polen Credit Income Inst          | 9.25                   | NA    | 9.6                       | PIMCO CA FMI-A1           | 9.85                    | NA    | NA   | 2.9  |      |
|                          |      |       |       |       |                       | Other Domestic Taxable Bond Funds |                        |       |                           | PIMCO CA FMI-AInst        | 9.85                    | NA    | NA   | 3.4  |      |

**Borrowing Benchmarks** | [wsj.com/market-data/bonds/benchmarks](https://www.wsj.com/market-data/bonds/benchmarks)

## Money Rates

April 18, 2025

Key annual interest rates paid to borrow or lend money in U.S. and international markets. Rates below are a guide to general levels but don't always represent actual transactions.

| Inflation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |           |       |       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------|-------|-------|
| March index                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Chg From (%) |           |       |       |
| level                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Feb. '25     | March '24 |       |       |
| U.S. consumer price index                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |           |       |       |
| All items                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 319,799      | 0.22      | 2.4   |       |
| Core                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 325,933      | 0.21      | 2.8   |       |
| International rates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |           |       |       |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Week ago     | -52-Week- |       |       |
| Latest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              | High      | Low   |       |
| Prime rates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |           |       |       |
| U.S.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 7.50         | 7.50      | 8.50  | 7.50  |
| Canada                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4.95         | 4.95      | 7.20  | 4.95  |
| Japan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1.875        | 1.875     | 1.875 | 1.475 |
| Policy Rates                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |           |       |       |
| Euro zone                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2.65         | 2.65      | 4.50  | 2.65  |
| Switzerland                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0.75         | 0.75      | 2.00  | 0.75  |
| Notes on data:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |           |       |       |
| U.S. <b>prime rate</b> is the base rate on corporate loans posted by at least 70% of the 10 largest U.S. banks, and is effective December 19, 2024. <b>Other prime rates</b> aren't directly comparable; lending practices vary widely by location. <b>Discount rate</b> is effective December 19, 2024. <b>Secured Overnight Financing Rate</b> is as of April 17, 2025. <b>DTCC GCF Repo Index</b> is Depository Trust & Clearing Corp.'s weighted average for overnight trades in applicable CUSIPs. Value traded is in billions of U.S. dollars. <b>Federal-funds rates</b> are Tullett Prebon rates as of 5:30 p.m. ET. |              |           |       |       |
| Sources: Federal Reserve; Bureau of Labor Statistics; DTCC; FactSet; Tullett Prebon Information Ltd.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |           |       |       |

|                       | Latest | Week ago | -52-Week-<br>High | Low    |
|-----------------------|--------|----------|-------------------|--------|
| Britain               | 4.50   | 4.50     | 5.25              | 4.50   |
| Australia             | 4.10   | 4.10     | 4.35              | 4.10   |
| Overnight repurchase  |        |          |                   |        |
| U.S.                  | 4.38   | 4.40     | 5.45              | 4.00   |
| U.S. government rates |        |          |                   |        |
| Discount              |        |          |                   |        |
|                       | 4.50   | 4.50     | 5.50              | 4.50   |
| Federal funds         |        |          |                   |        |
| Effective rate        | 4.3300 | 4.3300   | 5.3400            | 4.3300 |
| High                  | 4.5500 | 4.5500   | 5.6500            | 4.3700 |
| Low                   | 4.3200 | 4.3200   | 5.3300            | 4.3000 |
| Bid                   | 4.3200 | 4.3200   | 5.3300            | 4.3000 |
| Offer                 | 4.3300 | 4.3300   | 5.3600            | 4.3300 |
| Treasury bill auction |        |          |                   |        |
| 4 weeks               | 4.240  | 4.245    | 5.285             | 4.215  |
| 13 weeks              | 4.225  | 4.175    | 5.255             | 4.175  |

|          | Latest | Week ago | -52-Week-<br>high | low   |
|----------|--------|----------|-------------------|-------|
| 26 weeks | 4.060  | 4.000    | 5.170             | 4.000 |

| Other short-term rates           |        |                 |                   |       |
|----------------------------------|--------|-----------------|-------------------|-------|
|                                  | Latest | Week ago        | -52-Week-<br>high | low   |
| Call money                       |        |                 |                   |       |
|                                  | 6.25   | 6.25            | 7.25              | 6.25  |
| Commercial paper (AA financial)  |        |                 |                   |       |
| 90 days                          | n.a.   | 4.35            | 5.37              | 4.26  |
| Secured Overnight Financing Rate |        |                 |                   |       |
|                                  | 4.31   | 4.37            | 5.40              | 4.27  |
|                                  | Latest | Value<br>Traded | -52-Week-<br>High | Low   |
| DTCC GCF Repo Index              |        |                 |                   |       |
| Treasury                         | 4.384  | 62.550          | 5.471             | 4.286 |
| MBS                              | 4.417  | 89.826          | 5.526             | 4.294 |
| Weekly survey                    |        |                 |                   |       |
|                                  | Latest | Week ago        | Year ago          |       |
| Freddie Mac                      |        |                 |                   |       |
| 30-year fixed                    | 6.83   | 6.62            | 7.10              |       |
| 15-year fixed                    | 6.03   | 5.82            | 6.39              |       |

**Cash Prices** | [wsj.com/market-data/commodities](https://www.wsj.com/market-data/commodities)

Friday, April 18, 2025

These prices reflect buying and selling of a variety of actual or “physical” commodities in the marketplace—separate from the futures price on an exchange, which reflects what the commodity might be worth in future months.

| Friday                             |             | Friday                                              |          | Friday                                 |        |
|------------------------------------|-------------|-----------------------------------------------------|----------|----------------------------------------|--------|
| <b>Energy</b>                      |             | <b>n.a.</b>                                         |          | <b>n.a.</b>                            |        |
| Coal,CAPic,12500Btu1,2502-r,w      | n.a.        | Copper,Cornex spot                                  | n.a.     | Wheat,No.2 soft red,St.Louis-u         | n.a.   |
| Coal,PwdrRvrBsn,8800Btu,0.8502-r,w | n.a.        | Iron Ore, 62% Fe CFR China, n.a.                    | *\$99.9  | Wheat - Hard - CK (USDA) \$ per bu-u   | n.a.   |
|                                    |             | Steel, HRC USA, FOB Midwest Mill-s                  | *\$20.0  | Wheat,No.1 soft, white,Portld,0-R-u    | n.a.   |
| <b>Metals</b>                      |             | <b>Battery/EV metals</b>                            |          | <b>Food</b>                            |        |
| <b>Gold, per troy oz</b>           |             | BMI Lithium Hydroxide, EXW China ~99.5% v-w         | 9725     | Beef,carcass equiv, index              |        |
| Engelhard industrial               | n.a.        | BMI Lithium Carbonate, EXW China ~56.5% v-w         | 9375     | choice 1-3,600-900 lbs.-u              | 319.26 |
| Handy & Harman base                | n.a.        | BMI Cobalt sulphate, EXW China >20% v-k             | 6645     | select 1-3,600-900 lbs.-u              | 301.41 |
| Handy & Harman fabricated          | n.a.        | BMI Nickel Sulphate, EXW China >22% v-k             | 3769     | Broilers, National comp wtd. avg.-u w  | 1.3542 |
| LBMA Gold Price AM                 | *\$3224.35  | BMI Flake Graphite, FOB China, ~100 Mesh, 94.9% v-w | 435      | Butter,AA Chicago-d                    |        |
| LBMA Gold Price PM                 | *\$3305.65  |                                                     |          | Cheddar cheese,bbl,Chicago-d           | n.a.   |
| Kruggerand,wholesale-e             | n.a.        |                                                     |          | Cheddar cheese,bbl, Chicago-d          | n.a.   |
| Maple Leaf-e                       | n.a.        | <b>Fibers and Textiles</b>                          |          | Milk,Nonfat dry, Chicago lb-d          | n.a.   |
| American Eagle-e                   | n.a.        | Burlap,10-oz,40-inch NY yd-v-n                      | 0.8150   | Coffee,Brazilian,Comp-y                | 3.7937 |
| Mexican peso-e                     | n.a.        | Cotton '11' 16 std lw-mdMphs-u                      | 0.6538   | Coffee,Colombian, NY-y                 | 3.9432 |
| Austria crown-e                    | n.a.        | Cotton/AJ Index                                     | *\$77.60 | Eggs,Large white,Chicago-u,w           | 3.6050 |
| Austria phil-e                     | n.a.        | Hides,hvy native steers piece fob-u                 | n.a.     | Flour,hard winter KC-p                 | n.a.   |
|                                    |             | Wool,64s, staple, Terr del-u-w                      | n.a.     | Hams,17-20 lbs,Mid-US fob-u            | n.a.   |
|                                    |             |                                                     |          | Hogs,lowa,so. Minnesota-u              | 84.60  |
| <b>Silver, troy oz.</b>            |             | <b>Grains and Feeds</b>                             |          | Pork,bellies,12-14 lb MidUS-u          | 1.6185 |
| Engelhard industrial               | n.a.        | Bran,wheat middlings, KC-u-w                        | 85       | Pork loins,13-19 lb MidUS-u            | 0.9807 |
| Handy & Harman base                | n.a.        | Corn No. 2,yellow,Cent IL-bp-u                      | n.a.     | Steers,Text.-Okla. Choice-u            | 209.56 |
| Handy & Harman fabricated          | n.a.        | Corn gluten feed,Midwest-u-w                        | 105.4    | Steers,feeder,Okla. City-u-w           | 361.00 |
| LBMA spot price                    | *\$2,440.00 | Corn gluten meal,Midwest-u-w                        | 435.4    |                                        |        |
| (U.S.\$ equivalent)                | *\$2,310.00 | Cottonseed meal-u-w                                 | n.a.     |                                        |        |
| Coins,wholesale \$1,000 face-a     | 24695       | Hominy feed,Cent IL-u-w                             | 115      |                                        |        |
|                                    |             | Meat-bone meal,50% pr Mnpls-u,w                     | 250      | <b>Fats and Oils</b>                   |        |
| <b>Other metals</b>                |             | Oats,No.2 milling,Mpls-u                            | n.a.     | Degummed corn oil, crude wtd. avg.-u w | n.a.   |
| LBMA Platinum Price PM             | *\$959.0    | Rice, Long Grain Milled, No. 2 AR-u-w               | 34.88    | Grease,choic white,Chicago-h           | 0.5100 |
| LBMA Palladium Price PM            | *\$951.0    | Sorghum,(Milo) No.2 Gulf-u                          | n.a.     | Lard,Chicago-u                         | n.a.   |
| Platinum,Engelhard industrial      | n.a.        | Soybean Meal,Cent IL,rail ton48%-u w                | 297.20   | Soybean oil,crude,Cent IL-u-w          | 0.4698 |
| Palladium,Engelhard industrial     | n.a.        | Meats,No.1 yllw ll-bp-u                             | n.a.     | Tallow,bleach,Chicago-h                | 0.5350 |
| Aluminum, LME, \$ per metric ton   | *\$237.5    | Wheat,Spring14%-pro Mnpls-u                         | n.a.     | Tallow,edible,Chicago-u                | n.a.   |

KEY TO CODES: A=ask; B=bid; BP=country elevator bids to producers; C=corrected; D=CME; E=Manfra, Tordella & Brookes; H=American Commodities Brokerage Co; K=bi-weekly; M=monthly; N=nominal; n.a.=not quoted or not available; P=Sosland Publishing; R=S/NI Energy; S=Platts-TSI; T=Cotlook Limited; U=USDA; V=Benchmark Mineral Intelligence; W=weekly; Y=International Coffee Organization; Z=not quoted. \*Data as of 4/17

Source: Dow Jones Market Data



FINANCE & MARKETS

# Blackstone Sees Opportunity As Tariffs Roil Global Markets

Asset manager cites its \$177 billion cash pile as an advantage in its hunt to invest

By Luis Garcia

**Blackstone** sees investment opportunities in tariff-driven market turbulence, as the private-equity giant sits on a \$177 billion cash pile while increasingly raising capital from individual investors.

“Uncertainty around tariffs and their potential impact on economic growth and inflation has dramatically impacted investor sentiment,” said Blackstone Chief Executive Stephen Schwarzman. He spoke Thursday during a call with analysts to discuss the New York-based asset manager’s results during this year’s first quarter.

“Our experience through many economic and market downturns has taught us some

of the best times to deploy capital are in a risk-off world when sentiment is most negative” he said.

Schwarzman said that it is early to evaluate the “full implication of tariffs” as it will depend on trade negotiations between the U.S. and other countries.

He noted that the economy remains strong except in a few areas such as real estate, where rising construction costs are suppressing activity.

The “most important questions are how sustained will this period of uncertainty be and what are the second-order consequences, both domestically and for foreign countries,” he said.

The \$177.2 billion that Blackstone had in available capital to invest, or dry powder, at the end of March can help the firm “take advantage of the opportunities that arise,” he added.

Blackstone particularly sees opportunities to take publicly

traded companies private as their share prices fall, said President and Chief Operating Officer Jonathan Gray.

“When stock prices trade off, the receptivity from boards [of directors] to our prices may be better,” he said. “Having \$177 billion of dry powder and some real long-term conviction in the sectors we like...we are going to see this as an opportunity to put out more capital.”

As examples, he cited sectors such as digital infrastructure, energy and power, life sciences and commercial real estate, as well as markets such as India and Japan.

He added, however, that market volatility can increase the cost of deal financing and make it more difficult for private-equity firms to sell assets.

“The faster this tariff diplomacy can play out, the better it would be for the economy and markets,” Gray said.

Blackstone raised \$61.6 billion during the first quarter,

helping lift its total assets under management to \$1.17 trillion. Inflows included capital for its clean energy-focused Blackstone Energy Transition Partners IV fund, which wrapped up in February with \$5.6 billion.

The firm also recently closed its latest flagship private-equity fund with roughly \$21 billion, Gray said. Private-wealth strategies, or funds dedicated to individual investors, took in about \$11 billion during this year’s first quarter, nearly 40% more than in the year-earlier period and the highest level in three years, according to Gray.

Blackstone, which is a pioneer in private-equity firms’ increased efforts to attract individual investors, last week said that it joined with asset managers Vanguard and Wellington Management to offer individual investors access to portfolios comprising both public and private assets. Blackstone has four strategies dedicated to individ-



Blackstone’s first-quarter distributable earnings rose 11%.

ual investors, including private-equity, infrastructure, real-estate and credit funds that allowed for limited, periodic redemptions. A fifth, credit-focused strategy is in the works, Gray said.

Blackstone ended March with more than \$270 billion across its private wealth strategies, nearly one-quarter of the firm’s total assets under management, Schwarzman said. “Our vision in this area from the beginning was to provide individuals the same access to private-market [assets] that many institutions have enjoyed for decades,” he added.

Blackstone reported distributable earnings, or cash that could be returned to shareholders, of \$1.41 billion for the first quarter, or \$1.09 a share, about 11% more than the \$1.27 billion, or 98 cents a share, in the year-earlier period.

Net income dropped 27% to \$614.9 million, or 80 cents a share, compared with \$847.4 million, or \$1.11 a share, a year earlier.

Revenue decreased about 11%, to \$3.29 billion. But fee-related earnings, a metric highly watched by analysts, rose to \$1.26 billion, 8.8% higher than in last year’s first quarter.



Union Pacific reports results Thursday. Analysts expect earnings of \$2.74 a share.

## THE TICKER | MARKET EVENTS COMING THIS WEEK

### Monday

| Leading indicators |           |
|--------------------|-----------|
| Feb., prev.        | down 0.3% |
| March, exp.        | down 0.5% |
| Earnings expected  |           |
| Estimate/Year Ago  |           |
| AGNC Investment    | 0.41/0.58 |
| Comerica           | 1.16/0.98 |
| Equity LifeStyle   | 0.55/0.59 |
| Medpace Holdings   | 3.07/3.20 |
| W.R. Berkley       | 0.98/1.04 |
| Western Alliance   | 1.78/1.60 |

### Tuesday

| Earnings expected |  |
|-------------------|--|
|-------------------|--|

|                    |           |
|--------------------|-----------|
| Estimate/Year Ago  |           |
| Danaher            | 1.63/1.92 |
| GE Aerospace       | 1.27/0.82 |
| Intuitive Surgical | 1.74/1.50 |
| RTX                | 1.35/1.34 |
| Tesla              | 0.39/0.45 |
| Verizon            | 1.15/1.15 |

### Wednesday

| EIA status report                                |           |
|--------------------------------------------------|-----------|
| Previous change in stocks in millions of barrels |           |
| Crude oil                                        | up 0.5    |
| Gasoline                                         | down 2.0  |
| Distillates                                      | down 1.9  |
| Mort. bankers indexes                            |           |
| Purch., prev.                                    | down 5.0% |
| Refinan., previous                               | up 1.4%   |

| down 12.0%               |           |
|--------------------------|-----------|
| New-home sales           |           |
| Feb., previous           | 676,000   |
| March, exp.              | 680,000   |
| Earnings expected        |           |
| Estimate/Year Ago        |           |
| AT&T                     | 0.51/0.55 |
| Boston Scientific        | 0.67/0.56 |
| IBM                      | 1.42/1.68 |
| Philip Morris            | 1.61/1.50 |
| International ServiceNow | 3.83/3.41 |
| Thermo Fisher            | 5.10/5.11 |

### Thursday

| Durable-goods orders |         |
|----------------------|---------|
| Feb., previous       | up 1.0% |
| March, exp.          | up 1.4% |

| EIA report: natural gas                             |           |
|-----------------------------------------------------|-----------|
| Previous change in stocks in billions of cubic feet |           |
|                                                     | up 16     |
| Existing-home sales                                 |           |
| Feb., previous                                      | 4.26 mil. |
| March, exp.                                         | 4.10 mil. |
| Initial jobless claims survey                       |           |
| Previous                                            | 215,000   |
| Expected                                            | 220,000   |
| Freddie Mac mortgage                                |           |
| Previous weekly averages                            |           |
| 30-year fixed                                       | 6.83%     |
| 15-year fixed                                       | 6.03%     |
| Earnings expected                                   |           |
| Estimate/Year Ago                                   |           |
| Alphabet                                            | 2.01/1.89 |
| Merck                                               | 2.14/2.07 |
| PepsiCo                                             | 1.49/1.61 |
| Procter & Gamble                                    | 1.52/1.52 |
| T-Mobile US                                         | 2.47/2.00 |
| Union Pacific                                       | 2.74/2.69 |

### Friday

| U.Mich. consumer index                                         |           |
|----------------------------------------------------------------|-----------|
| April, prelim.                                                 | 50.8      |
| April, final                                                   | 50.8      |
| Short-selling reports                                          |           |
| Ratio, days of trading volume of current position, at March 31 |           |
| NYSE                                                           | 3.5       |
| Nasdaq                                                         | 2.2       |
| Earnings expected                                              |           |
| Estimate/Year Ago                                              |           |
| AbbVie                                                         | 2.37/2.31 |
| Aon                                                            | 6.01/5.66 |
| Charter Communications                                         | 8.44/7.55 |
| Colgate-Palmolive                                              | 0.86/0.86 |
| HCA Healthcare                                                 | 5.75/5.36 |
| Schlumberger                                                   | 0.73/0.75 |

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# HEARD ON THE STREET

FINANCIAL ANALYSIS & COMMENTARY

## Hospital Chains’ Medicaid Profits at Risk

Republicans looking for cost savings could hurt results at Universal Health Services and others

**Universal Health Services** has a billionaire controlling shareholder and a lucrative Medicaid cash stream. That plays into the hands of congressional Republicans eyeing large budget cuts while looking to show they aren’t just targeting the poor.

The stakes, and downside risks, are enormous for UHS investors.

Last year, 68% of UHS’s pretax income came from government outlays commonly known as supplemental Medicaid payment programs, including so-called state-directed payments. While the federal government funds these payments, state agencies direct where they go.

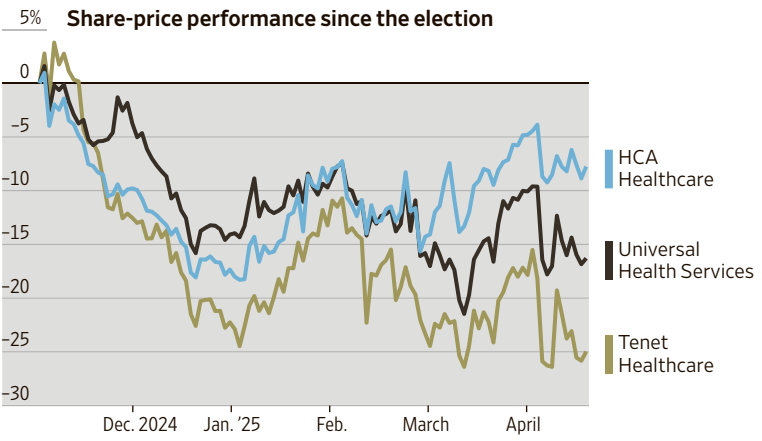
UHS netted \$1 billion of these payments last year, up from \$629 million a year earlier. Those payments contributed directly to pretax earnings because there was no incremental expense associated with them.

UHS has estimated this year’s payments would be almost as large as last year’s. Likewise, analysts’ earnings estimates for next year and beyond have remained largely the same since Election Day.

In essence, both the company and Wall Street are making a bullish call on federal spending at a time when congressional Republicans and the Trump administration are pushing austerity. While nothing is certain in Washington, especially these days, that is a gutsy bet.

Concerns over Medicaid spending have weighed on UHS and other hospital stocks. But their situation could easily get worse—depending on the political climate, far worse. The stocks currently represent a wager on political outcomes as much as they do on company fundamentals.

UHS is controlled by its billionaire founder and executive chairman, Alan Miller, and members of



\*Ebitda is earnings before interest, taxes, depreciation and amortization. Sources: FactSet (performance); KeyBanc Capital Markets estimates, the companies (medicaid)



Concerns over Medicaid spending have weighed on hospital stocks.

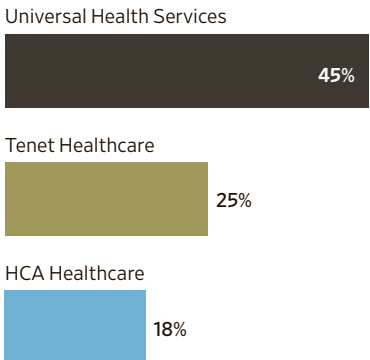
his family, including his son Marc Miller, UHS’s chief executive. Other big hospital companies that rely on supplemental payments for significant portions of their earnings include **Tenet Healthcare** and **HCA Healthcare**, although their dependence is less than UHS’s.

Even before last year’s presidential election, the supplemental Medicaid payment programs had drawn criticism across party lines for their largess and limited over-

sight. The payment arrangements have grown substantially in their usage by states and in their spending amounts.

The programs have often pushed payment rates above Medicare rates and up to commercial rates, according to an October report by Macpac, a nonpartisan policy adviser to Congress on issues affecting Medicaid and the government’s health-insurance program for children.

### Medicaid supplemental payments as a percentage of Ebitda, 2024\*



The states and healthcare providers have developed elaborate ways to maximize the payments providers receive. The states collect taxes from the providers and use those amounts to qualify for higher federal matching funds. Then the states return the taxes to the providers, along with higher Medicaid payments made possible by the higher matching funds. The “provider taxes” are taxes in name only.

Brian Blase, a former economic adviser to President Trump, called existing Medicaid financing “legalized money laundering” in a March report released by Paragon Health Institute, a conservative policy group he leads. “Congress created Medicaid to help finance health care services for the most vulnerable—not to serve as a slush fund for states and politically connected providers,” Paragon’s report said.

Whether one agrees with that characterization, the point is that the current system’s proponents face a serious challenge.

On UHS’s latest earnings call, finance chief Steve Filton said the supplemental payments “are really meant to provide for frankly,

what’s been inadequate Medicaid reimbursement for many years.” He said “the immediate impact is a significant boost to our earnings, but it is really making up, in our minds, for deficient earnings in the past.”

Recompense for past outlays is likely to hold little weight with lawmakers eager for future budget savings.

Last year, Medicaid outlays were \$618 billion, or 9% of the federal budget. The recent budget blueprint approved by the House of Representatives and Senate tasked the House Energy and Commerce Committee with finding at least \$880 billion of deficit reductions over 10 years in areas it oversees.

Most probably it would come from slowing Medicaid’s growth. Options for Congress could include clamping down on the circular provider taxes or capping Medicaid payment rates so they don’t exceed Medicare rates.

The haggling among Republicans over how to proceed could last months. The GOP has thin majorities in the House and Senate.

Some Republican lawmakers want Medicaid left alone. It covers 79 million people, lots of Trump supporters rely on it and hospitals that benefit are a powerful lobby.

But the Trump administration will have trouble hitting its stated deficit-cutting goals without targeting Medicaid. Even if Congress trims funding only a little, the Trump administration could find creative ways to slash actual spending later.

In UHS and the Miller family, Republican budget hawks may have a poster child to make their case that the provider-tax system needs an overhaul. If they succeed, investors in UHS and other large hospital operators could be in a world of hurt.

—Jonathan Weil

## MARKETS



STREETWISE | By James Mackintosh

## The Cost of Trump’s Economic Goals

Explaining what President Trump really wants has become a thriving industry in its own right, often proved wrong as soon as it is published. Two things are clear about his tariff policy, however: He wants a lower trade deficit and he wants investment to rebuild U.S. manufacturing. True believers who think he might achieve those goals should think through what else has to happen as a result.

The starting point is the balance of payments, the broadest measure of trade and investment in and out of the economy. Its two sides have to balance: the current account, which tallies up trade flows and some other stuff not worth getting into; and the capital and financial accounts, which measure capital and money flowing in and out to buy things such as stocks and bonds and investments in factories.

For years, Americans have imported way more than they exported, thus the trade deficit in the current-account part of the equation. For the balance of payments to balance, there needs to be a corresponding inflow of capital. That has largely come from foreigners buying assets, most prominently stocks and government debt in the form of Treasuries.

Trump’s plan will disrupt that dynamic. Smaller trade deficits mean smaller inflows of capital.

Trump’s obsession is the goods deficit—and there are two ways it can come down.

The first is that the overall goods-and-services deficit remains unchanged, but services—about which Trump doesn’t seem to care and in which the U.S. runs a surplus—are sacrificed for manufacturing. This, though,

would need shifts in domestic tax and regulation.

The second way for the goods deficit to shrink is to reduce the overall trade deficit. That will mean less foreign money coming in. Combine that with more investment in manufacturing—because imported goods are made less competitive by tariffs—and it will mean America has to provide more of the savings to finance new assembly plants, clean rooms and sweatshops.

But domestically things have to balance, too. More savings means less consumption. The flip side of America relying on foreign savings all these years is that it has been able to consume far more.

At the risk of joining the Trumpsplainers, I’ve long thought of Trump as focused on people as workers rather than as consumers.

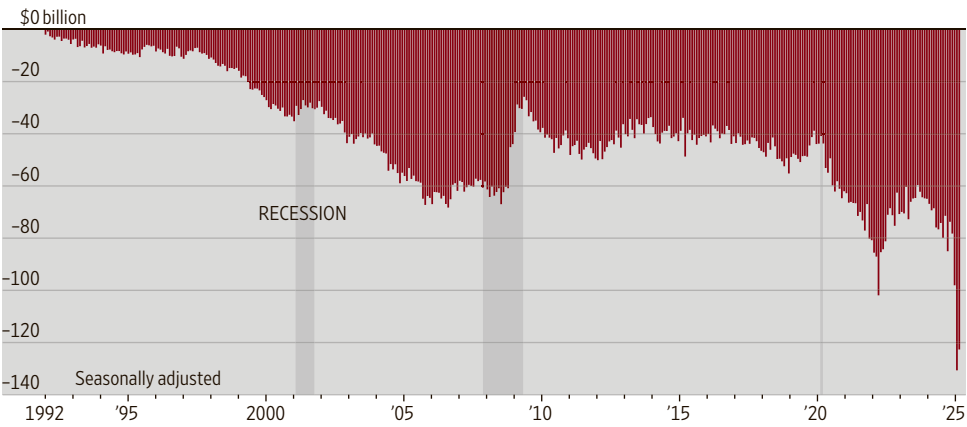
The existing system is focused on delivering stuff to satisfy consumer wants, and let jobs fall where they will, even if that is outside the U.S., rather than delivering jobs and supplying only the stuff that ends up being produced.

Markets are trying to figure out the implications of upending this system. Here are four:

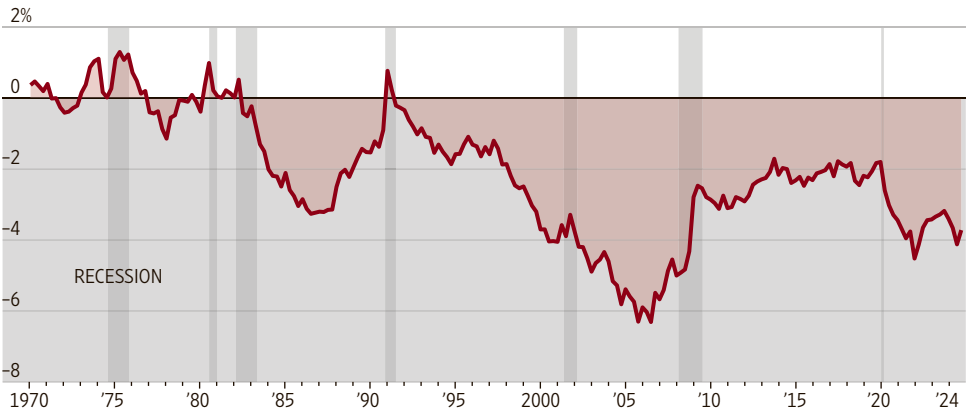
**More-expensive stuff.** and less choice of stuff. Increasing saving means reducing consumption. The tariffs amount to the largest tax increase in decades, which counts as government “saving”—as well as pushing up the price for almost everything imported.

**Higher interest rates.** The capital inflows that offset the trade deficit help fund a big chunk of federal government borrowing. Slash the trade deficit and the net inflow of

### U.S. trade in goods-and-services deficit, nominal dollars, monthly



### U.S. current-account deficit as share of GDP, quarterly



Sources: U.S. Census Bureau via St. Louis Fed (trade); U.S. Bureau of Economic Analysis via St. Louis Fed (current-account deficit)

foreign money dries up. Bond yields will need to rise to attract domestic savers to buy Treasuries instead of stocks or corporate bonds, which will hit share prices and raise the cost of borrowing for companies.

**Lower stock prices.** Only a small chunk of foreign investments goes into building factories. If there were more foreign direct investment, it could finance at least some of the reconstruction of manufacturing. But we’ve assumed Trump succeeds in shrinking the trade and current-account deficits, so there will be less foreign money coming in. So more

foreign factory building means less foreign buying of stocks and bonds, so lower stock prices.

**A weaker dollar.** In economic theory the dollar is the variable that moves when savings and investment don’t balance. If the U.S. saves too little to cover its investment, the dollar should weaken to make U.S. investments more attractive to foreigners.

In practice the dollar has been in demand for foreign reserves and for use in trade, as well as a safe place to stash the world’s savings. All three are now being questioned.

Questions over the inde-

pendence of the Federal Reserve, and Trump’s personal attacks on its chairman, Jerome Powell, have the potential to scare off buyers of both the dollar and Treasuries.

The dollar also should be weaker because the U.S. economy should be weaker.

America spent a century at the forefront of technology, gradually abandoning low-wage, low-productivity industries in favor of increasingly complex production and high-value-added services such as chip design.

Bringing back those low-

productivity jobs is possible if tariffs are high enough, but will reduce America’s economic lead over the rest of the world. Does America really want to bring back sewing jobs from Bangladesh or Cambodia? It looks like it, with the “reciprocal” tariffs set for those nations at 37% and 49%, respectively. Lower productivity, though, should mean a weaker currency, all else equal.

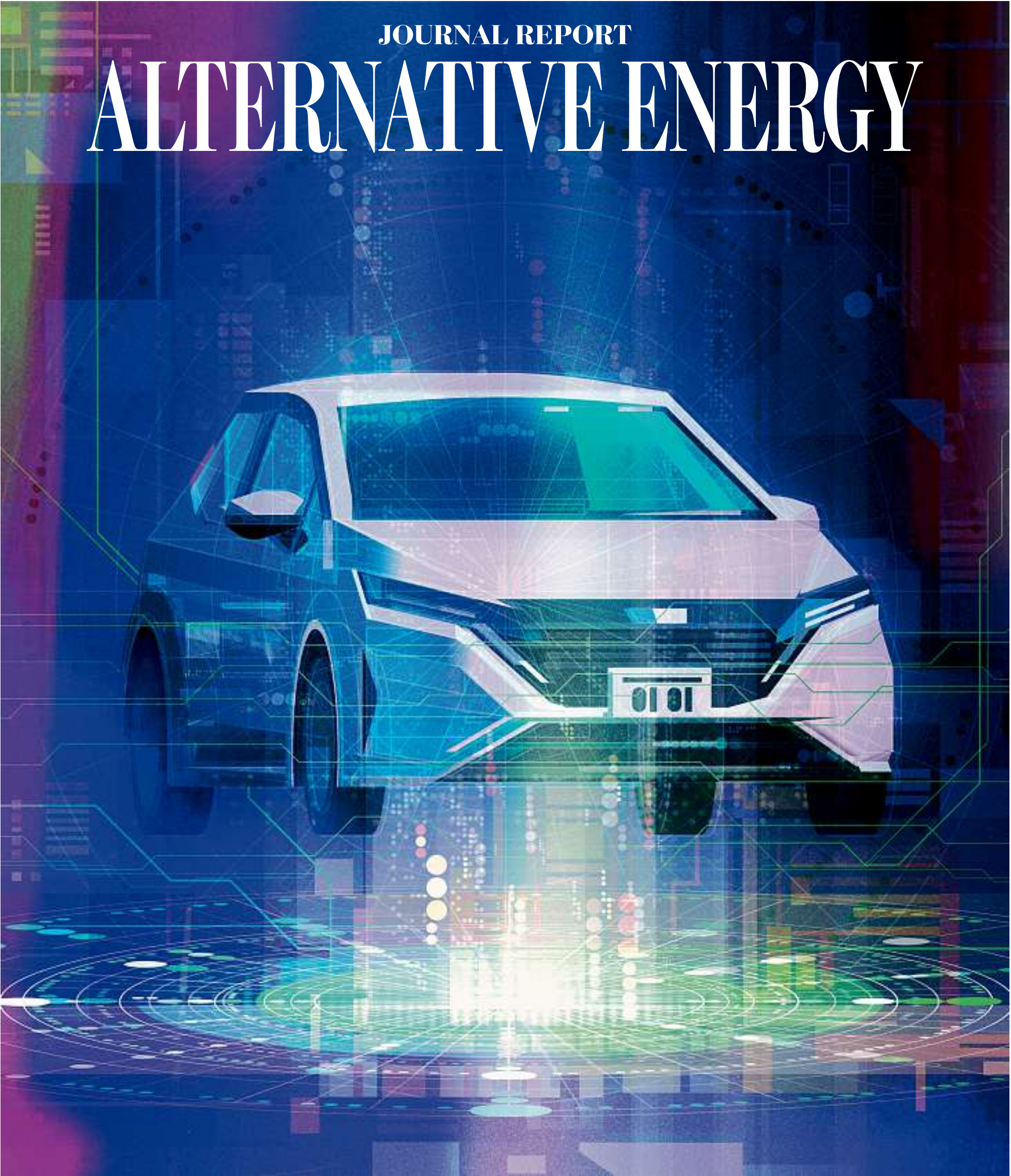
This simplifies somewhat. Interest rates, the dollar and the economy interact, so we get higher rates for any given level of the dollar and growth—with complex interactions as all three move.

Before Trump’s economists start spluttering into their coffee, a really good outcome also is imaginable—but seems to me highly unlikely. The deficit could drop because exports rise as demand in other countries rises, especially countries that have suppressed demand. Trump’s extreme approach on tariffs and defense shocked Germany into abandoning austerity, and part of China’s response to tariffs is to try to boost domestic consumption.

I’m skeptical that Germany and China will quickly welcome consumerism. But even if they did, there is no reason to think they would want to buy American manufactured goods, rather than the U.S.’s highly competitive services, energy and agricultural exports—and that is if they were willing to buy anything American at all, after the insults and unreliability of the past three months.

I doubt that Trump’s tariffs will bring much manufacturing back to America. If they do, investors and consumers will suffer.





JAMES GILLEARD

# I Really Like Electric Cars.

## *These Technologies Would Make Me Love Them.*

I’ve been buying EVs for more than 10 years. But **innovations** on the horizon will make a big difference to me—and lots of other skeptical consumers.

BY AMY MYERS JAFFE

**A**S AN ENERGY and sustainability expert who has been driving EVs since 2014, I am committed to electric cars. But I won’t lie: My experience as an early adopter hasn’t been a totally smooth one. In fact, a few events almost made me swear off electric cars. My first-generation hybrid came with lighter tires to boost

fuel efficiency—but they were also pothole sensitive and kept blowing out. When I bought a second-generation EV, everything was going great until a faulty software update made the car sporadically reject level-two chargers, with an annoying “car isn’t connected” message. Then there was the day this year that was so cold the car wouldn’t charge.

*Please turn to page R4*

Amy Myers Jaffe is director of the Energy, Climate Justice and Sustainability Lab and a research professor at New York University’s School of Professional Studies, and author of “Energy’s Digital Future.” She can be reached at [reports@wsj.com](mailto:reports@wsj.com).

## How Trump’s Policies Will Slow—but Not Kill—A Cleaner-Energy Transition

The administration has pledged to cut support for renewable-energy initiatives. Here’s how it could shake out.

BY KATHERINE BLUNT

**T**HE TRUMP administration has pledged to boost the oil-and-gas industry, in part by cutting support for renewable-energy and emissions-reduction initiatives. A transition to cleaner energy will continue regardless, many executives and analysts say, but legis-

lative uncertainty and policy changes are expected to slow the changeover.

Here’s what to know about how the Trump administration’s decisions could affect the industry:

**1. Tariffs are expected to drive up project costs**  
President Trump’s sweeping tar-

iffs are expected to affect almost every material needed to build wind and solar farms, battery-storage facilities, electric vehicles and other projects meant to reduce carbon emissions. Levies on imported steel, aluminum, solar modules and minerals will drive up construction and development costs.

“It’s premature to give an assessment of the policy, but there’s obviously deep concern, not just within our industry, but across the U.S. economy,” says Frank Macchiarola, chief advocacy officer for the American Clean Power Association trade group.

The U.S. is largely reliant on Asia for production of solar modules, the building blocks of large-scale solar farms as well as rooftop solar installations. The U.S. imported roughly 55 gigawatts

*Please turn to page R2*

## Inside

### Tailpipe Invention

Can the heat from your car’s tailpipe be turned into electricity? There’s a prototype. **R2**



### Plug-In Home Batteries

Prices have dropped for appliance batteries used in power outages. Here are four to consider. **R4**

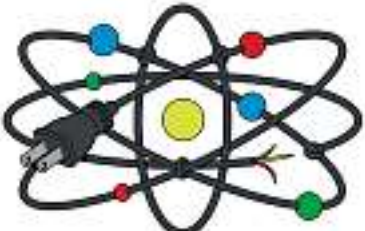
### ‘Waste Heat’

We flush a lot of energy down the drain (hot showers and sewers). Can more of it be recaptured? **R5**



### Energy-Efficient Homes

Straw-bale insulation...solar heat...hens for eggs: What conservation experts do on their own homes. **R6**



**PLUS:** A quiz on nuclear history. [WSJ.com/alternativeenergyreport](https://www.wsj.com/alternativeenergyreport)



# The Heat Coming Out of Your Car’s Tailpipe? Some Can Be Turned Into Electricity

Researchers have created a prototype that can be fitted onto the tailpipes of cars, helicopters and unmanned aerial vehicles

BY LISA WARD

ABOUT THREE-quarters of the energy produced by a typical combustion engine is lost through heat and released through a car’s exhaust system. But what if some of that heat could be captured and turned into electricity?

In a recent study, researchers created a prototype of a device, called a thermoelectric generator, designed to do just that. When inserted into an exhaust pipe, the device can turn waste heat into a continuous source of electricity for vehicles such as cars, helicopters and unmanned aerial vehicles, making them more efficient, the researchers say.

Converting even a fraction of the waste heat produced by such vehicles into electricity could reduce fuel consumption and battery usage, they say, leading to lower carbon-dioxide emissions.

The study was partly funded by the U.S. Army with unmanned aerial vehicles in mind, but the researchers believe the device has

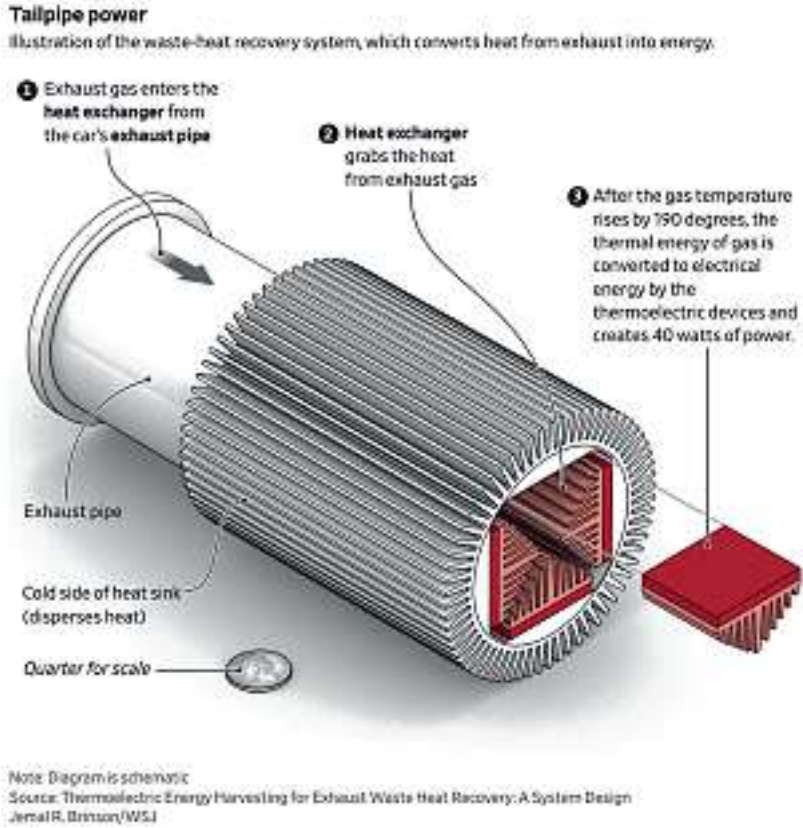
the potential to be used much more broadly. That’s because the thermoelectric generator can be installed into tailpipes of different vehicles without changing how they operate, says Bed Poudel, a research professor at Penn State’s College of Earth and Material Sciences and one of the study’s authors.

The Army says the next step is to independently evaluate the prototype in its own lab, which it plans to do.

### How it works

Thermoelectric generators rely on temperature differences to produce electricity. They have a hot and cold side, and when heat is transferred from one side to the other, electrical power is generated. Bigger temperature changes generate more power.

These generators have been around for a while, used in industrial settings and to provide power on space missions, among other things. But previous iterations weighed a lot and often relied on coolant or water, making them difficult to use in aerial vehicles and cumbersome for automobiles. The researchers say their prototype eliminates the need for coolant. Instead they use the temperature difference within the exhaust pipe and the outside air. “Objects moving at high speeds have natural cool surfaces that don’t need addi-



tional cooling,” says Poudel.

The researchers tested their prototype in a lab and found that with a change in temperature of 190 degrees, their thermoelectric generator produced 40 watts of power, enough to charge a phone or a few LED lights and act as a supplementary source of energy for a vehicle’s battery.

Perhaps more critically, it converted about 15% of exhaust heat

into electricity. Other commercial thermoelectric generators typically convert 6% to 8% of heat into additional electricity, says Wenjie Li, an associate research professor at Penn State’s College of Earth and Material Sciences who also was part of the study.

### Flying high

When the researchers mathemat-

ically modeled how the prototype might work outside the lab, they found that additional power could be generated if a vehicle was flying at high elevations because temperatures would drop with altitude and higher air-flow rates would cool the outside surface of the exhaust pipe.

Their model predicted that placing their thermoelectric generator in a helicopter’s exhaust pipe would generate about 146 watts of continuous power. Placing it in a car’s exhaust would provide 56 watts. The researchers estimate that 25% of the battery in either a car or helicopter could be powered through the thermoelectric generator, leading to greater fuel economy.

“Keep in mind that the thermoelectric generator is providing continuous electricity,” says Christopher Rahn, the J. Lee Everett professor of mechanical engineering at Penn State’s College of Engineering and another co-author. Finding sources of electricity is likely to become increasingly important as vehicles rely more on screens, sensors and computerized systems, he says.

Lisa Ward is a writer in Vermont. She can be reached at [reports@wsj.com](mailto:reports@wsj.com).



FirstEnergy’s coal-fired Harrison Power Station in Haywood, W.Va.

## How U.S. Policies On Cleaner Energy Could Shake Out

Continued from page R1

of solar modules last year, according to analysts at investment bank Jefferies. Panel inventories in the U.S. are relatively strong, the analysts say, but project delays and cost increases are likely imminent as developers draw from those stockpiles.

### 2. Parts of the Inflation Reduction Act are at risk

The Inflation Reduction Act, the Biden administration’s signature climate bill, contains numerous clean-energy tax credits. Trump has talked about repealing the legislation, though executives and analysts expect him to take a more nuanced

approach to dismantling parts of it. “Thanks to President Trump, America is leading the way in lowering costs by removing red tape and unleashing affordable, abundant and reliable American energy,” an Energy Department spokesperson says.

The Inflation Reduction Act includes tax credits for wind, solar and battery projects, as well as electric vehicles, clean fuel production and energy-efficiency initiatives. The credits have helped spur more than \$100 billion in manufacturing investments, many of which have benefited Republican-voting states.

Last month, 21 House Republicans wrote a public letter in support of preserving the tax credits, saying that they have helped to increase domestic manufacturing and keep utility costs down.

“Both our constituencies and the energy industry alike remain concerned about disruptive changes to our nation’s energy tax structure,” they wrote.

### 3. Development of wind, solar and battery storage may slow

Electricity demand in the U.S. is ex-

pected to surge after decades of remaining mostly flat. A study last month by S&P Global Commodity Insights found demand is likely to increase by 35% to 50% by 2040 as a result of domestic manufacturing growth, data-center development, electric-vehicle adoption and other efforts to phase out the use of fossil fuels.

The push to develop AI has turbocharged demand projections as tech companies race to invest billions of dollars in power-hungry data centers. Some centers consume as much power as midsize cities, setting off a scramble among power and utility companies to figure out how to accommodate them. While some utilities have incorporated more gas into their planning than before, they still expect alternative energy will be needed to meet the demand.

Tech companies, meanwhile, have pledged to continue the clean-energy push. Each of the largest tech companies has pledged to substantially reduce its emissions in the coming years, in part by working with power producers to speed up the development of wind and solar farms and other forms of

clean power.

“We’re still seeing a lot of new clean-energy generation coming online, and we very much expect that to continue,” says David Grumhaus, president and chief investment officer of Duff & Phelps Investment Management. “That investment just isn’t going to shut off.”

### 4. Electric-vehicle adoption faces roadblocks

The Trump administration earlier this year halted federal funding for expanding the nation’s network of electric-vehicle charging stations. The administration suspended approval of state plans for spending about \$5 billion in grants to add EV fast chargers along highway exits, meant to help drivers feel more comfortable driving long distances without fear of their batteries running out of juice.

Congress had approved the funding as part of the 2021 Infrastructure Investment and Jobs Act.

Tariffs are also expected to drive up the cost of electric vehicles, which are already more expensive than gasoline-powered ones. The U.S. is largely reliant on China for

imports of lithium-ion batteries used to power electric vehicles and to stabilize the power grid as more wind and solar projects come on line. Many U.S. automakers have been pulling back on plans to sell new EVs.

“The EV market is extremely volatile and uncertain,” says Vanessa Miller, a litigation partner with Foley & Lardner LLP who leads the firm’s national automotive team. “We’re trying to protect U.S. jobs, but we’re going to make vehicles cost prohibitive at the same time.”

### 5. Fewer power plants may close if Trump repeals legislation

Coal-fired power plants in the U.S. have been under siege for years as wind and solar farms, as well as gas-fired plants, became cheaper sources of electricity. Dozens of coal plants have closed in recent years. Utility companies and power producers plan to retire 8.1 gigawatts of coal-fired capacity this year, or 4.7% of the total U.S. coal fleet, according to the Energy Information Administration.

The Biden administration last year implemented rules requiring coal-plant operators to significantly reduce emissions and pollutants with carbon capture and other technologies. The Environmental Protection Agency this year said it would reconsider aspects of those rules, and established an email address for companies seeking certain exemptions.

Trump this month signed an executive order meant to boost support for the U.S. coal industry, in part by lifting certain barriers to mining it and using it to generate power.

While Trump’s changes aren’t expected to lead to the building of new coal plants, they will likely slow the closures. A study by Rhodium Group found that a rollback of the Biden-era rules, combined with a repeal of the Inflation Reduction Act, could more than halve coal-plant retirements by 2035.

“The biggest tool in the toolbox is just saying, ‘OK, you don’t actually need to comply with those regulations,’” says Ben King, who co-wrote the study. “The other thing you can do is make alternatives to coal more expensive, and that’s where [the Inflation Reduction Act] tax credits come into this.”

Katherine Blunt is a Wall Street Journal renewable-energy and utilities reporter in San Francisco. Email her at [katherine.blunt@wsj.com](mailto:katherine.blunt@wsj.com).



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Rechargeable Batteries to Keep Appliances Running: A Look at Some Options

Prices have dropped for batteries that can be used in case of power outages. Here are four to consider.

By Heidi Mitchell

IT HAS BECOME much less expensive in recent years for consumers to protect their homes against power outages. According to research company Statista, the price of rechargeable lithium-ion batteries—the kind that can keep home appliances and electronic devices humming in a blackout—dropped 20% last year and was down 39% over the past five years.

“Innovations in chemistry and materials science have resulted in improvements in batteries” and are the main driver of lower costs, says Jessika Trancik, a professor at the Institute for Data, Systems and Society at the

Massachusetts Institute of Technology.

There are many options for consumers, from whole-home power stations that last for days to smaller battery packs that can get critical devices through a typical 5½-hour blackout. Here’s a look at some of them.

**EcoFlow Delta Pro 3**

The EcoFlow Delta Pro 3 costs \$3,699 (currently on sale on the company website for \$2,799) and is eligible for a 30% rebate under the Inflation Reduction Act. It provides 4 kilowatt-hours of power when fully charged. That’s enough to power essentials like a refrigerator or a sump pump, along with smaller devices such as a laptop, a cellphone and a fan, for about three days of typi-

cal use. Its capacity can be doubled or tripled with extra battery packs, for \$2,599 each.

The Delta Pro 3 is about the size of a desktop computer and weighs 100 pounds, but it has wheels and a telescoping handle that make it easy to move from room to room. It takes about an hour to charge. And it can be plugged into EcoFlow’s foldable solar 220-watt panel, starting at \$399, for recharging.

**BioLite Backup Complete**

The BioLite Backup Complete is a less bulky alternative, at \$2,999 with 3 kWh of capacity. It is eligible for the Inflation Reduction Act rebate. The Backup made its debut at the CES consumer-electronics trade show in January, and the company plans to start shipping batteries this fall. The unit is 2.8 inches thick per battery pack, making it relatively easy to store. The Complete package arrives with two battery packs and weighs about 75 pounds. It charges in 2½ hours and can power a refrigerator, microwave, lamp, smartphone and laptop for 30 to 60 hours.



Innovations That Could Make EVs Much More Attractive

Continued from page R1

Sometimes I run into hassles that don’t have anything to do with my own vehicle. Occasionally, a public charger’s credit-card payment system won’t work. On longer road trips, I might plan a recharging stop only to find that multiple chargers there are out of service.

Even with these challenges, I see innovations on the horizon that excite me and ensure I stick with EVs. Some advancements are already here for the next generation of cars or are coming out in the next year or two. Others remain a few years away. But when and if these features get here, they could make a big difference for most EV owners.

**Fast-charging, long-range batteries**

There is a race on to build better EV batteries. These projects use either cheaper, less-rare materials; battery systems that deliver more power, are lighter weight

or safer; or all of the above.

From where I sit, the real promise is the crop of solid-state batteries potentially coming to market in the next two or three years. These batteries could enable faster, safer, more-weather-proof charging. Based on test data I’ve seen, solid-state batteries have can get 600 to 700 miles per single charge, instead of the current 250 to 400 or so. That is farther than most traditional cars can go on a single tank of gas.

For me, and a lot of other people, a 600-mile range would be a game changer. Going by tests others have done, I could go from my house to any city in the Northeast and back without charging, even if there were unexpected (or expected) traffic.

Basically, solid-state batteries have greater energy density than current ones—in other words, more energy in a smaller package—and the battery isn’t as heavy as today’s, so it doesn’t eat up as much energy per mile. Carmakers have been competing to get solid-state battery technology off the ground for years, but Japan recently made a show of confidence in those efforts: The nation announced it was breaking ground on a plant to produce lithium sulfide, the raw material for solid-state batteries.

From my conversations with people involved, it could mean the batteries will be in cars within the next two to three years, although there have been delays in the past.

China is also working on solid-state batteries. Meanwhile, the nation has introduced yet another new battery technology—one that combines a superfast charging time with extended range. If solid-state efforts don’t pan out, the new Chinese vehicle might present formidable competition to current electric cars in some markets.

This month, **BYD**, the Chinese battery and auto manufacturer, began selling its new Han L vehicle, which can travel 250 miles on a single charge. The cars use a new type of battery and more-powerful chargers, and have two charging ports as well: If drivers use both, the batteries can charge in an incredible five minutes.

That said, the power requirements of the chargers might be a deal killer for places that don’t have a very stable electric grid.

**Recharging without human help**

Better batteries aside, my real fantasy is a car that will charge itself with less, or no, intervention on my part. That isn’t as pie in the sky as it might seem. On the horizon are at least three ways I could get this wish: wireless e-roadways, solar panels on the car or automated driving that allows the car to take itself to a charging pad and back when I need it. None of these options are commercially feasible now, but all are in someone’s lab or test-pilot demonstration, perhaps arriving in the coming decade.

• **A road that delivers a recharge.** Imagine if key highways near your house had special lanes with charging coils embedded under the road that recharged your car as you drove.

For a lot of people, it would end range anxiety and frustration with charging stations overnight. It would also eliminate planning for where and when to charge on longer trips. Eventually, it would pave the way for automated vehicles to charge themselves.

The basic technology of wireless charging of devices is already established: Think of the charging pads you can use to charge your cellphone without plugging in a power cable. A number of places—including Detroit, China and Sweden—have even tried wireless charging road demonstration projects.

But the commercial feasibility of e-roads hasn’t been proved as cost effective, which means they may be a decade away. To date, a significant portion of energy gets lost as it is transferred from the road to the car. It is also costly to add the equipment that a car to needs receive electricity through a road. Moreover, for some locations, like the pothole-ridden Northeast, maintaining the charging equipment in road asphalt could be expensive. Still, to a technophile like me, e-roads seem like a technology that might find its time.

• **Cars that drive themselves to charging stations.** Picture this: An autonomous car could learn to drive itself to a charging pad, park on it—and then come back to its owner when summoned. No need for a human hand to plug it in. It isn’t as far-fetched



FROM TOP: ECOFLOW; BIOLITE  
L-R: STEPHANE MAHE/REUTERS; PAUL SANCTA/ASSOCIATED PRESS



# We’re Flushing a Lot of Energy Down the Drain. Can We Capture It?

Wastewater heat-recovery systems are gaining momentum, as cities and building managers look to capture the energy from hot water

BY JACKIE SNOW

IT TOOK SEVEN years, but Jocelyn Hittle finally got what she wanted: a heating and cooling system fueled by wastewater at Colorado State University’s new Spur campus in Denver.

When Hittle finally moved into one of the buildings served by this system, the most surprising thing was how utterly unremarkable it felt.

No unusual odors. No strange noises. Just the same comfortable indoor climate she’d experienced in conventional buildings throughout her career.

“As an occupant of the building, everything feels exactly the same,” says Hittle, associate vice president at CSU for special projects and the Spur campus, a three-building complex at the National Western Center in Denver. “The only difference is that we get to tell that story and have our visitors understand that we’re doing something new and how it helps us to meet our energy and climate goals,” she says.

According to Energy Department estimates, Americans flush the equivalent of 350 billion kilowatt-hours of energy from hot water down their drains annually. A substantial portion of this thermal energy is recoverable, yet most municipalities and building managers have barely begun to tap this resource.

Wastewater heat-recovery systems, which have been widely adopted in Europe, are gaining momentum in North America as the technology becomes more accessible and cost-effective. Major projects in Denver; Vancouver, British Columbia; New York state; and King County, Wash., are either expanding existing systems or breaking new ground.

And dozens of smaller installations in individual buildings are happening around the country.

### Several benefits

The technology is relatively straightforward. The tempera-

hands with warm water, run the dishwasher—all of that thermal energy is literally just flowing down our drain and into the sewer system,” says Derek Pope, who oversees a wastewater heat-recovery system in Vancouver.

In the National Western Center’s system in Denver, the cooled wastewater flows into the city’s sewer system. That reduces the amount of water, time and energy that sewage treatment plants need to cool wastewater,

the continent in 2010.

### Olympian task

The system started with providing heat to nine buildings in the 2010 Winter Olympic Village and has grown to serve 47 buildings housing about 10,000 residents, along with a community center, science center, university and various office and retail spaces.

The utility now delivers 70% of its energy to customers using

developed extensive district heating networks decades ago, making it easier to integrate wastewater heat recovery. These centralized systems already have the pipes in place to circulate recovered heat throughout neighborhoods.

“In Sweden, any area that is densely populated is served with district heating,” says Semida Silveira, professor of energy systems planning at Cornell University. “District heating is a very efficient way of heating homes.”

In contrast, most U.S. buildings rely on individual heating systems, which can still recover heat but require new infrastructure for each building. That works, but not as efficiently as centralized solutions, according to Silveira.

However, these projects on a smaller scale are gaining traction, thanks to new systems sized specifically for multifamily properties. In New York state, construction of a wastewater heat-recovery system has begun at Whitney Young Manor, a multifamily housing development in Westchester County, and a system is being designed for the Amalgamated Housing Cooperative in the Bronx, both with support from Nysesda, the New York State Energy Research and Development Authority.

Nysesda is receiving inquiries from other building owners interested in the possibility of replicating the approach, says Michael Reed, the agency’s acting head of large buildings. But they need some questions answered before diving in.

“When there’s a new product, people are interested, but they’re also skeptical,” Reed says. “They need their preferred engineering consultant to really dig into the tech and come back to them and say, ‘Yes, this makes sense, and it can work.’”

Jackie Snow is a writer in Los Angeles. She can be reached at [reports@wsj.com](mailto:reports@wsj.com).



Vancouver, British Columbia’s False Creek Energy Centre.

ture of wastewater is typically between 54 and 67 degrees Fahrenheit, making it a source of thermal energy. Heat-pump systems can recover that energy and use it to help warm buildings and heat water, and to cool buildings in warm weather. That helps cut carbon emissions and boost energy efficiency by replacing fossil fuels.

“Sewage is warm because we take hot showers and wash our

which they do to avoid ecological damage when the treated water is released into waterways.

“The system saves the equivalent of six million passenger vehicles of emissions and five Olympic pools worth of water a year,” says Brad Buchanan, former planning director for the city of Denver, who now heads the National Western Center Authority.

Vancouver was an early adopter of this approach in North America. The city’s False Creek Neighbourhood Energy Utility began operating the first sewage heat-recovery system on

renewable energy, with wastewater heat recovery being the “primary” piece of it, Pope says. That has reduced carbon emissions by around 7,000 tons a year compared with natural gas, according to Pope.

### Gaining traction

Despite its efficiency and proven record, wastewater heat recovery remains relatively uncommon in the U.S., while European cities have widely implemented these systems. The difference comes down to infrastructure. Countries like Sweden and Denmark



as it sounds. Automation in some existing EVs allows the vehicles to park themselves. Two of my current cars can do that with a push of a button, if I drive to the lot or open curb. The twofold of having the car park itself and charge in the same operation would kill off my interest in a gasoline car that will never be able to fuel itself.

● **Charging with sunlight.** You think you use your car all the time, but on average, cars are parked for 95% of their useful life. My own car sits outside in a driveway most of the day, so I

can mostly keep it plugged in at home, much the way you set your cellphone on a charging pad when you aren’t using it. But if my EV had some kind of efficient solar panels on its roof, something that is close to feasible, it could always be charging and not just when it was plugged in. Emergency solar-charging kits that fit on your roof rack and integrate with a car’s charging port are already on the market. Some sellers say their solar emergency kits can get you a dozen miles or more, presumably helping you get to a full-on charging station.

Left, BYD’s new model travels 250 miles on a single charge. Above, an in-road wireless charging coil before being installed in Detroit.

Questions remain about whether there would ever be sufficient surface area to practically be useful. On the one hand, experts say that even if the average car’s entire surface were a solar panel, with today’s technology, the charge would likely not take you more than 25 to 30 miles, possibly less. On the other hand,

how many miles do people need regularly? When I had a hybrid electric car that had a 25-mile battery and a home charger, I found that was sufficient battery charge to keep me all electric almost every day (if I plugged in routinely after each local trip). Most Americans don’t travel more than 30 to 35 miles a day, so a solarized vehicle that could go 30 miles might be enough to be useful.

### Turning your car into a generator

Technology that is getting more popular lets your car supply power to your home if you want it—say, during a blackout—and allow it to send some energy back to your utility when the grid is under stress.

Already, some EVs on the market let you run an appliance or two on your car’s battery, simply by plugging into the car using a customized adapter. The next generation will make the car battery part of a home-management system: Instead of just plugging in individual appliances, these cars will be able to essentially act as a power supply store for your whole house.

A typical U.S. home uses around 30 kilowatt-hours of electricity a day. At this amount, a good EV battery could power a house for a couple of days. It could keep a refrigerator on for several months.

Eventually, cars will likely

have a capability that could turn the vehicle into a profit center instead of a drain on household finances. With this technology, I would be able to sell my utility some of the power from my car battery to help the grid when demand was highest (late afternoon and early evening). I would then be able to recharge the car during off-peak hours when the rates were lower, at nighttime and on weekends. Already, many utilities across the country are launching initiatives called virtual power-plant programs that work along those lines.

Whether payments programs will ever take off at a wide scale will depend on several details, such as regulatory frameworks and financial incentives. The big question is how much will utilities be willing to pay residential customers for the electricity stored in their cars.

Right now, the level of payments for batteries in many of the locations I have studied probably won’t deliver a quick payback on the hardware a car owner needs to make the program work (such as wires, connectors and adapters).

But, as these programs go mainstream, utilities may start offering customers more money as a lure. EV owners may also be more interested in these deals as battery capacity goes up: The more power your battery can store, the more likely that you would be willing to sell some of it to the utility.



JOURNAL REPORT | ALTERNATIVE ENERGY



# How Do Conservation Experts Make Their Own Homes More Energy Efficient?

BY WILLIAM BOSTON

**WHEN IT** comes to energy efficiency, the people who design buildings and promote conservation have all sorts of insights and advice. But how do they apply that advice to their own homes?

We asked architects, policy analysts and sustainability experts to show us how they walk the talk in their own lives. And they didn't disappoint, detailing the energy-saving strategies they've adopted at home—everything from solar retrofits to low-carbon building materials to passive design approaches such as extra insulation, shading and ventilation that aim to minimize upfront energy consumption.

Here is a closer look at some of their responses:

**A home that makes more than it takes**  
*Barry Alan Yoakum, partner and CEO of architecture-and-design firm Archimania*

**MY WIFE** and I decided in 2017 to invest in a world of energy efficiency when designing our new home near downtown Memphis. We began by employing as many passive design strategies as possible before resorting to more-expensive building systems and interventions. This approach helped pave the way for a faster return on investment.

The home is 2,798 square feet with four bedrooms and shared spaces such as the kitchen and living areas arranged around a courtyard. Sited on the Mississippi River with its high water table, it employs a geothermal system for heating and cooling. The home has a robust energy-efficient envelope, meaning there is little air exchange between the inside and outside, thanks to insulated panels and other barriers such as coated aluminum panels on the exterior that reflect the bright summer sun. Adjustable sun shades provide additional protection. Add five dozen solar panels and a large battery backup and you have more than a reasonable return on investment for many years to come, driven in part by selling back excess energy to the utility company.

The home, which we sold a few weeks ago for \$1.35 million, produced 13% more energy than consumed in its first year and 9% the following year. It was recognized as the first zero-energy and zero-carbon single-family home (or building) in the world, which means it consumes less energy than it produces and has little net CO<sub>2</sub> emissions.

From solar panels to airtight windows to geothermal systems and more, these experts are building and remodeling their homes with sustainability in mind

**Hens and a solar cooker**  
*Richard Heinberg, senior fellow at the Post Carbon Institute*

**WHEN MY** wife and I bought our house in Santa Rosa, Calif., 25 years ago, our first strategy was insulation. Next came solar panels and battery backup. Our solar hot-water heater uses almost no energy during sunnier months and is super-efficient the rest of the year. Skylights give us free daytime lighting.

Food is our most important daily energy source. We keep four hens for eggs—which aren't cheap because we don't skimp on animal care—and the chickens are endless sources of amusement and companionship. We have gardens for veggies and beauty, including fruit and nut trees, which provide lots of nutrition for little effort.

Our most recent additions are a heat pump for heating and cooling and an electric car; both are big money and energy savers.

Our most surprising success was a solar cooker—a simple device consisting of a glass-lidded box and reflective surfaces that cooks our summertime dinners outside with no fuel or electricity.

We use a quarter of the energy of a typical California household.

**A step-by-step approach**  
*John Grable, architect*

**OVER THE** past decade, our 67-year-old home in San Antonio has become a living experiment in energy efficiency—proof that transformation doesn't happen overnight, but step by step, layer by layer.

The most significant shift came with the installation of solar panels. The investment paid for itself in eight years, demonstrating how sustainability is as practical as it is essential. We added deep overhangs to shade our large glass walls, improved insulation and installed steel windows that open and close to provide natural cross ventilation of breezes and fresh air. The windows are made with low-E glass, designed to be resistant to both extreme hot and cold temperatures, mounted in steel frames, which are strong enough to hold large glass panes and are more durable than wood or plastic. The addition of ceiling fans, solar shades that block direct sunlight and LED lighting further reduce our footprint.

The simplest choices make a significant impact: a gas fireplace, efficient yet effortless, replaces mechanical heating on



colder days. A tankless water heater, low-flow fixtures and an insulated garage refine efficiency at every turn. We planted native trees to shade and cool the lot, and upgraded to a heating, ventilation and air-conditioning system that uses durable rigid ducts to prevent obstruction of air flow for better air quality.

The next step is a rainscreen system. This construction leaves a small gap between the home's exterior walls and the outside cladding to allow for drainage and ventilation. This prevents moisture buildup and penetration, protecting the building from rot and improving its ability to maintain comfortable temperatures with more-efficient energy use.

**Airtight seals**  
*Michelle Krochmal, owner of Filament Architecture Studio*

**WHEN I** renovated my home in Brooklyn a few years back, it was an opportunity to create a very energy-efficient space using so-called Passive House principles.

First, I opened up the walls and ceilings to add insulation to the entire townhouse with more than the typical amount required. This so-called super insulation pays for itself after a few years through less expensive utility bills. I also air-sealed the home to get rid of leaks, added triple-pane windows and created

energy-efficient air circulation with an energy-recovery-ventilator system consisting of a fan with ductwork that pushes stale air out and fresh filtered air in while preserving the temperature inside. I also added Energy Star appliances and often air-dry my family's clothes.

I installed window boxes at the front of the home, to create a little bit of shade in the summer and a tiny bit of extra insulation. The AC and heat system is a mini-split heat pump. Finally, I have a solar array on top of our roof deck that also doubles as a shade structure. This solar power provides more electricity than we use during most of the year.

To capture natural light, which is more energy efficient and a more pleasant space to live and work in, I strategically located skylights above the bathrooms and the stairs, and located our windows and doors to maximize light and views.

My family was most surprised by how effective extra insulation and tight seals on the windows and doors are. They can sit right up against the window and not feel a draft. They love it!

**Straw wall panels**  
*Chris Magwood, manager, carbon-free buildings, at Rocky Mountain Institute*

**AFTER 20** years of helping clients achieve energy efficiency

**From top left:** John Grable's home in San Antonio, Richard Heinberg's home in Santa Rosa, Calif., construction on Chris Magwood's home with straw-bale insulation, and the solar array/shade structure on Michelle Krochmal's home.

and climate-impact goals as home designers and builders, my partner and I decided to put everything we had learned into the creation of a new home for ourselves in Peterborough, Ontario.

We started by finding an urban infill lot—an undeveloped property within an urban area—for the house. This type of site helps prevent urban sprawl and makes use of existing infrastructure like sewer, water and electricity. We also knew that living in an area where transit, walking and biking are feasible is an effective way to reduce our long-term emissions.

Next came better-than-code levels of insulation, triple-pane windows and airtight construction powered by efficient, all-electric systems. With 5 kilowatts of photovoltaic panels on the roof and solar hot-water collectors shading our south windows, the house is very close to net zero in its energy consumption.

Equally important, we minimized the upfront emissions that arise from the production of building materials (the "embodied carbon" of a building) by using low-carbon and carbon-storing materials throughout the house, including waste-wood foundation blocks and cellulose insulation. Our upfront carbon footprint was 85% less than a typical home.

We used prefabricated straw wall panels, which are load-bearing wall structures that use a combination of light wood framing and straw-bale insulation with a lime-based stucco coating. Straw bales are a sturdy renewable material with high-insulation value. They also are naturally fire-retardant and more affordable than many conventional materials.

**All in on electric**  
*Graham Irwin, principal of Essential Habitat Architecture*

**ONE IMPORTANT** thing I've done in my home in Fairfax, Calif., is to move away from natural gas toward all-electric operation any time I'm replacing something or installing new. Almost everything is better, though I've yet to find good electric substitutions for grilling, fireplaces or fire pits.

I've found induction cooking to be outstanding—as responsive and more widely adjustable than gas, without the emissions. Heat-pump condensing dryers have come into their own. Not needing to worry about installing and cleaning a dryer vent is a huge benefit, and our clothes enjoy not being roasted, which is literally what a gas dryer does.

*William Boston is a writer in Berlin. He can be reached at reports@wsj.com.*